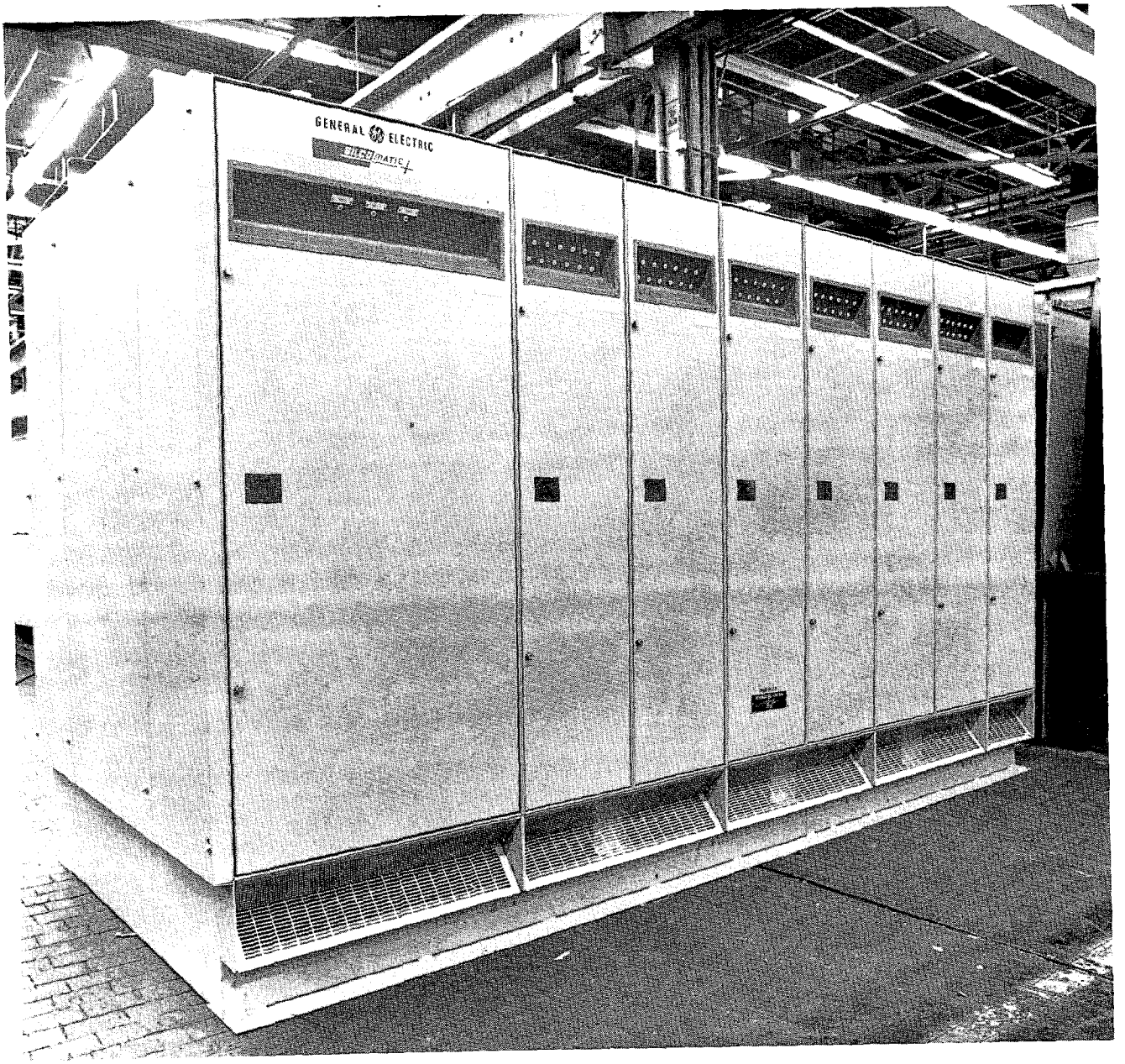
INSTRUCTIONS

GEK-36694

SILCOMATIC® PLUS THYRISTOR (SCR) POWER EQUIPMENTS

REVERSING AND NONREVERSING TYPES



These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC

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THYRISTOR (SCR) POWER CONVERTER EQUIPMENTS
REVERSING AND NON-REVERSING TYPES

INTRODUCTION

Although rectifier circuit technology on which these equipments are based is not new and has been treated at great length in many publications, application and design details for the thyristors are so new that little instructional information is thus-far available covering power conversion equipment in the range of ratings for industrial mill drives and auxiliaries. Therefore, it is the purpose of this instruction to present in as much detail as feasible, information which shoud be of assistance to purchasers and installers of these equipments.

These instructions apply to standard Silcomatic* Plus, 6-pulse or 12-pulse, reversing, or non-reversing-type power converters. Only where special purpose designs are involved or where large ratings require paralleling of converter line-ups should it be necessary to supplement or deviate from the standard design approach presented herein. In the case of single-direction converters, certain functions which are necessary for reversing converters are not normally required for single-direction drives. Thus, when installing a non-reversing drive, the reader should ascertain which portions of these instructions do apply by examination of the specific elementary diagram.

Since the instructions herein are universal for Silcomatic Plus they will constitute a Section of the composite converter instructions for each specific requisition. The over-all instruction book will include all references, identifications, ratings, settings and other useful data for the equipments under the subject requisition. A list of all necessary component elementary drawings and individual instructions will be included with the main elementary diagram.

These instructions do not purport to cover all details of variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

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GENERAL INSTRUCTIONS

INSTALLATION

Installation consists of unpacking, setting-up the apparatus, and making connections between the various assemblies. A few suggestions are given here; however, the actual methods used are dependent upon the local conditions existing at the time of installation.

RECEIVING, UNPACKING AND STORAGE

This equipment is assembled, tested, and packed with care to enable the purchaser to install and place it in operation with a minimum of time and labor. Immediately upon receiving the equipment, it should be carefully checked against the memorandum of shipment. If any parts are found damaged or missing, the purchaser should immediately present a claim to the transportation company and notify the nearest sales office of the General Electric Company.

It is good practice to place the equipment as near as possible to its permanent location before unpacking. The standard type of unpacking tools should be used. A nail puller and a slug-hammer can be especially useful. Careless unpacking methods will invariably result in parts being damaged or marred. Some parts, such as bolts and screws, are packed in special containers to keep them together. Since they may become separated, packing material should not be discarded until it is certain that all parts have been removed. Equipment should be thoroughly cleaned to remove particles of packing material or foreign substances which may have become lodged in, or between, any of the parts. Apparatus not immediately installed should be labeled and set aside in a clean, dry place of moderate temperature and protected from injury.

Refer to outline drawing for proper handling, moving and setting in place.

LOCATIONS AND CONNECTIONS

When equipments are installed, ample space should be provided around the various parts to permit easy inspection, adjustment, and repair. Whenever possible, all control devices are mounted at the factory; interconnections are made and leads are brought out to numbered terminals, making it necessary only to install the component pieces and connect the numbered terminals to the various pieces of apparatus.

When the equipment has been set up, it should be thoroughly inspected and checked to be sure that all connections are complete and tight, that the relays and instruments are in good mechanical condition, that they have been thoroughly cleaned, and that all of the movable parts work freely. All devices should make good contact and have clean contact surfaces.

MAINTENANCE

INSPECTION ROUTINE

A periodic routine of cleaning and inspection should be established. Such cleaning is necessary to prevent faulty operation of a device due to accumulation of dust and corrosion. Inspections should be more frequent if the demands on the equipment are exacting. Under normal conditions, the protective devices do not operate; therefore, these devices need careful inspection. If the equipment is in operation when being inspected and a short interruption of service can be permitted, the manual tripping of a protective device will afford an opportunity to observe the manner in which the equipment is functioning.

If at any time a device fails, the cause of the failure should be located and corrected before any remedy is applied and, if necessary, the part adjusted or replaced.

NOTE: When ordering parts, give a detailed description of the part and complete nameplate data on the device.

For specific maintenance information on the various devices, refer to the particular device instruction book.

CAUTION: UNDER NO CIRCUMSTANCES SHOULD AUTOMATICALLY CONTROLLED EQUIPMENT BE STARTED BY HAND UNTIL THE CAUSE OF THE FAILURE TO START AUTOMATICALLY IS DETERMINED AND CORRECTED.

RECORDS AND LOG BOOK

It is recommended that a regular inspection and maintenance program be employed, and that a log book be kept. This procedure will help detect malfunctioning equipment and determine the causes of the trouble. It is suggested that a form be made up and filled out at each regular inspection or when trouble is experienced. The form should contain the following data:

NORMAL OPERATION:

Operator
Date and Time
Converter DC Volts
Converter DC Amps
Maintenance Performed
Comments

ABNORMAL OPERATION – ADDITIONAL OBSERVATIONS:

Alarm Lamps
Tripped Breakers
Relay Targets and Positions
Circumstances at time of trouble, such as:

Starting Up
Shutting Down
Load
Test Operation
Disturbances
Abnormal Operations elsewhere but still associated with the Converter.

DEVICES

On Silcomatic Plus Converters, devices are designated in accordance with the industrial standards. The device nomenclature is similar to the system used in process-control elementaries and is explained in GET-3017 "Electrical Diagram Practices".

DETERMINATION OF PICK-UP AND DROP-OUT

The voltage or current at which a device will "pick-up" should be ascertained in the following manner: Connect the coil of the device across a proper source of voltage which can be varied from a value well below the operating range to the full operation voltage. Gradually increase the voltage across the coil (while reading the voltage or current applied as desired) until the device just operates and assumes its final energized position. The minimum voltage or current at which the device assumes the final energized position is the "pick-up" value of the device.

Similarly, determine the "drop-out" of the device by reducing the voltage from the full operating voltage until the device assumes the final de-energized position. The maximum voltage or current at which the device assumes the final de-energized position is the "drop-out" value of the device. The recommended "pick-up" voltage is given for a cold coil. The hot "pick-up" voltage will be slightly higher. Unless otherwise specified, a device with which an external resistor is used should be calibrated with the external resistor connected in series with the coil. The rating of the external resistor is given on the device nameplate.

EQUIPMENT CONSTRUCTION AND DESCRIPTION

These instructions will include the description of basically three different types of Silcomatic Plus Converters; the control and construction will vary accordingly:

1) 6-pulse auxiliary drives

2) 6-pulse main drives

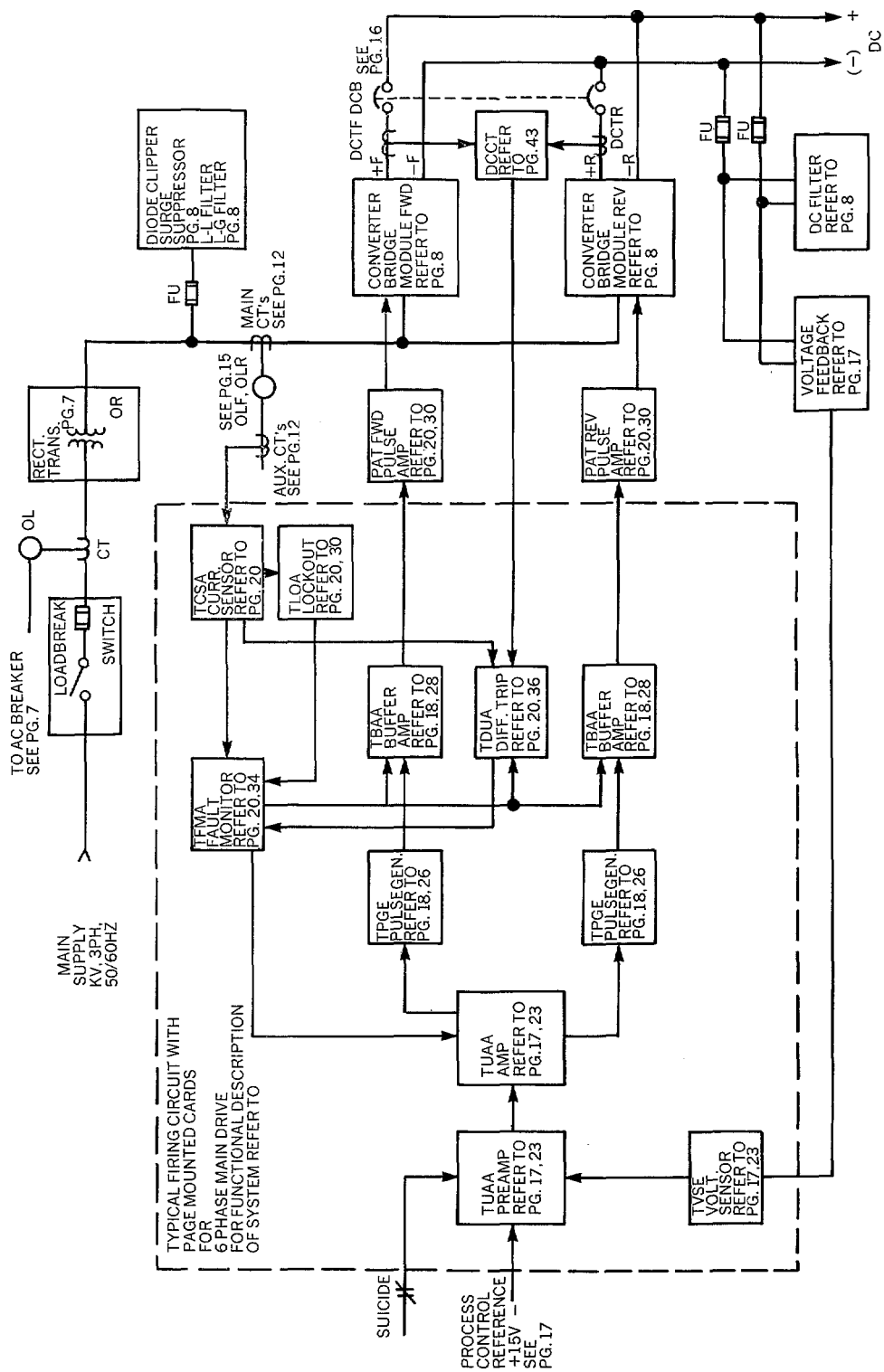
3) 12-pulse main drives.

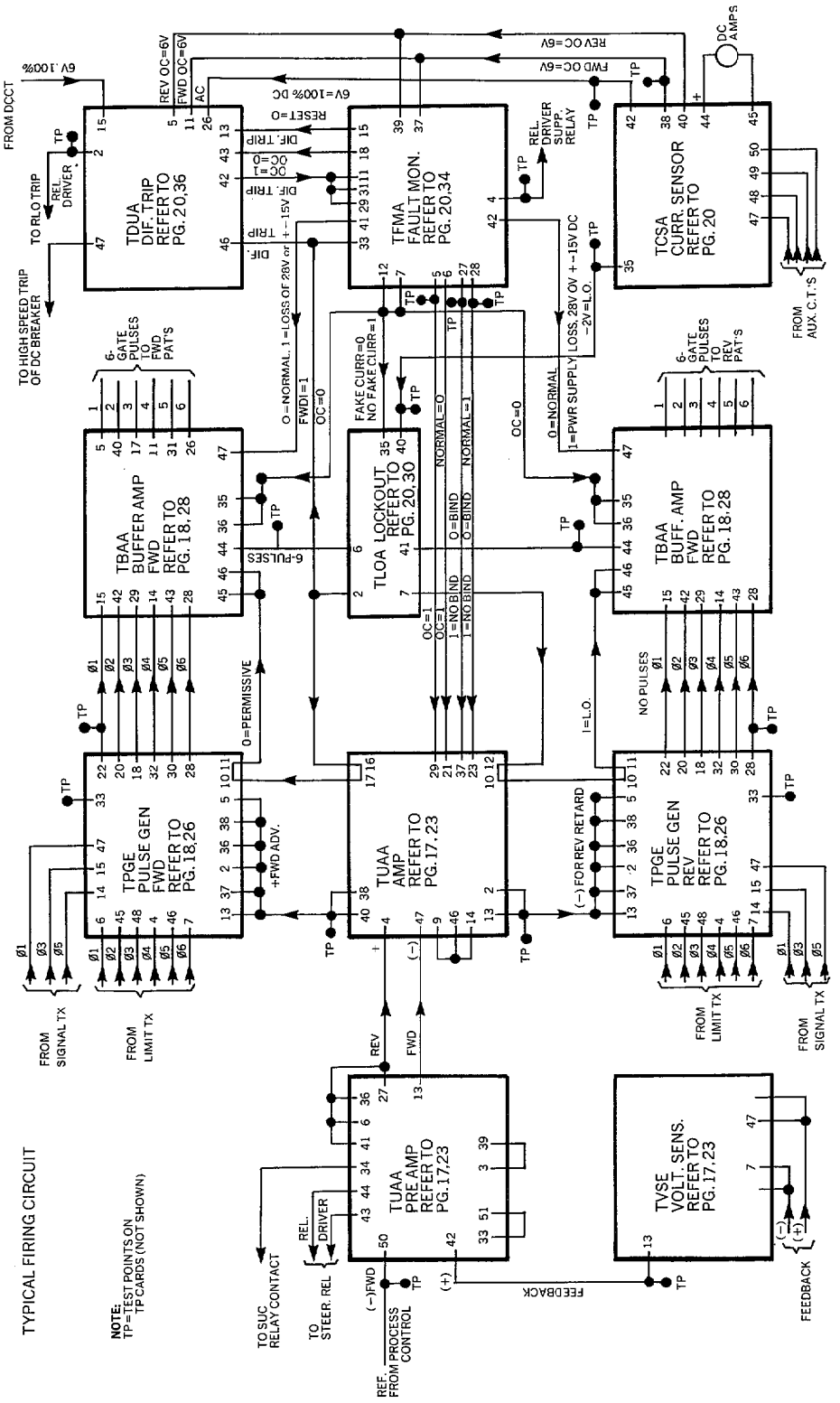
Every Silcomatic Plus power converter represents a complete rectifier system to convert a-c power to d-c power for a motor drive application, and includes:

1) AC power circuit breaker (when specified).

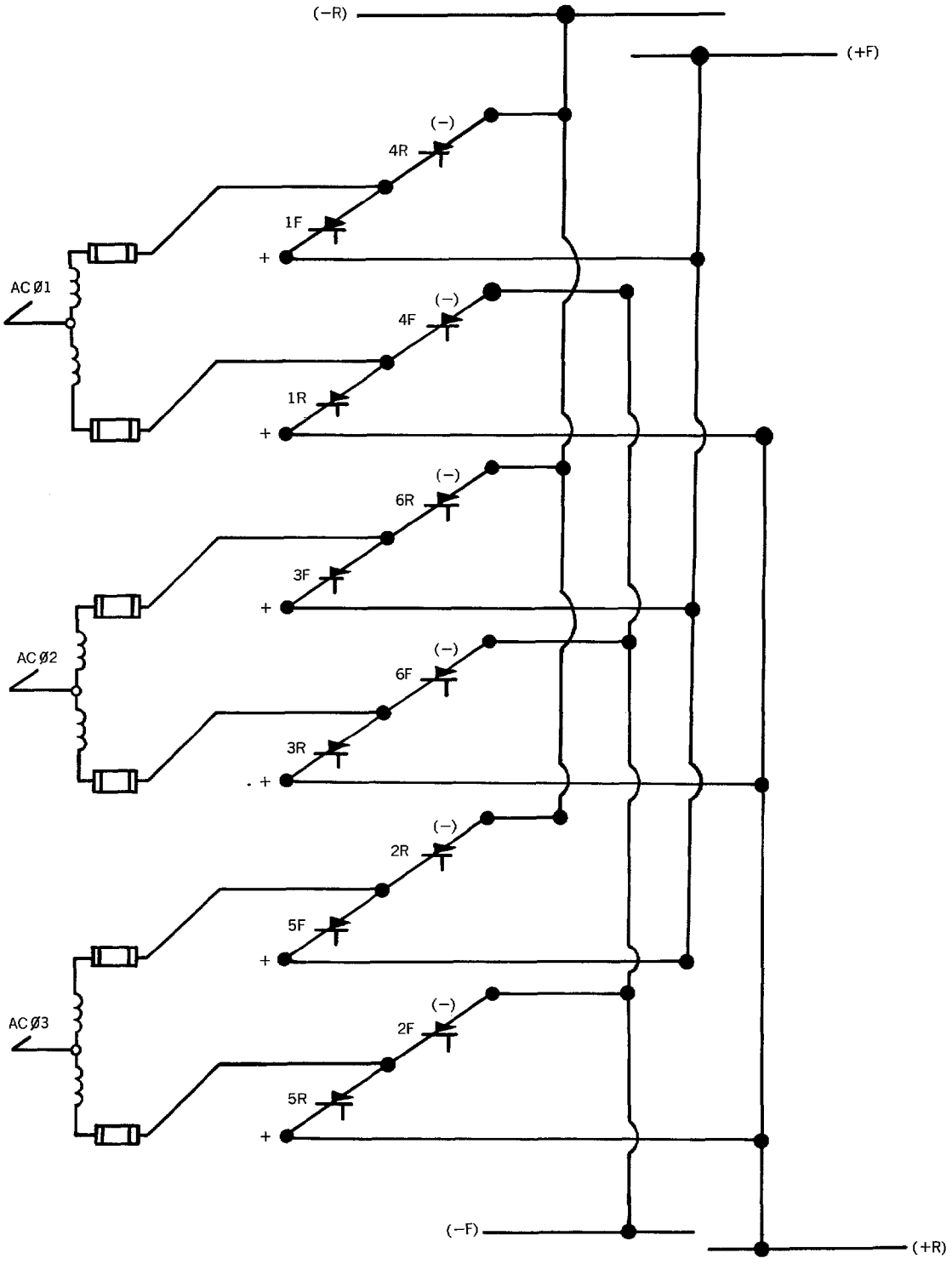
2) Converter transformer (when specified).

3) Filters and voltage surge suppressors.

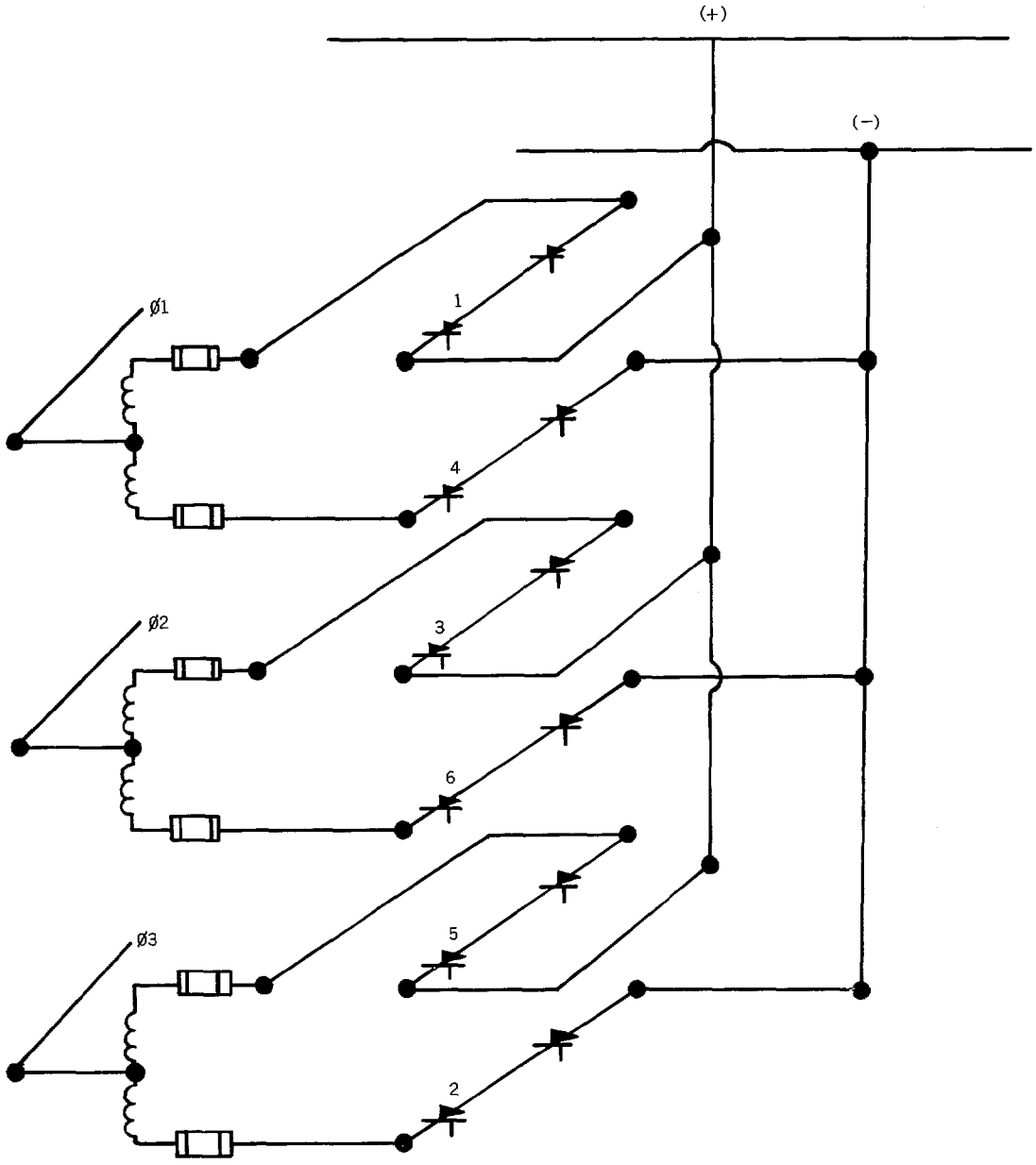

TYPICAL FIRING CIRCUIT WITH PAGE MOUNTED CARDS FOR 6 PHASE MAIN DRIVE FOR FUNCTIONAL DESCRIPTION OF SYSTEM REFER TO
MAIN SUPPLY KV, 3PH, 50/60HZ
TO AC BREAKER SEE PG.7
LOADBREAK SWITCH
OL
CT
RECT. TRANS. PG.7
OR
DIODE CLIPPER SURGE SUPPRESSOR PG.8 L-L FILTER L-G FILTER PG.8
FU
MAIN CT's SEE PG.12
SEE PG.15 OLF, OLR
AUX. CT's SEE PG.12
DCTF DCB SEE PG.16
+F
-F
+R
-R
DCTR
DCCT REFER TO PG.43
CONVERTER BRIDGE MODULE FWD REFER TO PG.8
CONVERTER BRIDGE MODULE REV REFER TO PG.8
FU
FU
DC FILTER REFER TO PG.8
VOLTAGE FEEDBACK REFER TO PG.17
+
(-) DC
PAT FWD PULSE AMP REFER TO PG.20,30
PAT REV PULSE AMP REFER TO PG.20,30
TCSA CURR. SENSOR REFER TO PG. 20
TLOA LOCKOUT REFER TO PG. 20, 30
TBAA BUFFER AMP REFER TO PG. 18,28
TDUA DIFF. TRIP REFER TO PG. 20,36
TBAA BUFFER AMP REFER TO PG. 18,28
TFMA FAULT MONITOR REFER TO PG. 20,34
TPGE PULSEGEN. REFER TO PG. 18,26
TPGE PULSEGEN. REFER TO PG. 18,26
TUAA AMP REFER TO PG.17,23
TUAA PREAMP REFER TO PG. 17,23
TVSE VOLT. SENSOR REFER TO PG. 17,23
SUICIDE
PROCESS CONTROL REFERENCE +15V SEE PG.17


TYPICAL FIRING CIRCUIT
NOTE:
TP=TEST POINTS ON TP CARDS (NOT SHOWN)
FROM DCCT
6V.100%
TO HIGH SPEED TRIP OF DC BREAKER
TO RLO TRIP
REL. DRIVER
TDUA DIF. TRIP REFER TO PG. 20,36
TFMA FAULT MON. REFER TO PG. 20,34
TCSA CURR. SENSOR REFER TO PG. 20
REV OC=6V
FWD OC=6V
AC
6V=100% DC
RESET=0
DIF. TRIP
OC=1 OC=0
REL. DRIVER SUPP. RELAY
DC AMPS
FROM AUX. C.T.'S
TBAA BUFFER AMP FWD REFER TO PG. 18.28
TBAA BUFF. AMP FWD REFER TO PG. 18.28
TLOA LOCKOUT REFER TO 41 PG. 20,30
6-GATE PULSES TO FWD PAT'S
6-GATE PULSES TO REV PAT'S
6-PULSES
0=NORMAL, 1=LOSS OF 28V or +-15V
FWDI=1
OC=0
FAKE CURR=0 NO FAKE CURR=1
NORMAL=0
NORMAL=1
O=BIND
1=NO BIND
OC=1
0=NORMAL
1=PWR SUPPLY LOSS, 28V OV +-15V DC
-2V=L.O.
0=PERMISSIVE
1=L.O.
NO PULSES
TPGE PULSE GEN FWD REFER TO PG. 18,26
TPGE PULSE GEN REV REFER TO PG. 18,26
+FWD ADV.
(-) FOR REV RETARD
TUAA AMP REFER TO PG. 17.23
TUAA PRE AMP REFER TO PG. 17,23
TVSE VOLT. SENS. REFER TO PG. 17.23
REV
FWD
(-)FWD
(+)
FEEDBACK
FEEDBACK
REL. DRIVER
TO SUC. RELAY CONTACT
TO STEER. REL
REF. FROM PROCESS CONTROL
FROM SIGNAL TX
FROM LIMIT TX
FROM LIMIT TX
FROM SIGNAL TX
TP

REVERSING BRIDGE ARRANGEMENT



BRIDGE ARRANGEMENT, SINGLE DIRECTION
WITH 2-THYRISTOR CELLS IN SERIES



4) Thyristor power modules.

5) DC breaker (when required).

6) AC current transformers.

7) Converter excitation controls.

8) Differential trip circuits (when applicable).

9) Protective devices.

Power converters are arranged in free-standing lineups consisting of any number of power supplies per lineup, subject to desired locations within the mill. Refer to the tabulation under Introduction for details of each lineup and individual power supply covered under these instructions. Following is a design and functional description of the basic elements making up the Silcomatic converter package.

AC CIRCUIT BREAKER

For those converters fed from a low-voltage substation source, a power circuit breaker may be mounted either in the converter lineup or remotely, and may be either manually or electrically operated. It is equipped with series overcurrent trips, shunt, and/or undervoltage trip devices. In addition to isolation, it serves only as protection for primary faults, and is not tripped by converter protective functions other than a-c undervoltage or differential trip (when a-c current is greater than d-c current).

Primary protection for converters on a-c sources above 600 volts is provided by close-coupled or remote metal-clad type switchgear, with a control or permissive switch at the Silcomatic. The functions are essentially the same as on low-voltage primaries.

CONVERTER TRANSFORMER

An isolating-type converter transformer is used and is especially designed to coordinate with the converter for proper circuit and regulation characteristics and to limit available fault currents within the converter capability. For 6-pulse converters, the transformers are normally wound either delta-wye or delta-delta, for converter circuits ASA-23 or ASA-25 respectively. For 12-pulse converters, ASA-31 is normally used.

For drives of a large KW rating, a self-cooled liquid-filled transformer is applied, either close-connected with the converter lineup, or separate with bus duct connections. Standard features include pressure, liquid temperature, and liquid level indicators with switch contacts, pressure relief device, and a sudden pressure device.

For drives below approximately 1000 KW, a dry-type transformer is usually provided, and is mounted within the converter line-up. A thermal switch for overtemperature detection is normally imbedded in each winding during manufacture. Access to the transformer is by means of removable bolted panels at front and back.

FILTER AND VOLTAGE SURGE SUPPRESSOR

Voltage Surge Protection Devices are applied to the a-c secondary busses at the point of entry into the converter.

Line-to-frame capacitors are applied to suppress voltage spikes due to switching surges by providing a path between secondary windings of the transformer and ground.

A diode bridge and surge suppressor unit is usually used for higher voltage ratings which provides a completely static function designed to clip line-to-line surges at a value between peak voltage level of the circuit and the PRV rating of the thyristors. A loading resistor maintains the circuit in conductive or "ready" condition. The suppressor triggers or "breaks over" at a pre-set level to clip even the highest rate-of-rise spikes, and then extinguishes the follow-up arc after the surge has passed. These devices are fused, with indicating lamps monitoring the CLF fuses.

Line-to-line a-c filters are designed to prevent voltage oscillations which would result from switching of the thyristors as commutation progresses from one phase leg to the next. This effect becomes more severe as phase retard increases to the point of maximum ripple (between 60 degree and 90 degree retard, depending on actual inductance of the load circuit), where the switching change of voltage approaches peak line-to-line a-c volts ($2E_{L-L}$). Without these filters, overshoot would not only stress the inverse blocking capability of the cells, but ringing could occur sufficiently to switch off the cells after firing.

D-c filters, calculated for load and inductance values in the d-c bus loop, are applied to damp the d-c system and to aid commutation when the d-c current is discontinuous between successive conduction periods.

No adjustments for these devices is provided or intended. They are calculated, fixed values.

THYRISTOR POWER MODULE

The function of converting a-c power to adjustable voltage d-c power is performed by the converter and is made possible by the ability of the thyristor to block voltage in both directions, while the degree of current flow in one direction is controlled by "gating" the thyristors at adjustable times on the a-c voltage wave. The converter is modular in arrangement of both the basic power circuits and the gating control circuits, thereby lending itself to the building-block principle for flexibility and versatility in meeting varied ratings and applications.

The basic converter circuit is a three-phase bridge consisting of six leg (L-N) modules each containing a reactor, current limiting fuse and a thyristor in series, and a commutating snubber network connected across the thyristor. A complete set of components comprising the three-phase bridge power circuit along with pulse amplifier and transformer, PAT, Module (used to distribute gating pulses to the thyristors), are contained in a single power module. These power modules are paralleled to the extent required to assure current-carrying capability for a given rating including overload and fault duty.

In each leg module, snubbers consisting of series reactors and RC shunt networks are applied to each cell and conducting path to protect the cells against switching surges, ringing, and excessive rates-of-rise of voltage and current during the commutation period. Without this protection, the cells would tend

to conduct from dv/dt, receive excessive di/dt, be overstressed by transient voltages, and tend to switch off after firing due to ringing. Failure would be inevitable, usually from dv/dt. Careful visual inspection of these circuits should be made during installation inspection, and especially following failure of any cell where the cause is unknown.

A leg module for the Silcomatic Plus consists of a cell assembly, a pulse amplifier, and transformer (PAT) module, a reactor, and a current limiting fuse. Six leg modules are combined to form a full wave bridge. Additional bridges may be paralleled to increase the load current rating of the Silcomatic Plus Converter.

A cell assembly consists of the thyristor(s), the heat sinks, and the RC snubber circuit combined into an integral unit. There are three different types of cell assemblies available for use in the Silcomatic Plus depending on voltage rating and application of the conveter. The first type consists of two series thyristors. Type two consists of a single thyristor. The third type consists of one forward and one reverse thyristor. Refer to page 9 and 10.

A cell assembly may be removed from the front of the unit by removing the two or three bolts (depending on type of cell assembly) and unplugging the gate lead connections at the PAT Module. Since the cell assembly is an integral unit, field repair should not be attempted. All cell assemblies considered to be repairable should be returned to the factory for component replacement.

Each power module of 6-pulse and 12-pulse main-drive-type construction will consist of twelve thyristors, two per phase leg. The first cell failure of a leg will be indicated by an indicating light on the front door of the power module. If the second cell in the same leg should fail, TFMA card will retard or inhibit gate pulses to the thyristors, the d-c breaker or contactor will trip, and a process fault stop will be initiated. Again, indication of the second cell failure will be by means of an indicating light.

DC CIRCUIT BREAKERS

On those power supplies where a d-c breaker is applied, either a stationary d-c, Type AK-2-50, two-pole, medium-high-speed breaker or a high-speed breaker, Type AG-7, developed specifically for Silcomatic protection is used. Generally, the Type AG-7 is applied where the KW rating and number of modules in parallel is substantial.

The AK-2-50 breaker may be manually or electrically operated, and is controlled by a control or permissive switch at the Silcomatic. It includes series overcurrent trips and shunt trip or undervoltage trip. In addition to the control switch and overcurrent trips, the breaker is tripped by relaying following rectifier pulse retardation.

The AG-7 breaker consists of one to six poles on a common frame. High-speed tripping is accomplished by the flux cancellation principle. During normal operation, each pole is maintained closed by holding coil flux from an approximately 200 VDC source, with trip coils being rated at 250 VDC. Upon detection of a fault condition, a static tripping circuit delivers a pulse of current from a capacitor bank which bucks out the holding coil flux almost instantaneously, and the breaker opens at high speed. Interruption of substantial currents is completed one cycle from inception of the fault. A semi-high-speed trip can be achieved by de-energizing the breaker holding coils.

Lamps are provided to indicate open and closed pole positions, and auxiliary switches are provided for these and relaying functions.

INTERPHASE, CURRENT-LIMITING, AND SMOOTHING REACTORS

Power converters which are either fully reversing or which produce regenerative power require separate converter packages connected opposite in polarity or "back-to-back". In some cases, such as when ratings of the two packages are unequal, or where large ratings involve paralleling of lineups, reactors may be applied in the DC busses.

On 12-pulse systems where the two 6-pulse converter outputs are displaced from each other by 30 degrees, the resulting difference in peak-to-peak ripple drives the 12th harmonic circulating current between the converters and is limited only by impedance of the bus and commutating paths. Interphase reactors, applied in the DC bus, are usually required to augment the reactance already designed into the system in order to reduce this ripple current to a value within design criteria for the equipment. Figure No. 1 illustrates the ripple voltage wave shapes.

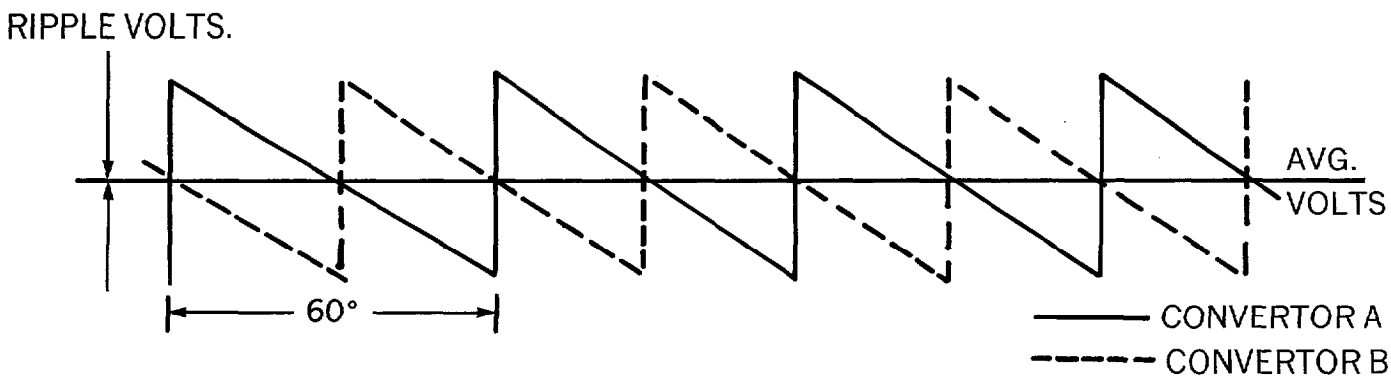


FIGURE #1 - RIPPLE VOLTAGE WAVE SHAPES

On systems where the "F" and "R" packages of a converter are unequally rated, but both powered from the same transformer winding, it is usually necessary to add reactance in the bus of the package with fewer paralleled power modules to limit fault current to within the surge rating of each individual power module. On 12-pulse systems this current-limiting function can usually be combined with the interphase reactor requirements into a single set of reactors.

On some 6-pulse applications, and applications with 12-pulse forward and 6-pulse reverse converters, a reactor, generally the reverse reactor, must fulfill the requirements of a current-limiting and/or a smoothing reactor. The smoothing reactor must sufficiently supplement the loop impedance to limit 6-pulse ripple current within commutation and heating limits dictated by motor design.

CURRENT TRANSFORMERS

Main a-c current transformers are applied on the a-c bus for regulation and protective sensing. The secondaries of these current transformers feed auxiliary current transformers as shown in Figure No. 2. The Auxiliary functions include:

Lockout

Load Balance (Between secondaries on 12-pulse rectifiers)

Detection of fault current

Overcurrent fault trip of AG-7 breakers

Differential fault trip of AG-7 breakers

LIST OF PROTECTIVE DEVICES AND FUNCTIONS

The following protective functions are provided as standard for the converter. Refer to the elementary and the section on test procedures (page 44) for details of operation.

ACW: Main AC power auxiliary relay

1) Provides a light indication
2) Permits closing of DC breaker
3) Permits pick-up of process control relay UVX

TTR: Transformer over-temperature switch (dry type transformer). Picks up auxiliary relay TRX.

26Q: Liquid temperature alarm contact and
71QL: Liquid level alarm contact. Both contacts pick up auxiliary relay TRX.

TRX: Transformer Auxiliary relay

1) Provides a white light indication
2) Causes a controlled stop (if required)

FSS: Cooling fan motor centrifugal switch

1) Provides an amber light indication
2) Drops out auxiliary relay FLX

FLX: Fan auxiliary relay

1) Initiates a process fault stop or controlled stop, depending on application.
2) Trips the DC breaker

SUC: Suicide relay

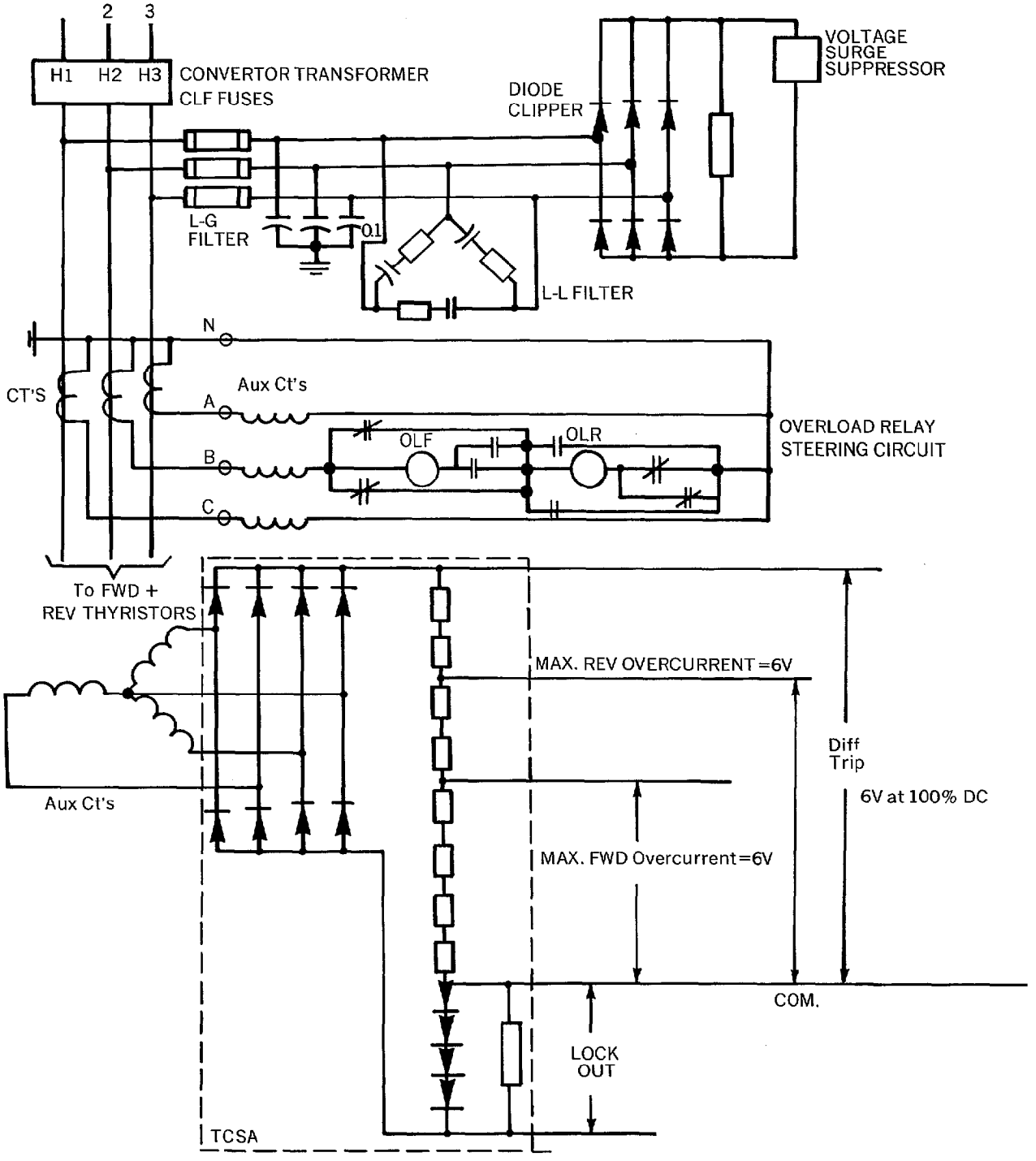
1) Provides suicide function on pre-amplifier board TUAA to obtain 0 volts converter output.

IOC: Instantaneous overcurrent relay, sensed by fault monitor card TFMA

1) Provides red light indication.
2) Fault stops process control through UVX relay
3) Slow trips DC breaker
4) Provides remote annunciation through DCW relay

FTU: Fault trip under-voltage relay which monitors the capacitor bank for high speed DC breaker trip.

1) Slow trips the DC breaker through DCT relay



AC CURRENT TRANSFORMERS AND CONNECTIONS FOR PROTECTIVE CIRCUITS

FIGURE No. 2

2) Fault stops process control through UVX relay
3) Provides remote annunciation through DCW relay

DCT: DC breaker tripping relay

1) Slow trips the DC breaker
2) Prevents reclosing of DC breaker

DCW: DC breaker auxiliary relay

1) Fault stops process control through UVX relay
2) Provides remote annunciation

UVX: Undervoltage fault relay

1) Fault stops process control
2) Provides remote annunciation through DCW relay

RLO: AC breaker lockout relay, operates when AC > DC as sensed by differential trip card TDUA

1) Provides white light indication
2) Picks up relay RLOX which trips primary AC breaker.

OLF, OLR: Converter overload inverse time relay

1) Picks up OLY auxiliary relay
2) Drops target for visible indication.

OLY: Overload auxiliary relay

1) Fault stops process control through UVX relay
2) Provides remote annunciation
3) Lights remote indicating light at operator's station
4) Picks up timing relay OLX

OLX: Timing relay set for 60 sec. to allow the converter to cool off after overload

1) Trips out OLY relay after time delay

STA, STX: Steering relays for overload relays OLF and OLR

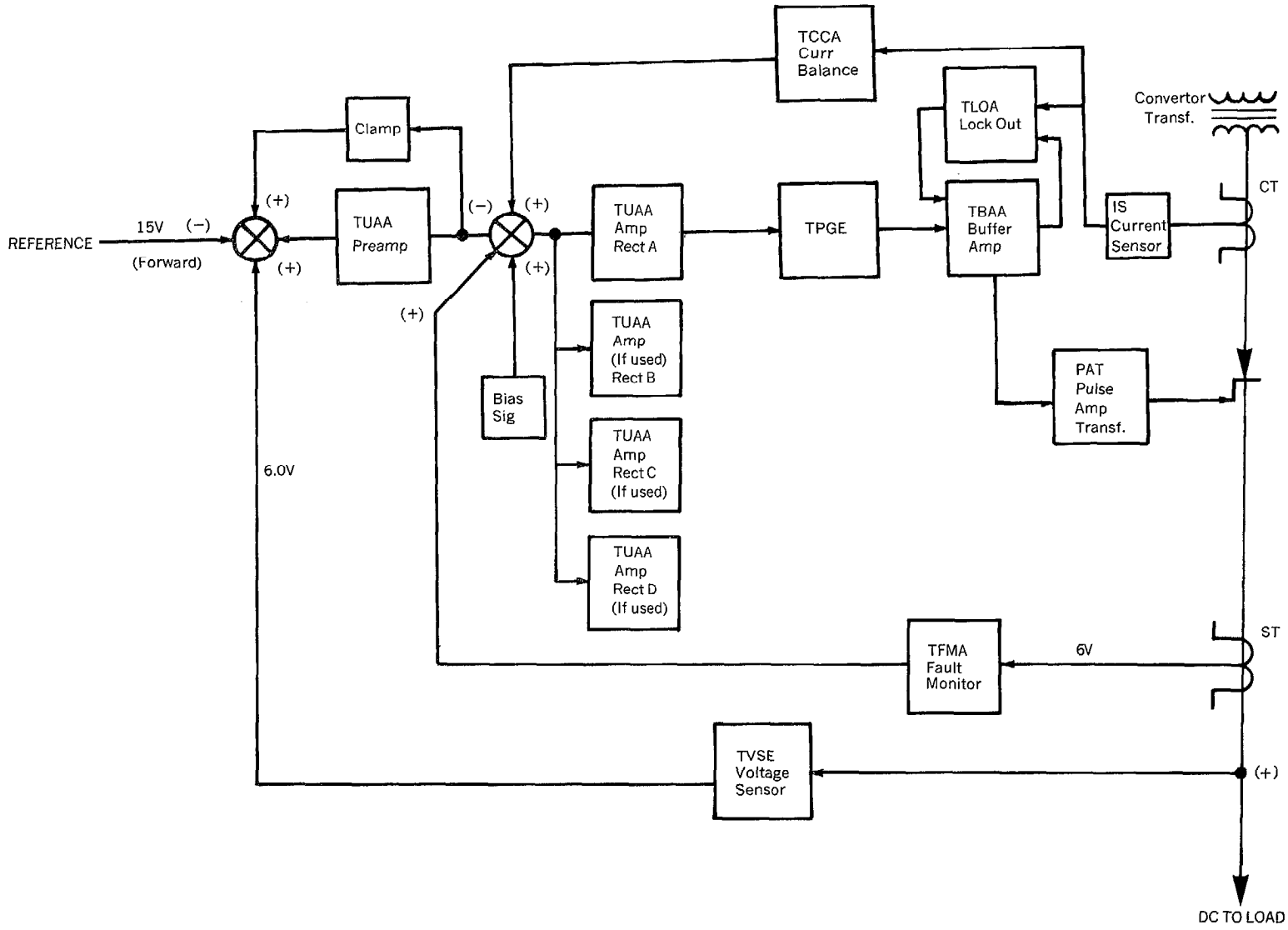
AKTR: DC breaker trip for AK breaker only. Used with signal from differential trip Card TDUA, if required.

Trips DC breaker through shunt trip

NOTE: Other relaying functions may be employed for special applications or on large drives.

SILCOMATIC PLUS REGULATING LOOP DESCRIPTION

The output d-c voltage of the converter is regulated by adjustment of the time at which the thyristors are gated with respect to the line-to-line voltage. This voltage regulating function is accomplished by the use of selected components mounted on printed circuit boards with each board having one or more functions. The Silcomatic will thereby serve as an internal voltage regulator loop within the system regulator provided by Process Control.



SILCOMATIC EXCITATION CONTROL LOOP BLOCK DIAGRAM

FIGURE No. 3

The Silcomatic Plus Voltage Regulating Loop Block Diagram is shown in Figure No. 3. A brief description of the function of each block is given below. A detailed description of the operation of each circuit will follow under the title of "Operation."

Starting at the DC Power Bus, the Regulating Loop Flow is as follows:

1) DC VOLTAGE SENSOR CIRCUIT

The DC Voltage Sensor, TVSE, Printed Circuit Board in combination with its external voltage dividing resistor assemblies VFBK1 and VFBK2, is used to attenuate, isolate, and accurately reproduce the DC bus voltage waveshape. This circuit forms the feedback path of the voltage regulating control loop.

2) PREAMPLIFIER CIRCUIT

The Preamplifier, TUAA, Printed Circuit Board is used for summing the process control voltage reference with the TVSE feedback signal, amplifying the resultant error signal, and preventing it from over-driving and possibly saturating the next stage. The Silcomatic Voltage Regulating Loop is stabilized at this point in the circuit by the introduction of components selected to form the necessary loop time constant. Also the signal is split into two equal but opposite outputs which further on in the regulating circuit act to simultaneously phase on the thyristors in one direction and phase back those in the opposite direction.

3) AMPLIFIER CIRCUIT

The Amplifier, TUAA, Printed Circuit Board is used to convert the two preamp board outputs into signals capable of driving the gate pulse generator circuits. This is also the point in the excitation control circuit where the bias voltage, current (load) balance, and fixed angle of retard signals are introduced.

4) BIAS VOLTAGE SIGNAL

The Bias Voltage Signal is used to prevent the gating of one set of thyristors without first phasing off the opposite set. Under conditions of zero process control reference signal, both amplifiers are biased to gate their respective rectifiers off. If the reference is moved from zero, either plus or minus, one converter is phased further off and its pulses cease by the time the opposite direction package is advanced sufficiently to develop converter output volts.

5) CURRENT (LOAD) BALANCE SIGNAL

On 12-pulse rectifiers where two or more parallel amplifier circuits are used, a current balance signal is required to assure that each rectifier package carries its share of load current. The current balance signal is detected by current transformers on the AC bus of each package, rectified, and fed back to the summing junctions. The signal is of a polarity that will advance the gate pulses on the lightly loaded converter(s) and retard the gate pulses on the heavily loaded converter(s). The object is to achieve a current balance at full load to within 10% in all packages.

6) OVERCURRENT SIGNAL

The Overcurrent Signal originates at the Fault Monitor (TFMA) Printed Circuit Board and is initiated when the current through the ACCT's reaches a preset level. The signal reaching the amplifier summing junction is of a polarity that will phase retard the gate pulses. The amount of phase retard will be minus 140 degrees on both converters.

7) FIXED ANGLE OF RETARD SIGNAL

The Fixed Angle of Retard Signal originates at the Fault Monitor (TFMA) Printed Circuit Board and is initiated when an overcurrent signal is present. The signal is applied to the forward and reverse amplifiers to prevent driving the gate pulses further than 140 degree retard under abnormal operating conditions. During DC fault conditions, for example, retarding the gate pulses allows a controlled rate of current rise which limits the amount of current carried by the faulted converter.

8) GATE PULSE CIRCUIT

The Gate Pulse Generator, TPGE, Printed Circuit Board is used to produce a complete set of gate pulses every 360 degrees. These pulses, numbered one to six, are applied to their respective thyristors as shown in Figure 4.

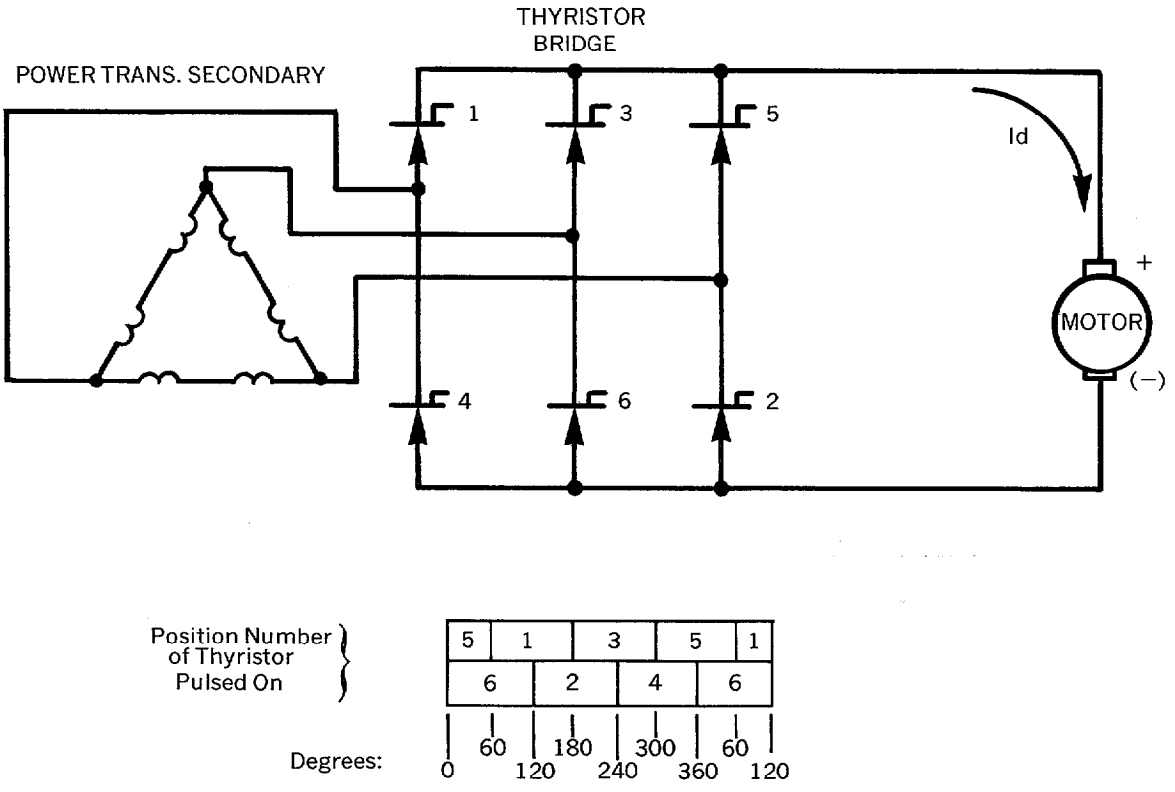
The phase relationship between the pulses and the 3-phase power applied to the bridge allows for each thyristor to receive a gating signal during the proper half of its phase voltage cycle. A second pulse is received, during the cycle whenever a complementary thyristor undergoes commutation (approximately 60 degrees later).

The time during the AC power wave that a gate pulse is generated depends upon the DC level of the Amplifier output signal. A maximum positive signal, about + 6 VDC for 60 HERTZ, will allow pulses to occur at the full advance point, while a maximum negative signal, about - 6 VDC, will retard the pulses to 180 degrees where they will cease. Voltages in between the two extremes will give continuous adjustment in time for the pulses to be generated. An open or short circuited input signal will cause pulses to form at 90 degrees phase retard.

One gate pulse generator printed circuit card is provided for each direction of operation.

9) BUFFER AMPLIFIER CIRCUIT

The Buffer Amplifier, TBAA, Printed Circuit Board is used to reshape, amplify, and increase the pulse width of the output from the Gate Pulse Generator Circuit. A signal is also fed into this circuit from the Lockout Board whenever complete pulse suppression is required. The Lockout Board, in turn, receives a signal from the TBAA Board to enable steering of the Lockout signal function during normal operating conditions. One TBAA Board is provided for each direction of Converter Operation.



THYRISTOR FULL WAVE BRIDGE CURRENT

COMMUTATION SEQUENCE

FIGURE No. 4

10) PULSE AMPLIFIER TRANSFORMER CIRCUIT

The Pulse Amplifier Transformer, PAT, Module is used to provide isolation between control and power circuits and to provide gate pulses to thyristors with fast rate of rise and adequate pulse amplitude. One PAT is provided in each thyristor Power Leg Module.

11) CURRENT SENSOR CIRCUIT

The Current Sensor Board, TCSA, is used to convert the current signals from the axuiliary current transformers, into a direct current voltage form that is usable for protective circuit input. Although this board has several variations in the number of output signals, the 12-pulse circuit shown in Figure No. 2 is typical of the maximum number of outputs that can be expected.

12) LOCKOUT CIRCUIT

The Lockout, TLOA, Printed Circuit Board, is used to prevent disturbances, instability, or other causes of abnormal operating conditions from initiating simultaneous conduction of opposite polarity converter packages. This is accomplished by detecting the flow of converter current while it is still at a low level and identifying the converter package that is receiving actual Gate Pulse Signals. The output of the Lockout Circuit is then used to inhibit the operation of the Buffer Amplifier for the opposite polarity converter package.

13) FAULT MONITOR CIRCUIT

The Fault Monitor, TFMA, Printed Circuit Board is used to monitor the level of converter Load Current, and if conditions warrant to initiate a fixed angle of phase retard. On main drive converters this circuit can also effect a Differential Trip, DC Breaker Trip, and/or Process Control Undervoltage trip by sending signals to the Differential Trip Circuit.

14) DIFFERENTIAL TRIP CIRCUIT

The Differential Trip, TDUA Trip, Circuit is used on Silcomatic Main Drive Applications where fault currents are likely to be of high magnitude. The circuit operates under fault conditions as follows:

1) Trip the AC and DC breakers in the event that AC should exceed DC by a given value of current (generally 50 percent; however, refer to the information given on the specific elementary).

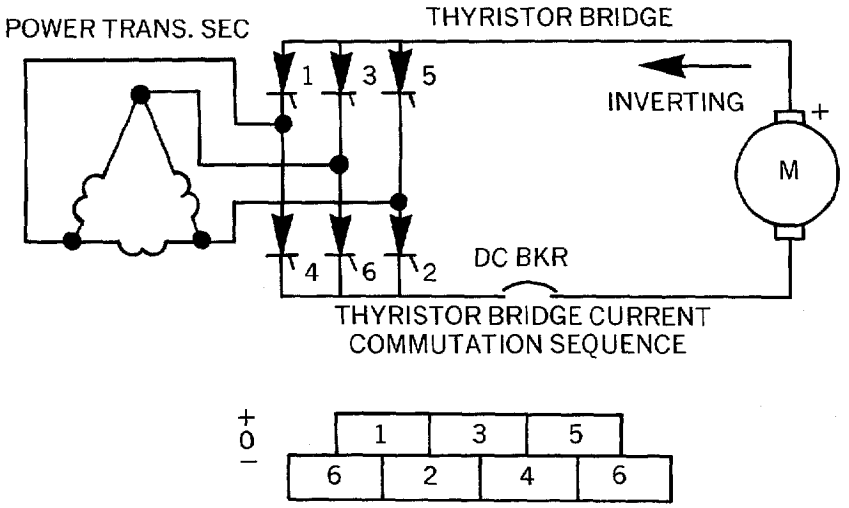
2) Trip the DC breaker in the event that DC should exceed AC by the given value of current.

The latter fault condition is called "Conduction–Through" and is explained below.

15) CONDUCTION–THROUGH

Conduction–Through occurs when one of the cells in the inverting package fails to commutate to its next cell. The failure may be caused by a suppressed gate signal, low AC input voltage, or simply the failure of the cell to turn on even though both gate signal and supply are satisfactory.

To illustrate a Conduction-Through (Refer to Figure No. 5), suppose cell No. 1 has failed to commutate to cell No. 3, cell No. 1 will then maintain conduction by virtue of the impressed motor volts and a Conduction-Through results. During a Conduction-Through, the transformer polarity progresses such that the fault becomes: motor voltage adds, through cell No. 1, to transformer voltage, through cell No. 2, back to the motor. Next, when cell No. 2 commutates to cell No. 4, the AC fault commutates out resulting in cells No. 1 and No. 4 shorting the motor counter EMF voltage. Since the AC commutates out, the suppression and overcurrent functions, which receive current signals from the AC current transformers and normally protect the converter, are ineffective during the Conduction-Through fault. Consequently, some means must be provided to interrupt the resulting fault current, thereby preventing damage to the power thyristors. The DC breaker (type AK-2-50) is tripped by direct-acting overcurrent devices, and the breaker (type AG-7) is tripped by the Differential Trip Circuit.



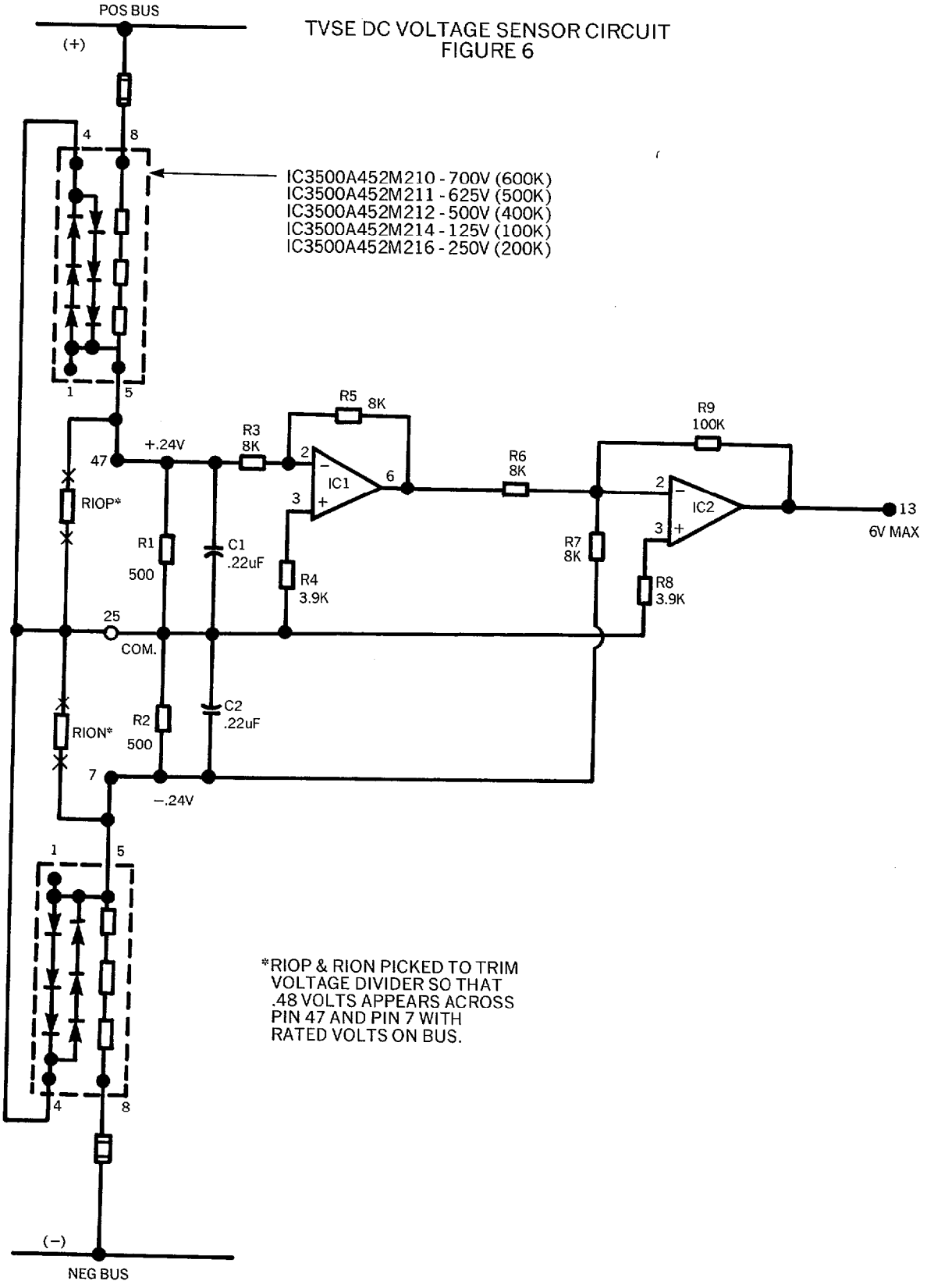
CONDUCTION-THROUGH ILLUSTRATED

FIGURE No. 5

16) VARIATIONS

On voltage regulated Silcomatics, the output signal of the Amplifier Circuit comprises the input control to the Gate Pulse Generator. On most non-reversing power converters, the high-speed voltage regulator is not required, and the gate pulse generators are controlled directly from the Process Control System Regulator.

TVSE DC VOLTAGE SENSOR CIRCUIT
FIGURE 6



*RIOP & RION PICKED TO TRIM VOLTAGE DIVIDER SO THAT .48 VOLTS APPEARS ACROSS PIN 47 AND PIN 7 WITH RATED VOLTS ON BUS.

TVSE DC VOLTAGE SENSOR CIRCUIT

FIGURE No. 6

On 12-pulse converters where control must be maintained over 6-pulse converters with ripple displaced by 30 degrees, the silcomatic voltage regulator will consist of one or more parallel amplifier circuits. The number of amplifiers will be equal to the number of 6-pulse converters controlled by the regulator. Only one voltage feedback is provided.

SILCOMATIC PLUS CIRCUIT OPERATION

1) DC VOLTAGE SENSOR CIRCUIT (TVSE and IC3500A452M21__)

The DC Voltage Sensor Circuit consists of a voltage dividing resistor network, an inverting amplifier, and a signal amplifier.

The DC bus voltage signal is attentuated to approximately 0.24 volts to ground across each of the two 500 ohm resistors (See Figure No. 6). The positive bus signal is the input to the unity-gain inverting amplifier (IC1). Both the output from IC1 and the negative bus signal are summed at the input to signal amplifier IC2. The signal amplifier has a gain of 12.5:1 producing an output voltage signal of 6 Volts with an input of twice 0.24 volts. Thus the output signal produced is proportional to the voltage difference between the positive and negative busses.

2) PREAMPLIFIER CIRCUIT (TUAA)

The Preamp circuit consists of an input summing junction, an inverting amplifier with an output clamp, and a unity gain inverting amplifier (See Figure No. 7).

The Process Control Reference Signal (Negative for forward drive) and the TVSE Feedback Signal are summed at the input to inverting-amplifier IC1. Input resistors convert reference and TVSE signals to 1.5 milliampere level at the current summing junction. The feedback resistor R13 is selected for a forward gain of 50 with TVSE signal. Positive and negative clamps at 10 volts are used to prevent amplifier saturation. Transistors Q2 (for positive limit), and Q1 (for negative limit), perform this clamping function by feeding back a signal opposite in polarity to the one trying to drive the output of IC1 above 10 Volts. Also, connected around IC1 is a parallel RC combination, R13 and C1, which are selected to insure stable operation of the voltage regulating loop.

Unity gain amplifier IC2 inverts the output of IC1 to produce a complementary output signal which is equal in amplitude but opposite in polarity.

Thus, any error signal present at the input to IC1 will create two output signals from the preamp printed circuit board. Ultimately, one of the signals phases on its gate pulse generator while the complementary signal phases off opposite gate pulse generator circuit.

3) AMPLIFIER CIRCUIT

The Amplifier Circuit consists of two identical inverting amplifiers each with a summing junction on the input and a clamping circuit on the output.

FIGURE No. 7: PRE-AMPLIFIER – TUAA

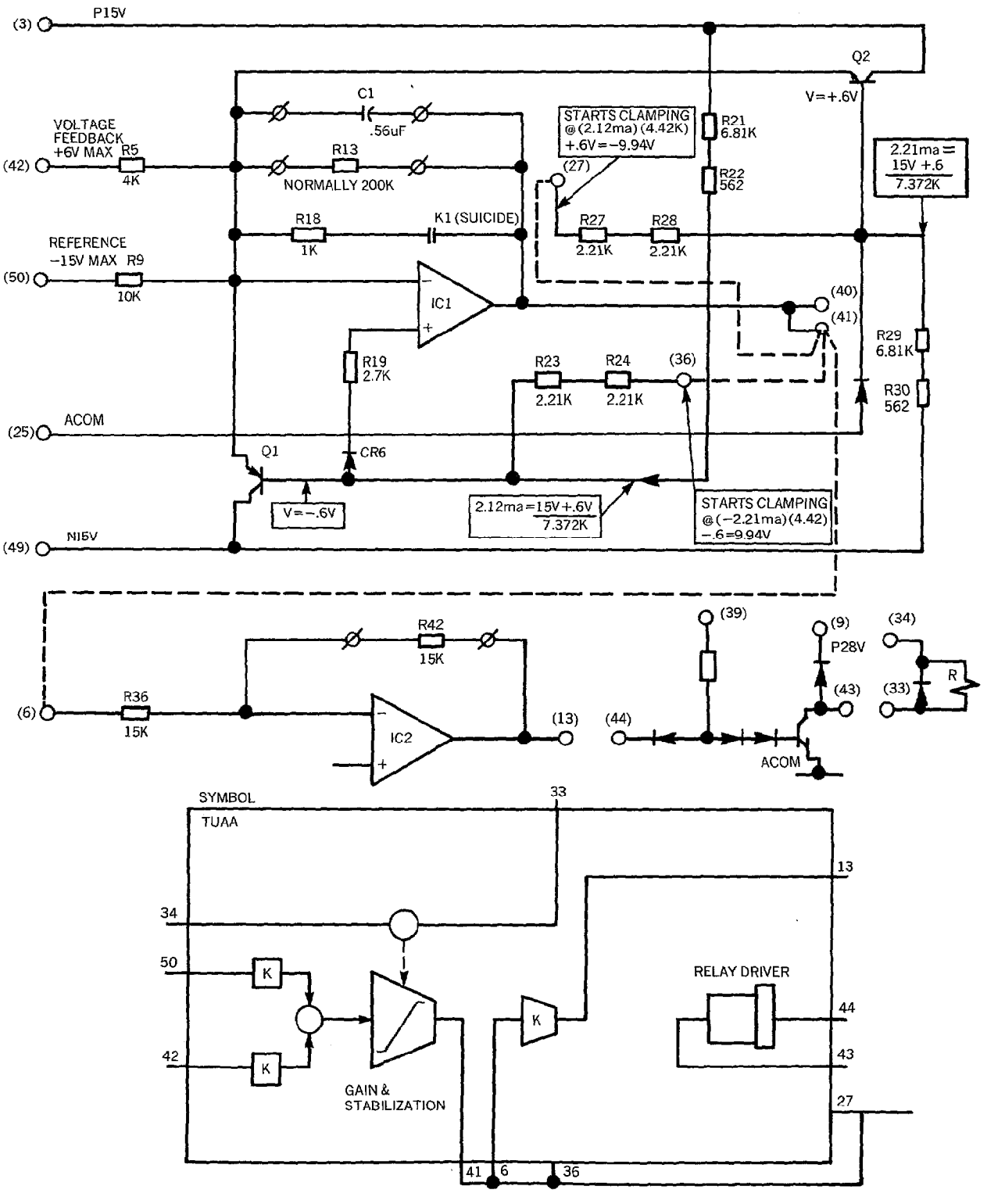
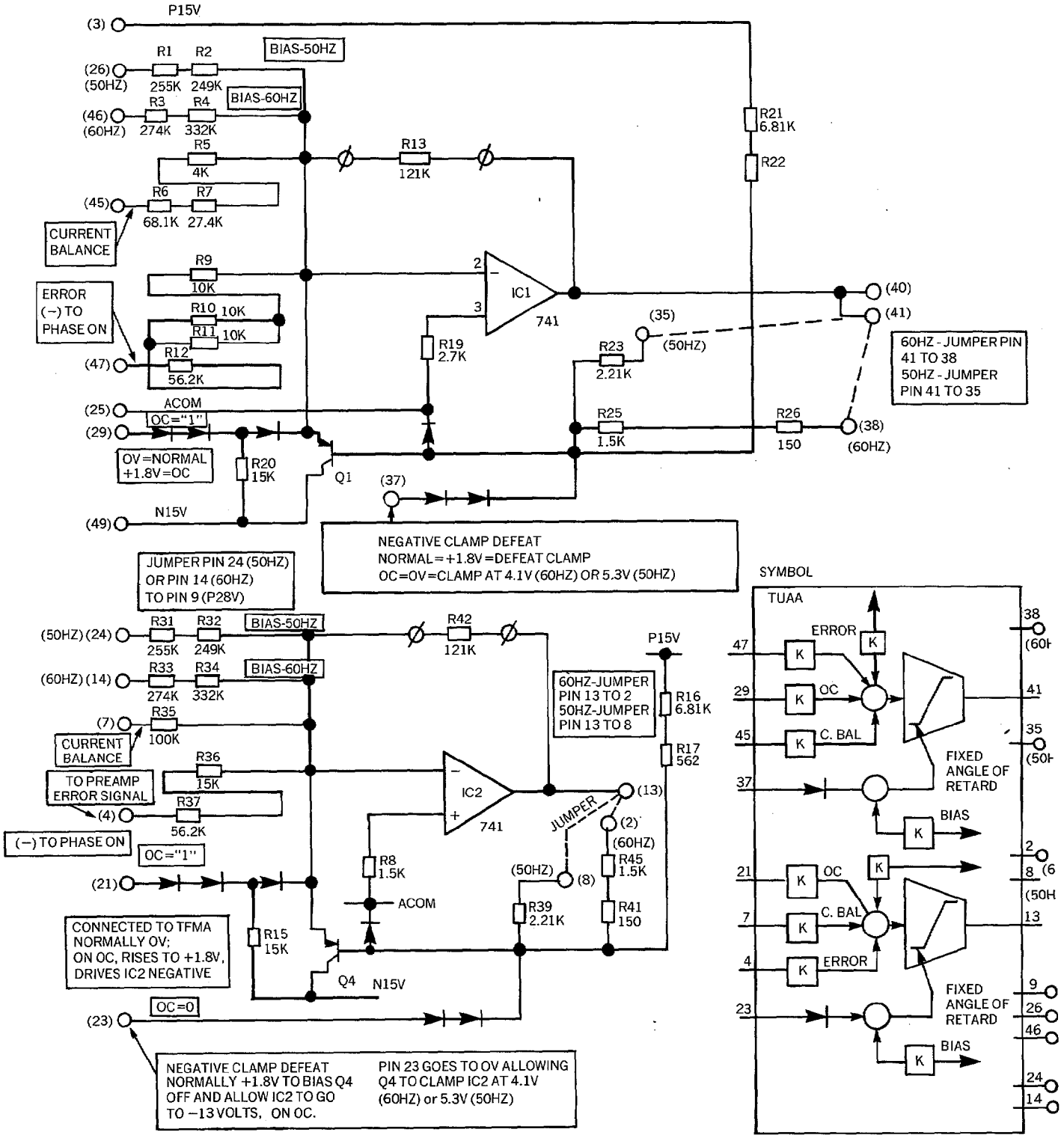


FIGURE No. 8: AMPLIFIER – TUAA



The outputs of the preamp board are routed to separate input pins on the amp board (See Figure No. 8) where they are summed with their respective bias, current balance (if used), overcurrent limit, and fixed angle of retard signals.

Whenever the bias signal is the only input to the summing junction (all others are zero), both amplifier outputs will be about - 5.6 Volts (-7 Volts on 50 Hz drives).

Under normal conditions a negative signal from the Preamp and/or current balance card drives the output of the amp card positive and ultimately phases the Silco pulses forward. Positive signals drive the amp negative and are limited to -12 V by saturation of IC1 and IC2. Under fault conditions the TFMA card injects positive current into PIN 29 (21) which drives IC1 (IC2) negative. At the same time the TFMA card enables the clamp circuit with a "0" at pin 37(23) the net result is that the amp clamps at -4.1V on 60 Hz drives and -5.3V on 50 Hz drives. This occurs in the forward and reverse amp cards at the same time, the result being that under fault conditions, both forward and reverse SCR's receive pulses at the "fixed angle of retard". On most types of faults, this places the pulses in a relationship with the AC voltage wave form such that no current can flow. On conduction-through faults where the motor CEMF is near rated voltage, the retard angle is such that the Silco can accept regenerating current. In this condition it is possible for the Silcomatic to continue commutating current from cell to cell, thus forcing all cells to share the fault current until the DC Breaker or line contactor can open.

4) GATE PULSE GENERATOR CIRCUIT (TPGE)

The Gate Pulse Generator Circuit consists of the TPGE Printed Circuit Board, the Limit Voltage Transformer, and the Signal Voltage Transformer.

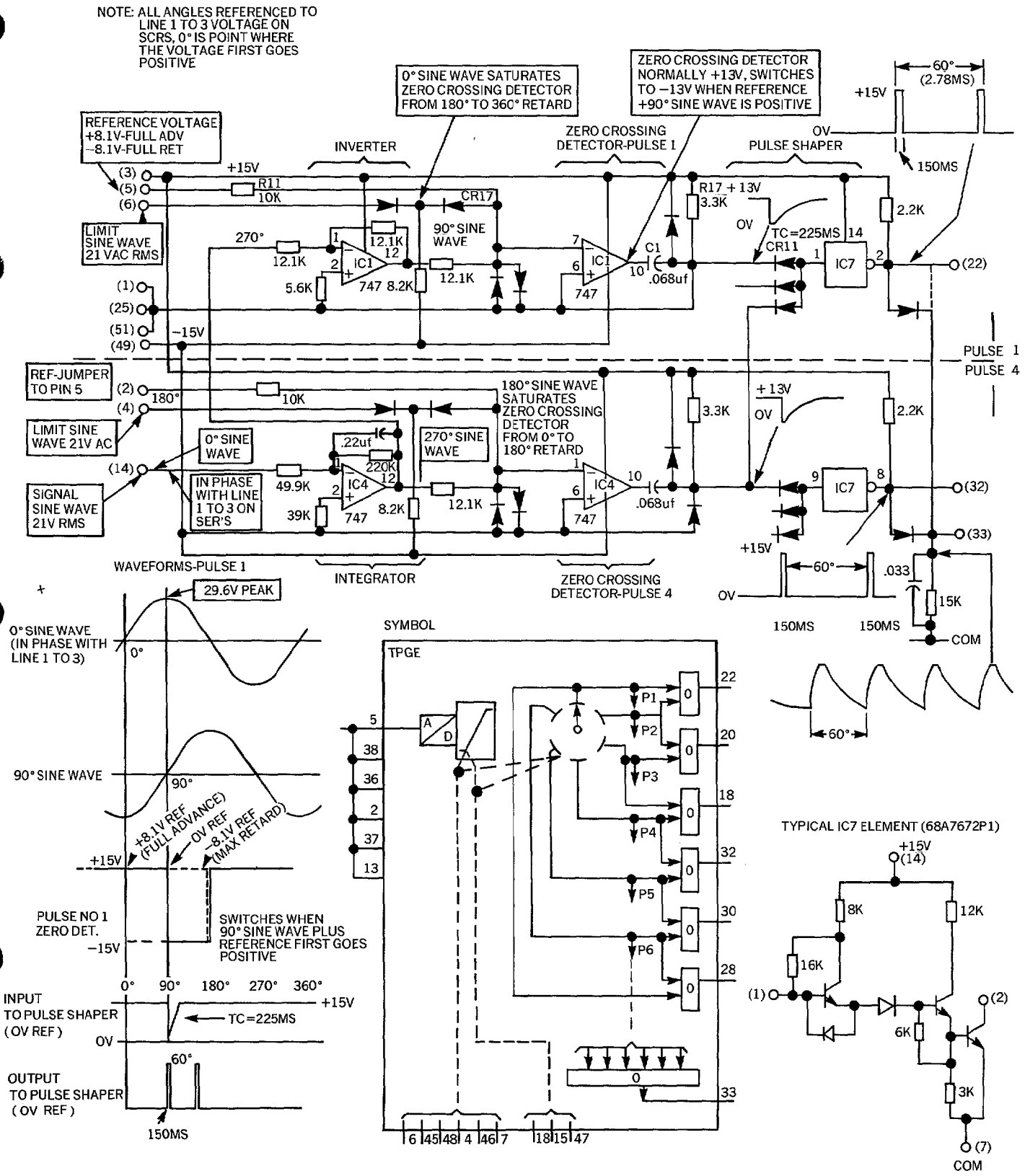
The Limit Voltage Transformer is used to supply an isolated AC line to neutral voltage in 60 degree increments. The Line No. 1 to neutral voltage of the star connected secondary is in phase with Line No. 1 to Line No. 3 voltage applied to the thyristor power bridge. The neutral on the secondary of this transformer is connected to zero volts which forms a common reference point for all phase related events. Provisions are made on the primary to allow for phase shifts in steps of 30 degrees to accommodate transformer connections of either delta or wye.

The Signal Voltage Transformer is used to supply an isolated AC line to neutral voltage that is phase related to the voltage applied to the thyristor power bridge. The neutral of this transformer is also tied to zero volts, the common reference point.

The TPGE Board contains three identical generator circuits, one for each of the phases applied to the power bridge. Each circuit consists of one integrating amplifier, one inverting amplifier, two zero crossing detection amplifiers, and two pulse shapers. The inverting amplifier and double circuitry (See Figure No. 9) provide for two sets of gate pulses per circuit. One set is for the thyristor conducting during the positive half of the AC wave and the other for the thyristor conducting dur the negative half of the wave. Figure No. 4 identifies the location of the six thyristor positions.

All three circuits on the TPGE Board share the DC control voltage from the AMP Board while each receives its own signal and limit voltages.

FIGURE No. 9: PULSE GENERATOR – TPGE



The Gate Pulse Generator Circuit shown in Figure No. 9 is for AC Phase No. 1 and it produces pulses for thyristor positions one and four. The other two circuits, for thyristor positions 2 and 5 and 3 and 6, operate in the same manner except for phase relationship and time differences.

The signal voltage, Phase No. 1 to zero, is applied to the input of inverting integrator IC4 pin 1 where the resulting output is rotated 270 degrees from the input. Unity gain amplifier IC1 pin 12 inverts the signal which is now 90 degrees behind limit voltage No. 1.

The DC control signal is added to the 90 degree signal through resistors at the input to zero — crossing detector IC1 pin 7.

When the sum is negative, output of IC1 pin 10 is about +12 volts (logic 1), CR11 blocks and IC7 pin 2 output is about 0.2 volts (logic 0). If the sum is positive, the output of IC1 pin 10 will be a logic 0 and will "sink" current from +15V through R17 and C1 until C1 is charged. During the charge time of C1 the output of IC7 pin 2 will be a logic 1. This pulse generation depends on limit voltage being positive which back-biases CR17.

If the limit voltage is negative CR17 is permitted to conduct and the input to IC1 pin 7 is forced to be negative which produces a logic "0" at the output of IC7 pin 2 and prevents the pulse. The pulse duration is determined by capacitor C1.

The magnitude of the reference voltage determines how soon after the limit voltage goes positive that the sum of the 90 degree signal plus the reference will also cross zero volts.

5) BUFFER AMPLIFIER CIRCUIT (TBAA)

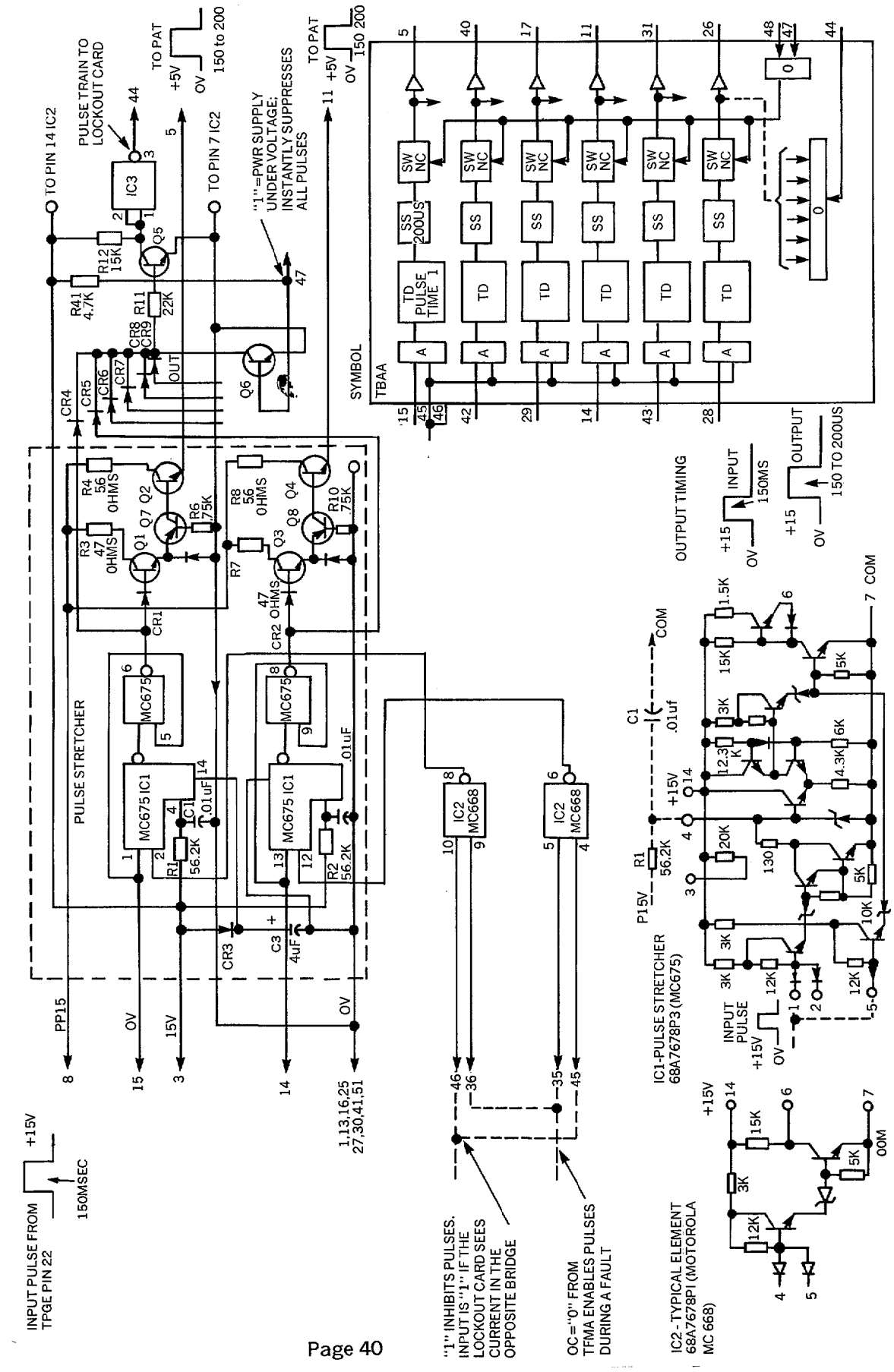
The Buffer Amplifier consists of six identical circuits, one for each thyristor position on the power bridge (See Figure No. 4). Each circuit consists of ½ of a dual monostable multivibrator and a two transistor signal amplifier. This board also contains IC's for pulse suppression, lockout steering, and DC power supply loss functions.

Monostable multivibrator IC1 (See Figure No. 11) accepts trigger pulses from its gate pulse generator circuit and produces an output pulse with a duration determined by the R-C time constant of resistor R1 and capacitor C1. The start of the output pulse is delayed until the input pulse goes to zero thus preventing any slight phase shifts on the AC line from causing pulses to occur before the full advance point.

Transistors Q1, Q2 and Q7 are supplied from an auxiliary 15-volt power supply, PP15. PP15 will droop with increasing PAT load but the output of TBAA will maintain a 5-V 200 microsecond pulse.

A signal from the lockout board is fed into inverting gate IC2 via an interlock on the amp board. During operation of the opposite polarity rectifier package or when the amp and/or lockout boards are unplugged, the signal is a logic "1" which is inverted to a logic "0" to inhibit operation of IC1. Under operating conditions requiring gate pulses from the TBAA board, a logic "0" (Zero Volts) is fed into IC2.

TBAA - BUFFER AMPLIFIER FIG. 11



An output from each of the multivibrators is fed through diodes CR4 through CR9 into transistor Q5. Gate IC3 is driven by Q5 to produce an output signal consisting of a train of output pulses that are routed to the lockout board to be used for steering the circuit.

A signal from the TFMA board is fed to transistor switch Q6. A logic "1" applied to the base of Q6 will turn it on and pull all of the multivibrator outputs to zero volts thus killing all gate pulses at the BA board.

6) PULSE AMPLIFIER TRANSFORMER CIRCUIT (TPAF, TPAG)

The Pulse Amplifier Transformer Circuit is contained in a potted assembly which also houses the voltage dropping resistors for the thyristor indicating lamps. The circuit consists of an energy storing capacitor, switching transistor, and a pulse transformer (See Figure No. 12).

Capacitor C1 is charged from the 28 VDC power supply through resistor R1 and transformer T1. A pulse signal from the buffer amplifier circuit switches transistor Q1 into the saturation region. Capacitor C1 is now a low impedance source for supplying pulse energy to transformer T1 via transistor Q1 back to zero volts. The charging current for C1 flowing through T1 resets the transformer core. Diodes CR1 and CR2 prevent excessive voltage build-up across T1 as pulse current returns to zero.

7) LOCKOUT (TLOA)

The lockout circuit incorporates the AC current transformers (ACCT), the TCSA board, and the lockout (TLOA) printed circuit board. The circuit consists of a resistor-diode network, a level detecting amplifier, and two identical sets of NAND gates, pulse stretchers, and inverting gates.

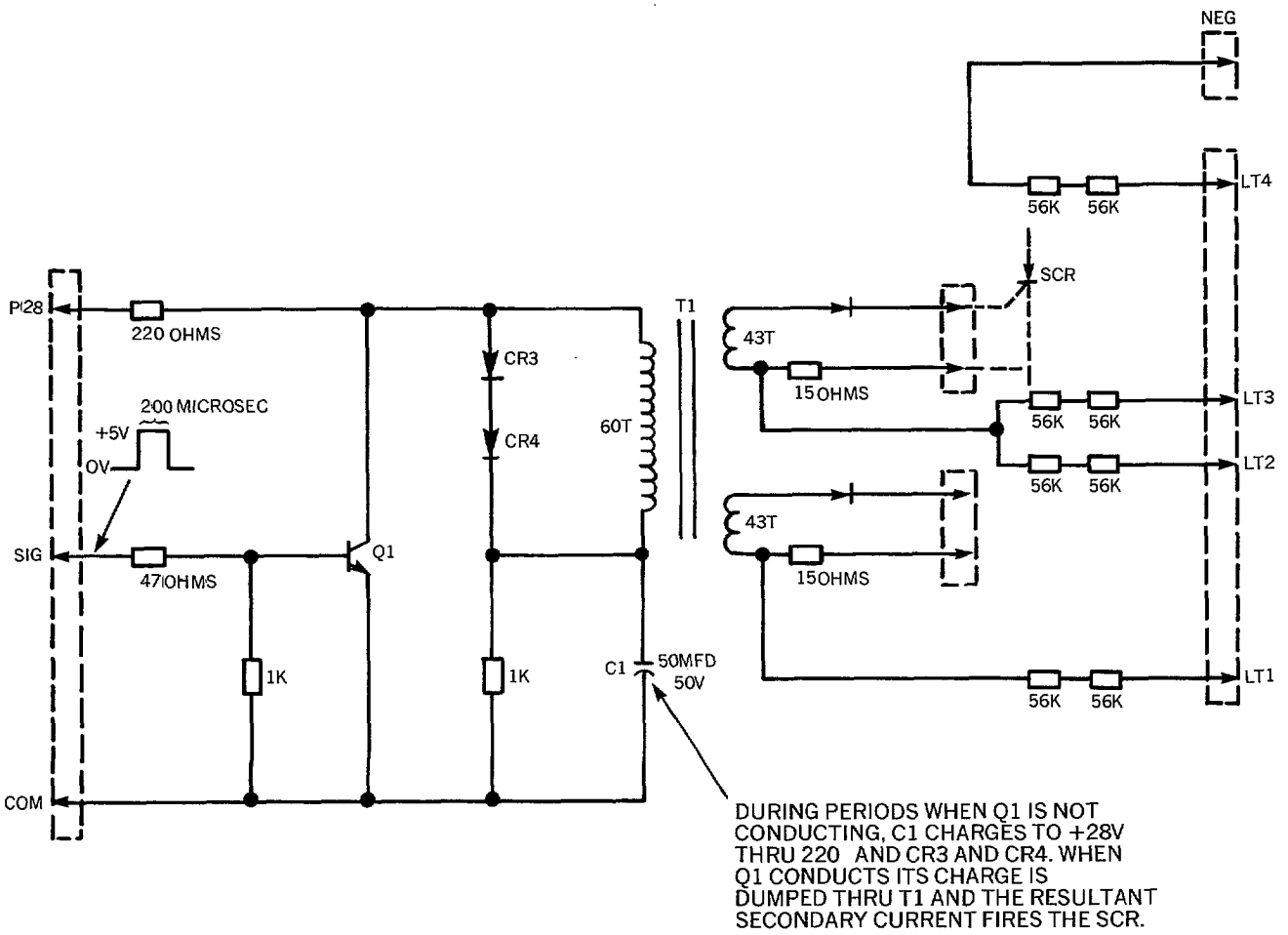
The lockout circuit as shown in Figure 13, receives gate pulse signals from both TBAA boards, current signals from the ACCT circuit, and a "fake current" signal from the TFMA board. Outputs are sent to each TBAA board to inhibit either set of gate pulses.

Lockout signals are generated in only one direction at a time, can only occur after gate pulses and converter load current signals are detected at the TLOA Board, and can only be reset by converter load current returning to zero. Under fault conditions the TFMA circuit sends a "fake current" signal to simulate the flow of load current after the converter power output has ceased. This keeps both TBAA circuits at a fixed angle of phase retard until the process control reference can be lowered to zero.

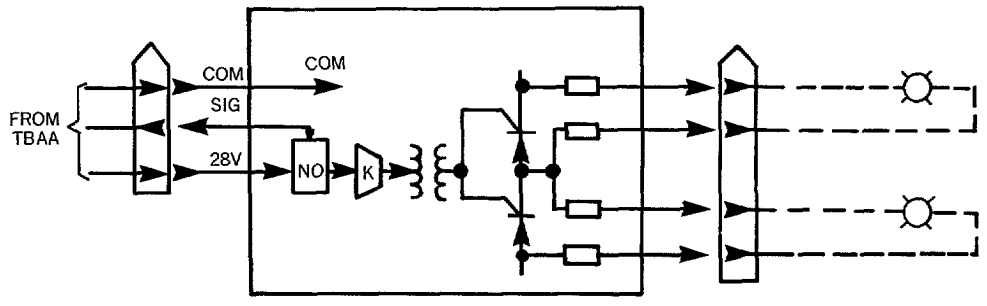
The logic of the Lock-out circuit depends on identification of current by pulses present: Forward pulses identify current as forward current and through the logic of the lockout circuit permit forward pulses to continue and prevent reverse pulses.

Referring to Figure 13, the initial state is the condition of no pulses and no power current which is identified as a circle. Lack of pulses provides a logic "0" signal to IC4 pin 1 and attempts to provide a logic "0" to pin 1 IC3. Lack of current provides a logic "1" at the input and output of level detector IC1 which forces the input to IC4 pin 11 to be a logic "0". This makes the input to IC3 pin 1 and the output of the circuit IC4 pin 6 to be a logic "0". The other half of

FIGURE No. 12: PAT – PULSE AMPLIFIER TRANSFORMER CIRCUIT



SYMBOL - LEG MODULE
(INCLUDES PAT & SCR'S)



the circuit is operating in identical manner with no pulses or current: logic "0" is provided to IC2 pin 9 which attempts a logic 0 at IC3 pin 12, lack of current provides a logic 1 to pin 13 of IC2 which forces a logic 0 at IC3 pin 12 and at the circuit output, IC4 pin 4. If pulses occur on either side without power current, no change in state occurs since pins 1 and 12 of IC3 are forced to remain at logic 0 by the IC2 pin 10 and 12 outputs which are logic "0" due to lack of power current signal.

The next state shown is Forward Pulses with Forward current identified by a square. Forward pulses make the input to IC4 pin 1 a logic "1" and attempt to make the input to IC3 pin 1 a logic "1"; the presence of current provides logic "0" at the input and output of IC1 which makes a logic "1" signal at IC3 pin 1 input and at the IC4 pin 6 circuit output to the Reverse TBAA. On the other half of the circuit, since no Reverse gate pulses are present, the input to IC2 pin 9 and IC3 pin 1 are a logic "0"; even with power current present which makes the input to IC2 pin 13 a logic 0, the input to IC2 pin 1 is clamped at a "0" by output of IC2 pin 4 due to lack of Reverse pulses. This provides a logic "0" at the IC4 pin 4 circuit output which permits forward pulses— the above logic "1" output from IC4 pin 6 circuit output locks out reverse pulses.

Following the condition of forward pulses and forward current, the next state is termination of forward pulses with forward current present — identified as an inverted triangle. This changes the condition at IC4 pin 1 input to a logic "0" which changes the IC4 pin 2 output to a logic "1" but the presence of current before this change made the output IC4 pin 12 a logic "1" which holds IC3 pin 1 input and the IC4 pin 6 output signal to the Reverse TBAA at a logic "1" for continued lockout. No changes occur in the rest of the lockout circuit board.

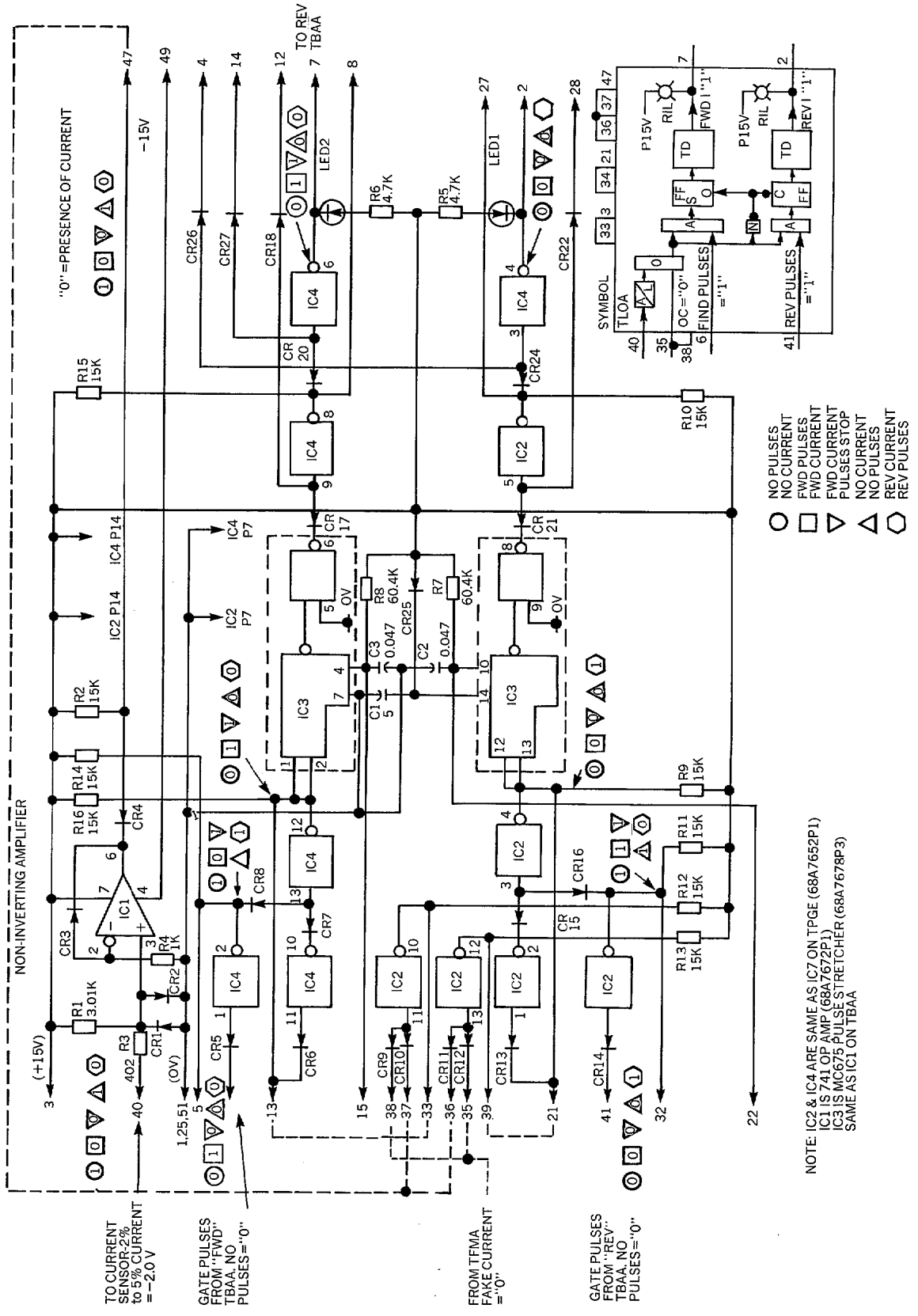
The fourth and last phase of operation for the lockout circuit starts as the power current decreases below the lockout threshold; this is identified on Figure 13 as a triangle. The input and output of IC1 current level detector are changed from logic "0" to "1" which clamps the IC4 pin 12 output to logic "0"; this changes IC4 pin 13 input from logic "0" to "1" and IC3 pin 1 input to logic "0". Since IC3 is a pulse stretcher, its output remains at logic "1" for about 800 microseconds and then switches to a logic "0". This delayed change from logic "1" to "0" also occurs at the IC4 pin 6 output of the card connected to Reverse TBAA. The delay is included to allow time for forward power current to decrease below lockout threshold level to zero before permitting the Reverse TBAA card to provide pulses for reverse power current.

The change in level detector IC1 output from logic "0" to "1", in addition to changes above, also changes IC2 pin 12 output from logic "1" to "0". No other change occurs in the circuit since the lack of Reverse pulses had forced a logic "0" output at IC2 pin 4 which clamps IC2 pin 12 output to a logic "0" and prevents any change.

The case of reverse pulses and reverse power current is also shown in Figure 13 identified as a hexagonal figure. Reverse pulses provide a logic "1" to IC2 pin 9 which attempts to make IC3 pin 12 input a logic "1". With power current present, IC1 input and output is a logic "0" and the IC2 pin 12 output is a logic "1", this permits the output of IC2 pin 4 to be a logic "1" as attempted by the reverse pulses. The output of the lockout circuit IC4 pin 4 which connects to the forward TBAA card becomes logic "1" and prevents forward pulses. Reverse pulses are permitted by logic "0" at IC4 pin 6 output.

As noted earlier, fault current which results in TFMA board trip will produce a logic "0" at IC2 pins 11 and 13 inputs. Since this signal is constant until reset, the lockout circuit remains in state of forward or reverse current until reset.

FIGURE No. 13: LOCKOUT CIRCUIT – TLOA



8) FAULT MONITOR (TFMA)

The Fault Monitor circuit incorporates the AC current transformers (ACCT), the resistor-diode signal board (TCSA), and the TFMA Printed Circuit Board. The circuit consists of a resistor-diode network, two level detecting amplifiers (one forward and one reverse to allow independent trip settings), seal-in circuits, NAND gates, and switching transistors.

The TFMA circuit as shown in Figure 14 receives current signals from the ACCT circuit and steering signals from the lockout circuit. Outputs are sent to the TBAA boards for signaling the loss of any of the DC power supplies, and to the amplifier board for fixed angle of retard signaling. In main drive applications, outputs are also sent to the differential trip board for AC circuit breaker and DC circuit breaker trip functions.

Loss of plus 15 volts, minus 15 volts, or plus 28 volts will send a logic "1" signal to the TBAA boards to switch off the transistors Q6 and inhibit all pulse outputs as described under Buffer Amplifier circuit of this instruction.

During a period of overcurrent the TFMA circuit provides signals to produce a fixed angle of phase retard on both converter packages. The circuit also provides signals to trip the process control undervoltage circuit, to trip the DC circuit breaker through the differential trip circuit, and to keep the lockout circuit sealed in ("fake current").

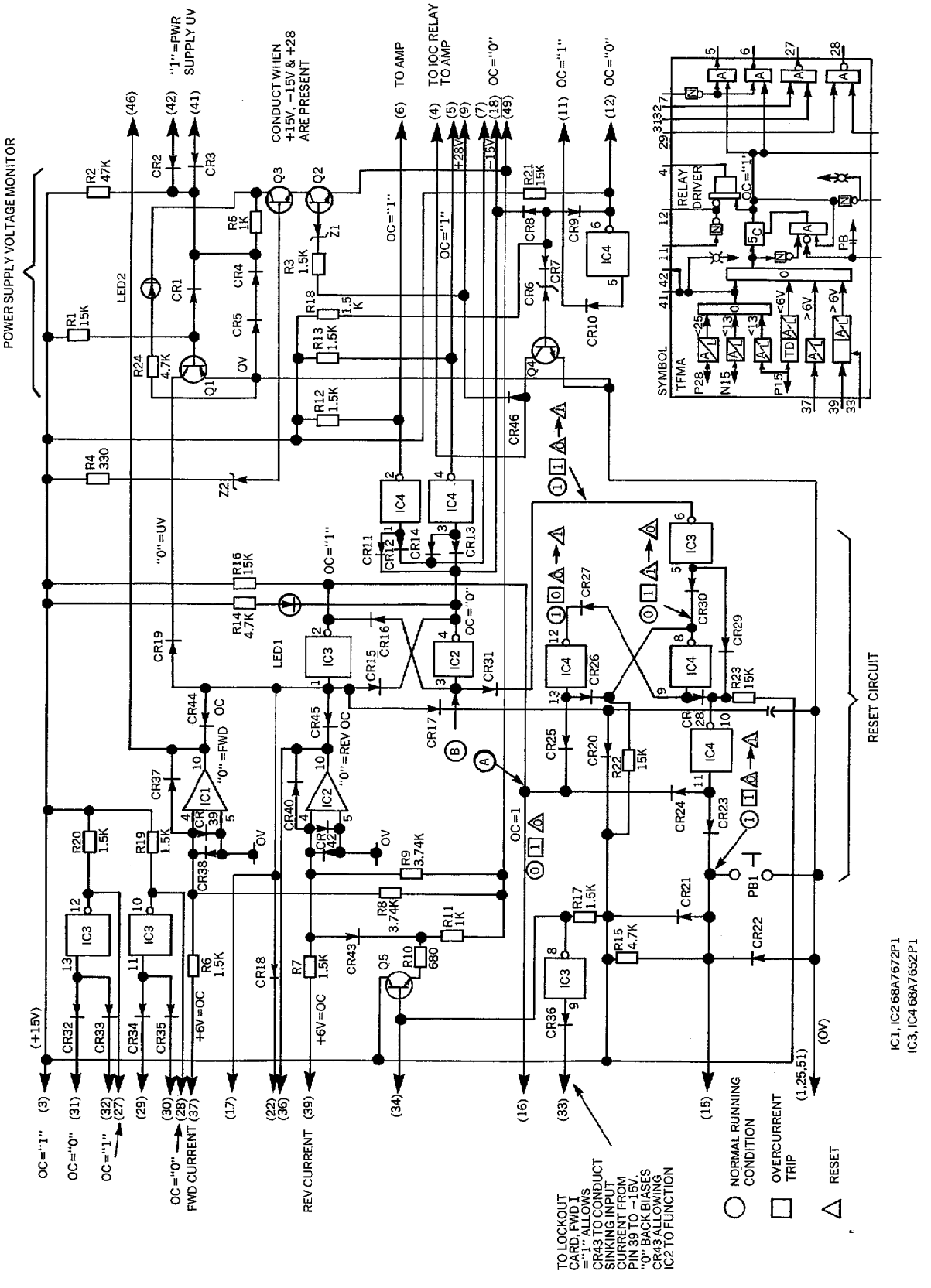
The TCSA portion of the circuit (See Figure 14) rectifies and converts the ACCT currents to DC voltage signals for input to the TFMA board's forward and reverse level detectors. The reverse level detector, IC2, also receives a signal from the lockout circuit to inhibit its operation whenever reverse lockout signals are produced (forward converter load current is flowing).

During forward converter operation inverting gate IC3 (See Figure 14) receives a logic "1" input from the reverse lockout circuit. The output of IC3 pin 8 places a logic "0" on the base of the switching transistor Q5 turning it off. Whenever Q5 is off, diode CR43 is biased to conduct from the input circuit of level detector IC2 into the minus 15 volt bus. Under this condition it will take about 60 volts or ten times maximum overload current to switch the output of IC2. During reverse rectifier operation, IC3 pin 9 receives a logic "0" from the reverse lockout circuit. A logic "1" output from IC3 pin 8 turns Q5 on to back bias CR43 and effectively turn it off. A signal input of 6 volts will now be able to switch the output of IC2.

The outputs of both level detectors IC1 and IC2 are fed through isolating diodes and paralleled. The signal present at this point in the circuit is used for the TFMA wired flip flop circuit.

The TFMA wired flip flop circuit consists of two NAND gates of IC3 and diode CR15. During an overcurrent condition either IC1 or IC2 will have a logic "0" output. The output of IC3 pin 2 will be driven to a logic "1" causing the output of IC3 pin 4 to go to a logic "0". Diode CR15 will hold the IC3 pin 1 input to a logic "0" until the circuit is reset. A zero will turn off transistor switch Q4 and drop out the IOC relay while light emitting diode Led 1 will become illuminated.

FIGURE No. 14: FAULT MONITOR – TFMA



The initial control excitation or temporary loss of plus 15 volts will also affect the seal in circuit when power returns because as capacitor C1 recharges its low impedance appears as a logic "0".

A signal for maximum phase retard is initiated when the output of IC3 pin 4 goes to logic "0" forcing both IC4 pins 2 and 4 outputs to a logic "1" level. The two outputs are fed to the pair of amplifier input summing junctions on the amp board. They appear as a plus signal and are only effective if the clamping transistor is turned off as explained under Amplifier Circuit section of this instruction.

A fixed angle of retard signal is initiated when the output of IC3 pin 2 goes to logic "1" and is applied to gate IC3 pin 11 and pin 13.

A fake current signal is fed from the output of IC3 pin 4 to the TLOA Board. A logic "0", when overcurrent occurs, is used to simulate the flow of rectifier load current as explained under Lockout Circuit of this instruction. Indication is the illumination of Led 1 on the TFMA Board and one Led on the TLOA Board.

The reset circuit is used to release the wired flip flop circuit when the reset pushbutton, PB1, is depressed. Normal operating condition (Converter running or ready to run) is shown on Figure 15 by circles. Points A and B are logic "0", the IC4 pin 10 and 12 outputs are logic "1" which make IC4 pin 8 logic "0" and locks IC4 pin 13 input to logic "0". The IC4 pin 8 output also forces IC3 pin 5 input to logic "0", pin 6 output is a logic "1".

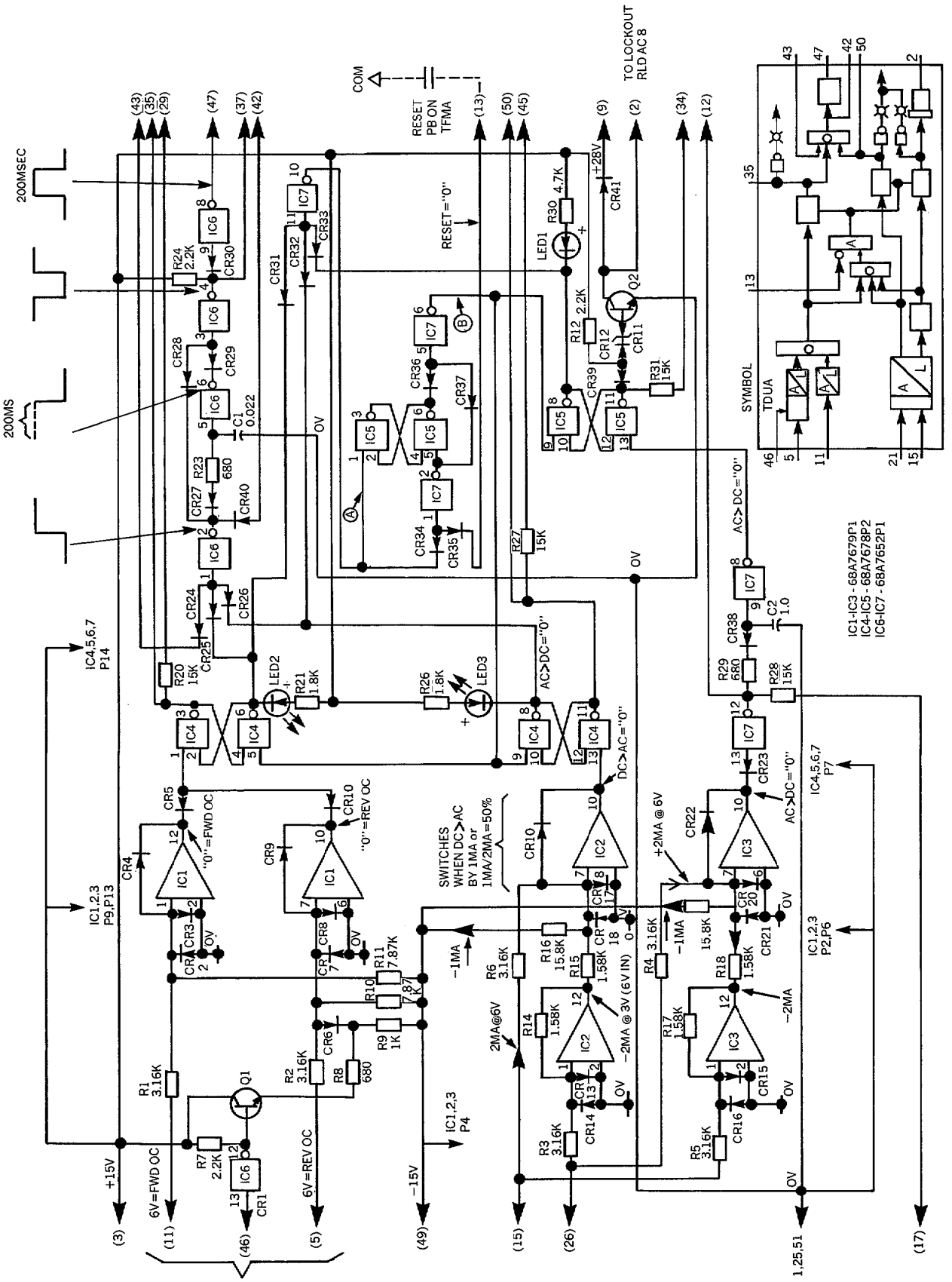
Overcurrent trip (shown as a square) causes both points A and B to be logic "1" which changes IC4 pin 10 output to logic "0" and pin 8 output to logic "1". This changes IC4 pin 12 output to logic "0" but IC3 pin 6 remains logic "1". The reset circuit will remain in this condition if the converter overcurrent drops to zero since points A and B remain at logic "1" due to the overcurrent wired flip flop shown on Figure 14. With this condition, reset to normal condition is now done only by the reset pushbutton.

Reset state is shown as a triangle on Figure 15 and makes the IC4 pin 11 input a logic "0" and pin 10 output a logic "1"; this changes the IC3 pin 6 output and point B to logic "0". Point B can reset the overcurrent wired flip flop shown on Figure 14 by changing IC3 pin 4 output to logic "1" and IC3 pin 2 output to logic "0". Since IC3 pin 2 is also point A of the reset circuit, IC4 pin 11 is forced to logic "0" if the pushbutton is a logic "0" or "1". Point A also connects to IC4 pin 13 which resets the IC4 pin 12 output to logic "1". This changes IC4 pin 8 to logic "0" and IC3 pin 6 to logic "1" which completes the reset operation.

9) DIFFERENTIAL TRIP CIRCUIT (TDUA)

The differential trip circuit consists of current reading circuits on the AC and DC converter power busses, the differential trip circuit board, breaker trip circuits for the AC and DC breakers and interlock circuits with the Fault Monitor (TFMA) Board. The function of the circuit is to compare AC and DC currents and to compare AC current to a pre-set level. Under fault conditions the circuit is to operate as follows:

FIGURE No. 16: DIFFERENTIAL TRIP CIRCUIT – TDUA



1. Trip the AC and DC breaker in the event that AC should exceed DC by a given value of rated converter current (usually 50%).

2. Trip the DC circuit breaker if DC current should exceed AC current by the stated value or if AC current should exceed the overcurrent value.

The AC bus currents are monitored by ACCT's and the TCSA, as in the TFMA circuit.

Several output taps are provided with calibration adjusted to supply a 6 volt output for rated current or reverse overcurrent trip or forward overcurrent trip value.

As indicated in Figure 16, the AC rated current output signal and the DC current transformer output signal are compared in the circuits of IC2 and IC3. If AC current exceeds DC current by 50% or more, the AC current signal through R4 will exceed the DC current signal through IC3 pin 12 and R18 and the bias current through R 19. This will make IC3 pin 7 positive and pin 10 negative which is a logic "0". The output of IC7 pin 12 will become logic "1", after the time delay for R29 and C2. The output of IC7 pin 8 will be logic "0" which will change the state of IC5 wired flip flop and turn Q2 transistor on to energize RLO relay and trip the AC breaker. The IC5 pin 8 output is connected to LED 1 to indicate the AC breaker trip to IC7 or gate which makes points A and B a logic "1". Points A and B connect to a reset circuit identical to that of the Fault Monitor card; both reset circuits operate as explained in the Fault Monitor circuit description.

If the DC bus current exceeds the AC bus current by 50% of rated current or more, the DC current signal through R6 is larger than the AC current signal through R15 and bias through R16. This makes IC2 pin 7 positive and pin 10 negative or logic "0". The wired flip flop connected to IC2 pin 10 is changed to make IC4 pin 8 output a logic "0" and IC6 pin 2 output a logic "1".

The IC6 pin 2 output is directly connected to IC6 pin 3 input and to IC6 pin 5 input through a delay R23 and C1. As IC6 pin 2 changes from logic "0" to "1" the IC6 pin 3 input is changed to logic "1" but IC6 pin 5 input is a logic "0" and remains "0" for the delay time of R23 and C1. This makes the pin 3 input to IC6 follow three states:

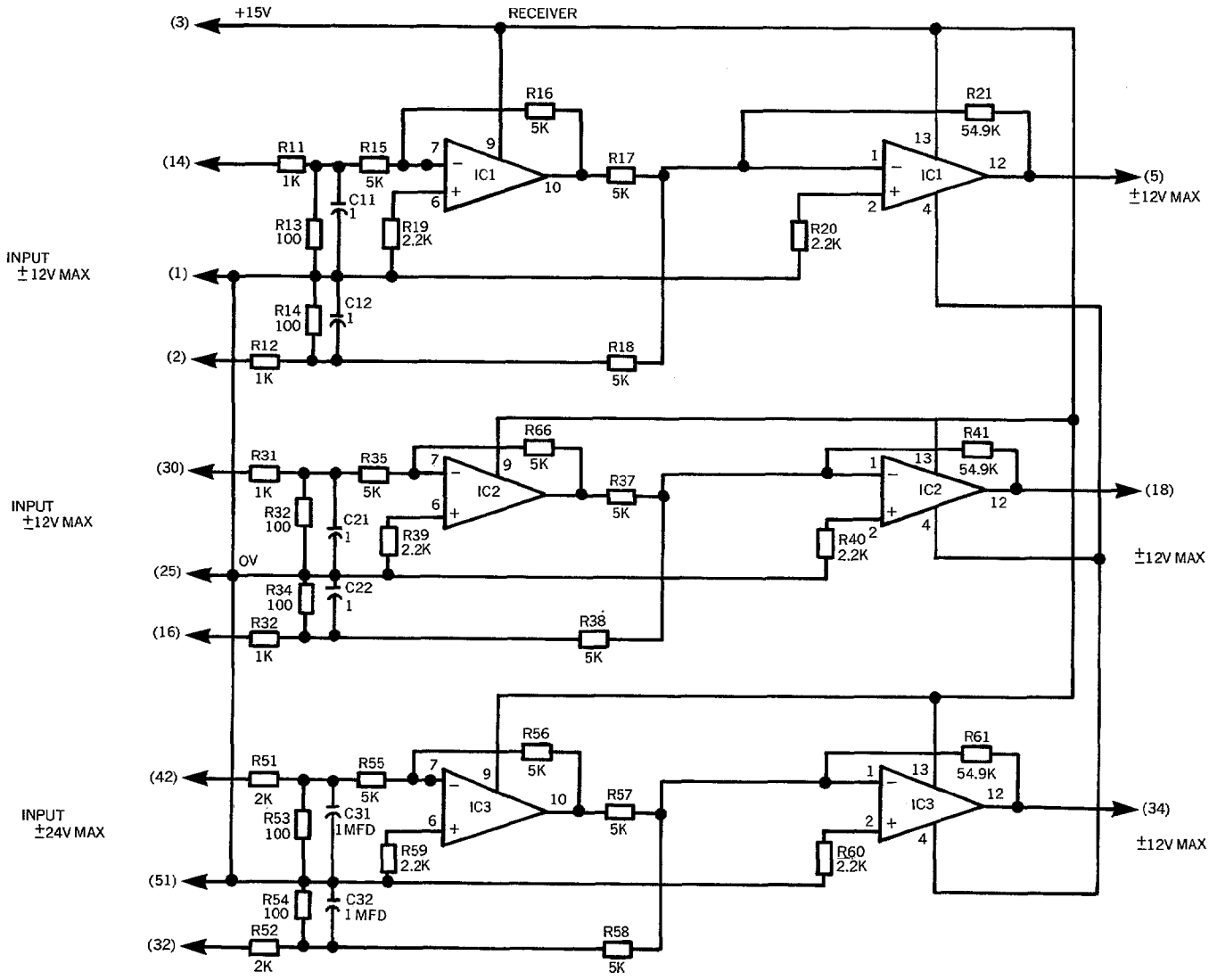
1. Initially set at logic "0" by IC6 pin 2 output.

2. Change to logic "1" as IC6 pin 2 changes.

3. Change again to a logic "0" after the time delay allows IC6 pin 5 input to change from logic "0" to "1", and the pin 6 output from "1" to "0".

This produces a pulse at IC6 pin 3 input and at IC6 pin 8 output to the DC breaker trip circuit. As in the preceding breaker trip circuit, the state of the wired flip flop is indicated by LED 2 and the flip flop is connected to the reset circuit through points A and B.

The third function, to sense an overcurrent in either the forward of reverse converter circuit, is done by two level detectors. This is identical to the two level detectors described in the Fault Monitor circuit including the calibration controls for reverse current. The presence of overcurrent produces a signal of 6 volts or more at pin 5 or pin 11 of the card and a logic "0" at the IC4 pin 1 input. This changes the state of the wired flip flop and IC4 pin 6 output becomes logic "0"



NOTE:
THE TOP TWO AMPLIFIERS ARE USED FOR APPLICATIONS REQUIRING AN INPUT VOLTAGE LESS THAN ±12 VOLTS, SUCH AS BETWEEN THE MASTER AND SLAVE OF A 12 PULSE CONVERTER. THE BOTTOM AMPLIFIER HAS A GAIN OF 1/2 AND CAN BE USED ON INPUTS OF ±24 VOLTS OR LESS SUCH AS ±15V FROM PROCESS CONTROL.

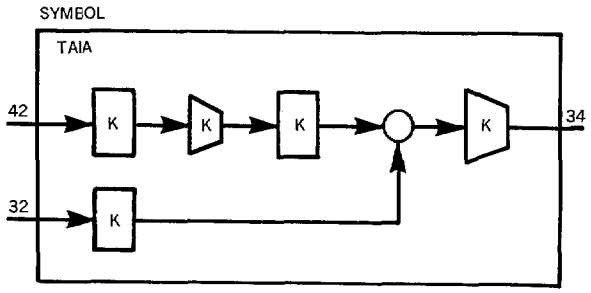


FIGURE No. 17: ANALOG INTERFACE – TAIA

which initiates the formation of a pulse to the DC breaker trip circuit. The logic "0" signal is impressed at the OR gate IC6 pin 1 input and produces a pulse signal at IC6 pin 6 output and turns on LED 3 to identify overcurrent trip.

10) ANALOG INTERFACE CIRCUIT (TAIA)

The Analog Interface Circuit consists of a voltage dividing resistor bridge, an inverting amplifier and a signal amplifier.

The function of the Analog Interface Circuit is to receive analog signals from distant equipment such as the Process Control and eliminate any common mode noise voltage that might have been picked up on the interconnecting wires. The signal arrives on a shielded twisted pair; the signal wire goes to pin 14 (or 30 or 42) the other wire (usually connected to ACOM at the other end) goes to pin 2 (or 16 or 32). The signal is bridged down by the input resistor divider and filtered at 10,000 radians, then inverted. The signal on the other wire of the twisted pair goes through a similar bridge and filter but is not inverted. The two signals are summed and amplified by the second amplifier, the output of which is essentially free of common mode noise. The common mode noise is of equal amplitude on each wire of the twisted pair but since one side is inverted and added to the other they cancel leaving a noise-free signal.

11) DIGITAL INTERFACE CIRCUIT (TDIA)

The Digital Interface Card contains three transmitter circuits and four receiver circuits. Each transmitter contains several high level logic gates connected according to the application. The output includes a buffer transistor to protect the gates.

The receiver circuit is a resistor capacitor network which provides a pull-up resistor and some filtering for a remote transmitter. A noise pulse of 7 volts at 7.2ma in the "1" state or 7 volts at 700ma in the "0" state is required to cause improper gating.

An input pulse width of at least 800 microseconds is required.

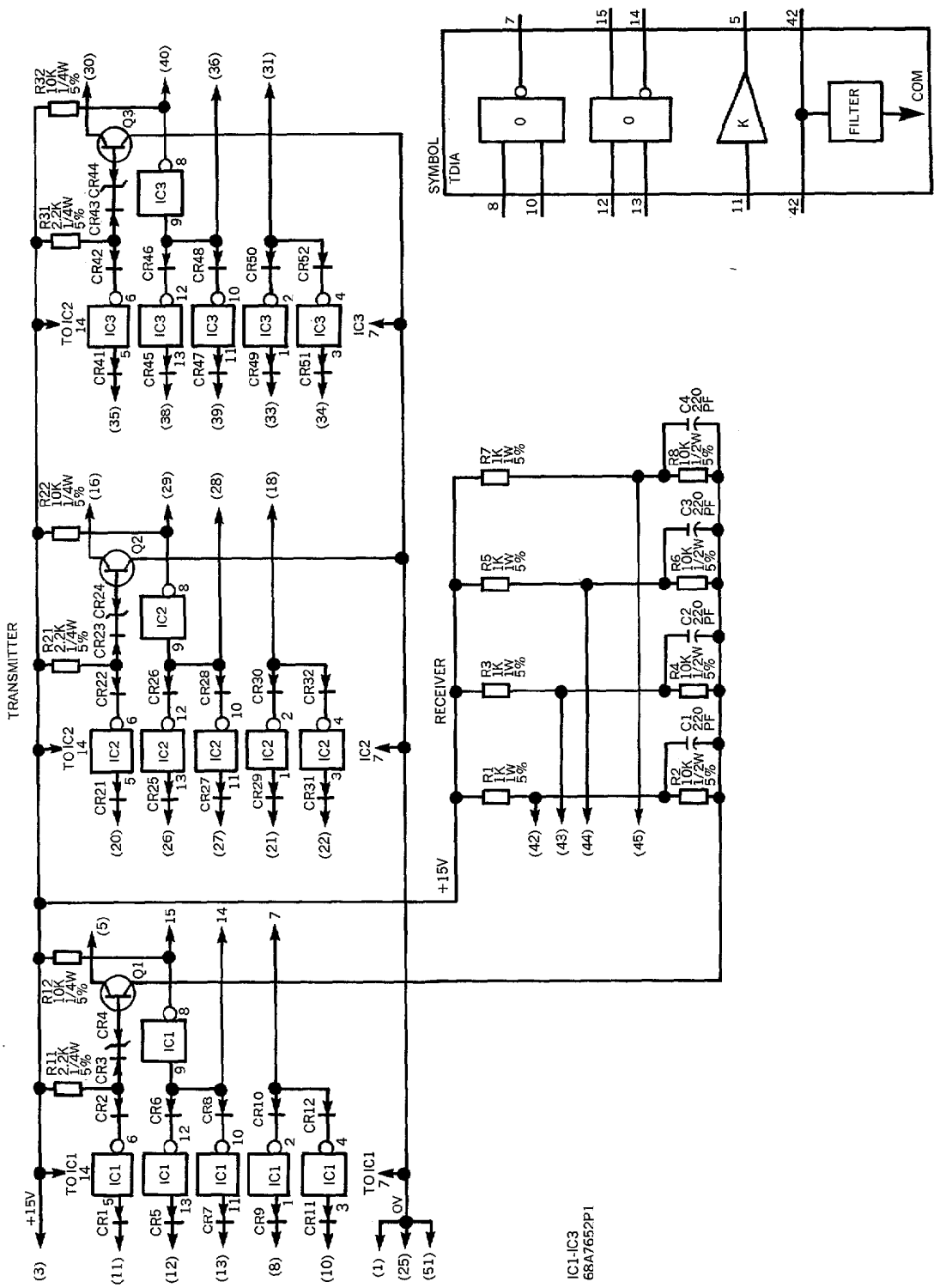
A typical application for the digital interface card is to transmit logic signals between the master and slave firing circuits on a 12-pulse converter.

12) CURRENT BALANCE (TCCA)

The Current Balance Circuit is used on 12- and 24-pulse converters to force the currents in the individual 6-pulse packages to match within 10% at full load. It consists of two sets of two amplifiers with summing and gain resistor networks. IC1 inverts the current signal from the package being balanced and IC2 sums this signal with the average of the current signals of all the converters including the one being balanced. The result is an error signal proportional to the difference between the current being balanced and its share of the total average current. This error signal goes to the amp card of the package being balanced where it acts as a vernier to adjust the output-voltage of the package in a direction to force its current to be proportional to the total average current.

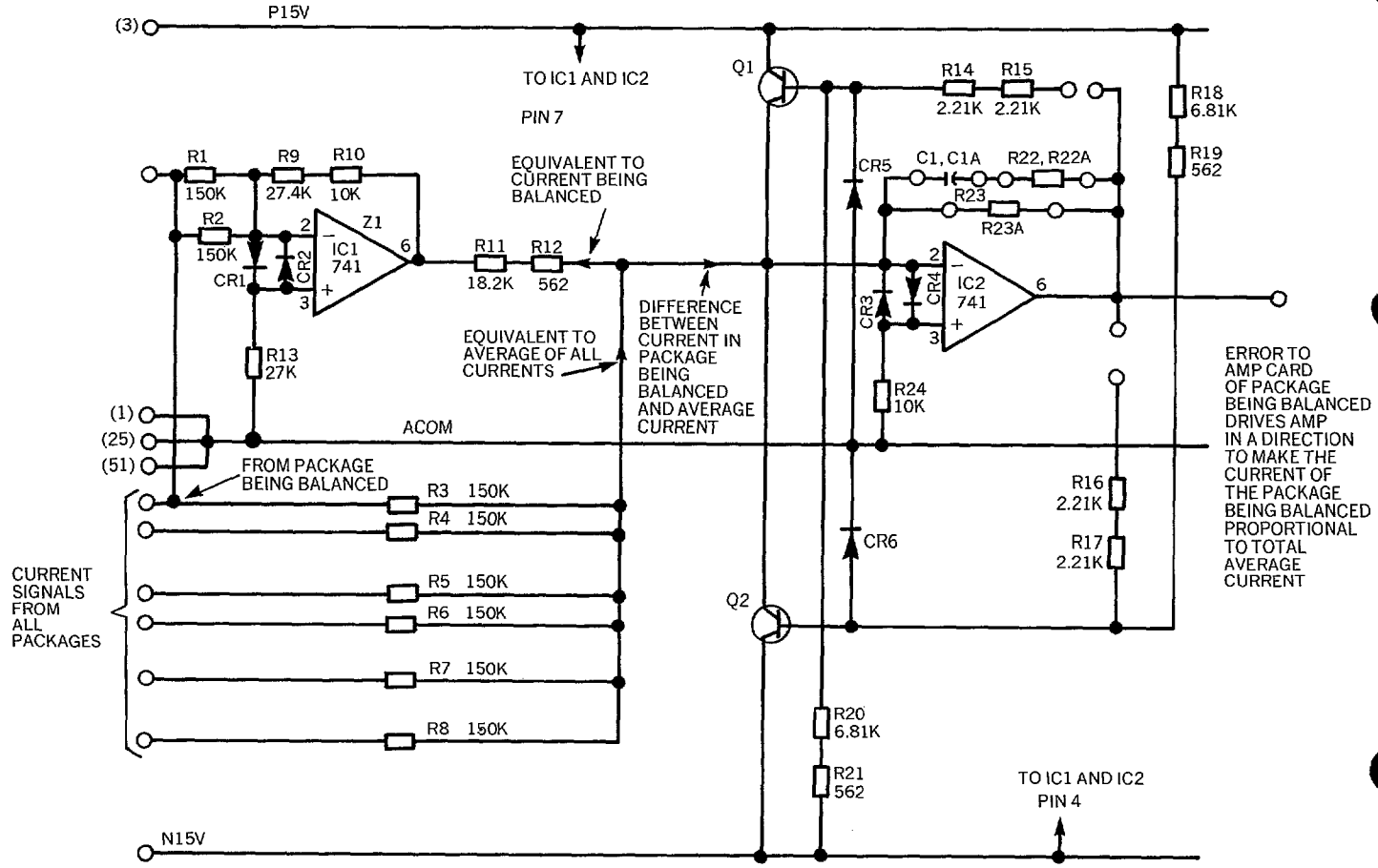
DIGITAL INTERFACE – TDIA

FIGURE No. 18

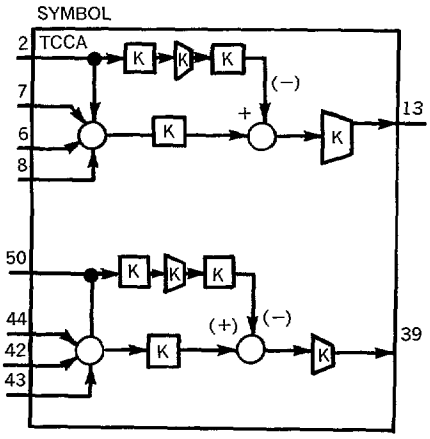


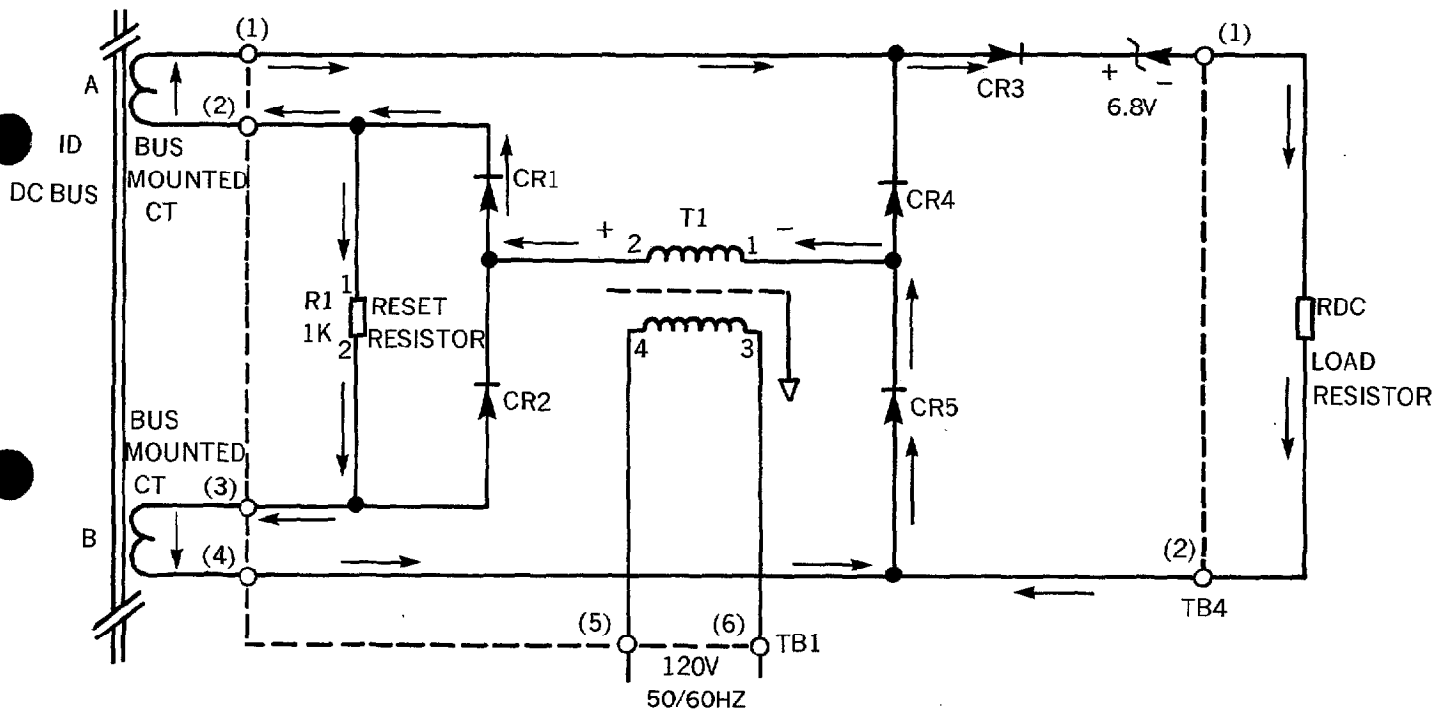
CURRENT BALANCE – TCCA

FIGURE No. 19



IC1, IC2 68A7672P1 (741 DP AMP)





OPERATION OF DCCT

1) Case 1 (T1 terminal 2 positive)

   (a) Current flows from T1 terminal 2 through CR1, CT(a), CR3, 6.8 V Zener, RDC, CR5 to T1 terminal 1. This excitation current flows in a direction opposite to DC load current and brings CT(a) out of saturation. In this area of operation, the CT acts as a true CT, forcing the current through RDC to be equal to DC load current divided by the CT ratio.

   (b) Since CR1 and CR5 are conducting the transformer T1 voltage appears across R1 and CT(b) in series. This current flows in same direction as the DC load current and forces the CT core into saturation. CT(b) is thus reset and is ready for operation when T1 reverses polarity.

2) Case 2 (T1 terminal 1 is positive)

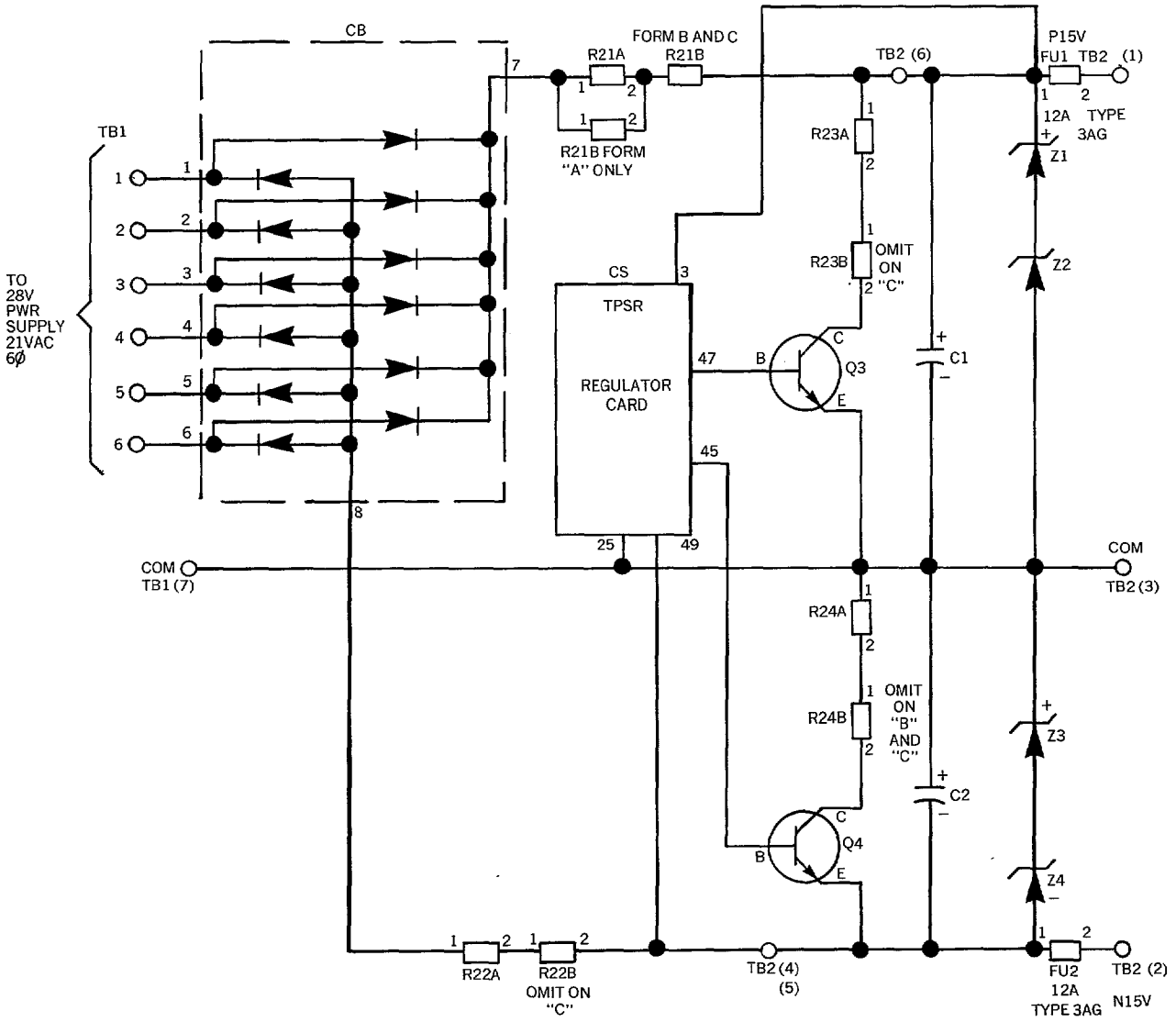
   (a) Current flows from T1 terminal 1 through CR4, CR3, 6.8V Zener, RDC, CT(b), CR2 to T1 terminal 2. Operation is as above except CT(b) is now unsaturated and acts as a CT.

   (b) Since CR4 and CR2 are conducting T1 voltage appears across CT(a) in series with R1, resetting CT(a).

DCCT

FIGURE No. 20

± 15 VOLT POWER SUPPLY – FIGURE No. 21



IC3601A191 POWER SUPPLY

Form A + 15 V @ 1.4 A, – 15 V @ .9A (12 Pulse and 24 Pulse Rev Drives)

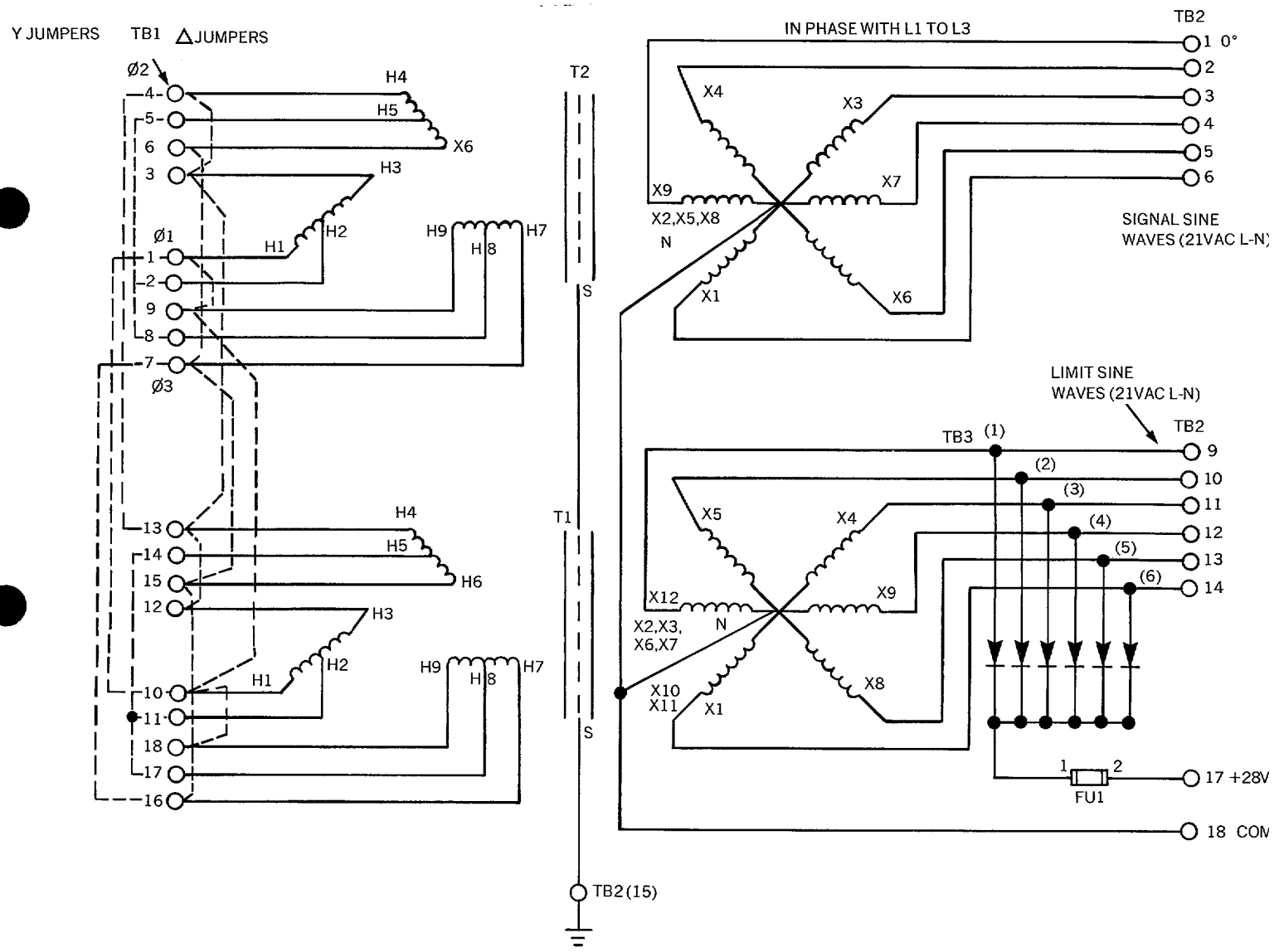
Form B + 15 V @ .9A, – 15 V @ .5A (12 Pulse non. Rev. and 6 Pulse Rev)

Form C + 15 V @ .5A, – 15 V @ .25A (6 Pulse non. Rev. or 12 Pulse slave)

Output voltage after a 25% voltage dip for .2 sec. is ± 13 V.

28 VOLT POWER SUPPLY AND PHASING SINE WAVES

FIGURE No. 22



IC3601A 192 POWER SUPPLY

Forms C and D have an additional transformer T3 with its primary in parallel to the primary of T1 and a secondary and diode bridge similar to T1 (+28 volt output on TB2 terminal 7).

INSPECTION AND TEST PROCEDURES

OBJECTIVE

The objective of this section is to:

1. Serve as a sequential guide for inspection and testing of the Silcomatic Plus, either in part or in its entirety.

2. Present test methods and functional descriptions to assist those performing maintenance and trouble-shooting tests on the Silcomatic Plus.

Basic procedures for proper checkout are stated in bold type and in numerical order, each followed by further discussions and clarification.

PRECAUTIONS:

The following precautionary advice is included to forewarn the tester against committing procedures which might tend to nullify important protective relationships that are designed into the system.

1. Do not under any circumstance remove gating signals from only a portion of a number of power modules paralleled to share load. The converter is designed to withstand all fault conditions, provided the full complement of paralleled modules is gated and all protective functions are operable. When directional isolation is desired, remove the gate pulse generator card.

2. Do not megger control modules and circuits in which solid state electronic devices are present, as they can be damaged easily. When meggering converter power circuits, short all busses together to avoid impressing potential across rectifying devices unnecessarily. Hipotting of rectifier power circuits is not considered advisable or necessary.

3. Use extreme care and caution when working with or around the rectifier power circuits and modules. They are by nature high capacity, low impedance paths, and a destructive arcing fault could be precipitated by careless procedure. When phasing out or otherwise observing the power circuit, be extremely careful to prevent an arcing fault. The connecting of oscilloscope leads to the DC bus without the proper probes is dangerous. Refer to the subsequent section on oscilloscope techniques.

4. Do not operate the converter with fault monitor or AC current transformer circuits open or ineffective. During inspection or trouble-shooting, circuits may be systematically opened or isolated as logical procedures, but care must be taken not to disable the suppression function or leave its CT's open-circuited.

5. Do not remove or insert printed circuit cards unless control power switch CPS is open.

6. Do not operate the converter full reversing on a regenerative load with the lockout, or differential trip circuits disabled.

7. Do not apply main power to the equipment unless the equipment frame is grounded and "C" common is grounded per the elementary diagram.

8. Do not connect gating signals to both packages of a reversing converter until each directional package has been tested independently. This especially applies to closed loop operation on the process control system regulator. Always test and stabilize the process controls on each single direction package before operating back-to-back.

OSCILLOSCOPE TECHNIQUES

Observations of voltage and current waveforms in power rectifier circuits require extra care in selection and use of oscilloscope equipment. The make-up of the rectifier bridge circuit, whether grounded or not, produces large voltages to ground at nearly all points. In observing voltages or currents within the rectifier, differential signal circuits in the oscilloscope are used to reject these common mode voltages and to display the desired signal correctly. In addition to more accurate observations, differential signal circuits are also safer to use since the oscilloscope chassis is grounded. The oscilloscope equipment should be provided with a calibrated vertical deflection system to allow interpretation of observations. Since the oscilloscope equipment must accommodate the highest voltage in the converter, voltage probes are used to extend the voltage rating. These probes should be matched in rating and performance for use in differential observations. Type 453 portable scope with suitable voltage probes has been useful to examine rectifier wave forms. For systems up to 600 volts use Type P6006-10X probes and for systems to 1500 volts use Type 6007 (100 to 1) probes. Type P6023 probes are often very useful to improve rejection of voltage signals to grounds for more accurate presentation of desired waveforms. The C-30 camera available for the Type 453 oscilloscope is also a very useful tool.

The sweep system should be calibrated in sweep speed, and should preferably be of a triggered type to allow reliable display of the signal. If waveforms are to be recorded on film, the sweep generator should also include a single sweep feature. Type 453 contains all those features.

The oscilloscope equipment suggested must be correctly adjusted to produce accurate results. First the pre-amplifier is adjusted for DC balance (set AC-GND-DC switch to GND, set sweep mode switches to auto trig) by varying the step attenuator control until no trace of motion is seen as the variable vertical calibration control is moved from 20MV to 5MV. Both voltage probes must be adjusted for correct frequency compensation using the calibrator as a square wave and adjusting the AC coarse compensation control in the termination box of the probe. Voltage calibration of the probe and pre-am amplifier is adjusted by the gain adjustment, using the calibrator signal as a standard. The above operations are done with one signal circuit at a time. The scope is changed to display the difference between channel done with one signal circuit at a time. The scope is changed to display the difference between channel 1 and 2 by setting mode switch on "Add" and inverting the channel 2 signal. Both probes are connected to the calibrator. Set calibrator on 1V. Turn trigger from normal to channel 1. Adjust gain control of channel 2 to obtain straight line with volts/div control set on 10MV per division for 10 to 1 probes. If adjustable P6023 probes are used, set gain control of both channels full clockwise and adjust the DC attenuator control and AC compensation fine adjustment of the probes for the minimum signal display. In final adjustment, common mode rejection can be checked by connecting both probes to the same circuit point to be tested. The display should remain a straight and stationary line.

SEQUENCE OF TESTS: STEPS 1 to 10

STEP 1 – PERFORM PRELIMINARY INSPECTION APPLICABLE TO ANY NEW EQUIPMENT

1. Proper assembly and connections, both mechanical and electrical.

2. Loose, missing or damaged parts.

3. Primary breaker and converter transformer inspection and tests. Set transformer taps as required to provide rated input volts to the rectifier.

4. Megger insulation tests. See Item 2 under Precautions above.

   CAUTION: DO NOT MEGGER CONTROL CIRCUITS BECAUSE OF POSSIBLE DAMAGE TO TRANSISTORS, DIODES, CAPACITORS, ETC.

5. Check ratings and circuit functions in the light of specifications to be certain the intent is met.

STEP 2 – APPLY AUXILIARY CONTROL POWER AND TEST OUT ALL LOGIC, PROTECTIVE RELAYING AND INTERCONNECTING CIRCUITRY PER ELEMENTARY DIAGRAM.

NOTE: The exact control logic will depend upon the application of the drive. The objective of the following is to present a general format for sequencing without excessive emphasis on detail.

1. Remove all control and excitation circuit fuses including the power supply fuses in the excitation modules. Disconnect the primaries of the transformers supplying the DC Breaker fault trip circuit and the primary of the control power transformer. Open the toggle switch which supplies in-phase power.

2. The control power source will be 460 volts, 3 phase, 60 hertz or 380 V, 3 phase, 50 hertz. Refer to the elementary for the specific drive.

3. Insert the control-power fuses associated with the control-power safety switch (CPS), open all fan-motor starters (FMS), and close the control-power safety switch. As each fan-motor starter is closed, the cooling fans should start, drawing air in at the bottom and exhausting air at the top (unless special ventilation arrangements have been applied).

4. Open the control-power safety switch and connect the primary of the control-power transformer. Close the control-power safety switch; the control-power on the secondary of the control-power transformer should be 115 volt, single phase. Alternately open and close the control-power safety switch and insert fuses in the control-power circuits as required for sequencing; refer to the following, however, before so doing.

5. With control-power applied, the converter transformer fault relay (TRX) should be energized. On 6-pulse applications with dry type transformers, relay TRX will be deenergized by overtemperature detectors, embeded in the transformer windings, one per phase. On 6-pulse applications with

liquid filled transformers, relay TRX will be deenergized by either the liquid overtemperature or the liquid level detector. A transformer fault will initiate a process fault, controlled stop, or annunciation, depending on the application. Fault indication will be by means of a white indicating light. Reset of an overtemperature will be automatic after time to cool.

On 12-pulse applications, relay TRX again will be deenergized by an overtemperature or low-liquid level. A transformer fault will initiate an annunciator signal when applicable. Indication at the Silcomatic will be by means of a white indicating light, and reset of an overtemperature will be automatic after time is allowed for cooling. A transformer fault pressure relay operating on internal transformer faults will trip and lockout the AC breaker in the switch gear.

6. The cooling fan motor centrifugal switches (FSS) should close and maintain relay FLX energized during normal operation. Loss of a cooling fan will initiate a process fault stop or a controlled stop, depending on the application. Fan loss indication is achieved by energizing an amber indicating light mounted on the door of the relay panel.

7. Relays OLF and OLR (suffixes often will exist for clarity) are inverse time overcurrent relays monitoring rectifier current by means of the AC current transformers. Operation of either relay, in conjunction with auxiliary relays, will cause a process fault stop or a controlled stop, depending upon the application. Reset of the overload is automatic after an adjustable time delay. Overload indication in all cases is by means of a target on the overload relays. A contact is also provided for a remote light at the operator's station.

   Set the tap and time dial of the IAC relays in accordance with the information given on the elementary. Sequence the relaying by rotating the induction disc on the IAC relay until the relay interlock closes. Check and adjust the reset time, when applicable, for 75 seconds at the OLX timing relay.

8. Retard of pulses is accompanied by dropout of relay IOC which energizes a red indicating light on the door of the relay panel and initiates a process fault stop. Retard Stop requires manual resetting by a pushbutton at the fault monitor card on the page. Check sequencing while making the overcurrent adjustments outlined in Step 3.

9. Sequence breaker-control circuits in accordance with the elementary.

10. Interlocking should assure the following coordination:

    a. It is preferable to have excitation control power before the converter transformer is energized.

    b. The DC load circuit cannot be closed until the converter is energized. Tripping of the AC breaker or DC breaker opens the undervoltage-trip circuit to the process control equipment.

    c. The undervoltage-trip circuit to the process control equipment cannot be closed until all tripping-type functions in the Silcomatic are reset and in normal operating condition.

    d. Appropriate indicating lamps are energized to identify faulty operation at the Silcomatic as follows:

(AIL) – AMBER – Loss of cooling air

(WIL) – WHITE – Transformer fault

(CIL) – CLEAR – Open voltage surge suppressor fuses (light goes out)(applicable only on 2400V, or higher main feeder voltages).

(CIL) – CLEAR – Failed thyristor (SCR) or isolating fuse in power modules (light goes out)

(RIL) – RED – Fixed angle of retard of pulses

e. Target indication on relays OCL, OLF, OLR, preclude use of lamps.

STEP 3 – TEST OUT PULSE RETARDS, FAULT OVERCURRENT, AND DIFFERENTIAL TRIP FEATURES OR EXCITATION CONTROLS PRIOR TO APPLYING THE MAIN POWER TO ENSURE THAT THE FAULT PROTECTION IS SATISFACTORY.

1. If the main power and phase sensitive control power are applied to the converter during the check-out period of the page, the auxiliary C.T.'s have to be disconnected from the main C.T.'s which in turn have to be shorted. Exercise extreme caution at exposed high voltage points!! If the main power is not applied while the phase sensitive control power is on, a source of adjustable 0 to 115 volts isolated control power can directly be connected to the secondary side of the main C.T.'s. (The inductance of the current transformer windings blocks current from circulating through them.) Figure 2 illustrates test connections at points A, B, C, and N.

2. With an oscilloscope connected across the overcurrent point on Test Point Board – (see the specific elementary diagram), observe the peak DC signal volts as the test signal is applied and varied at each terminal.

   The resultant waveform should be a full wave, 120 hertz.

   If any unbalance or missing waves are observed, investigate before proceeding. The cause may be wiring errors, component failure, or improper wave shape from the source.

3. The equation for calculating overcurrent circuit values is:

   $$E = \frac{I_D \times R \times OL_{pu} \times 1.2}{\text{(product of CT ratios)}},$$

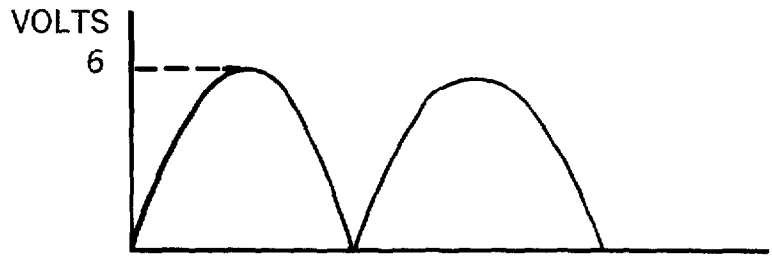
   I_D = rated DC for the converter

   R = value of the loading resistor in ohms

   OL_{pu} = maximum overload value permissible for the rectifier expressed in per unit.

   1.2 = factor for operating margin (20% above the overload rating)

Increase the test voltage until the DC signal (peak value) is 6 to 7 volts as indicated in Figure No. 23A.



D-C SIGNAL (Peak Value)

FIGURE No. 23A

Overcurrent trip should occur and relay IOC in the module should drop out. If not, adjust jumpers on Current Sensor.

Upon drop out of relay IOC, check indicator lights and associated relaying in accordance with the elementary.

4. With the gate pulse firing circuits functioning such that the pulses can be observed on an oscilloscope, ensure that IOC does retard all of the pulses instantaneously in both converters. Refer to paragraph 2 under Step 4 for the circuit location for observing pulses. Note that on a 12-pulse converter with A and B sections, an overcurrent trip on one lineup will cause a trip on the other lineup, and all DC breakers will be tripped. However, the PCB card lights indicate only on the unit with overcurrent.

5. In preparation for testing the fault overcurrent and the differential trip functions, the high-speed tripping circuit of the DC breaker must first be tested.

   CAUTION: HIGH VOLTAGES ARE PRESENT ON THE HIGH-SPEED TRIP CIRCUITS. BEFORE MAKING PHYSICAL CONTACT WITH THESE CIRCUITS, REMOVE THE CHARGING SUPPLY AND ALWAYS MEASURE THE VOLTAGE ON THE TRIPPING CAPACITORS TO ASSURE THE COMPLETE DISCHARGE OF THE CAPACITORS.

   a. Open the control-power safety switch, connect the primary of the fault trip control supply transformer, and then close the safety switch; the transformer secondary voltage and the steady-state DC voltage on the tripping capacitors should be as indicated on the elementary.

   b. Likewise, the pickup and dropout voltages of relay FTU should be as indicated on the elementary. As a safety precaution, FTU should drop out within 2 to 5 seconds after removal of the fault trip control AC supply voltage. As a result of this time delay, the dropout of FTU, which is adjustable by means of the 2-POT's can be readily checked by placing a voltmeter across the capacitors removing the AC supply voltage, and observing dropout. Charging of these capacitors is much more rapid, therefore confirmation of the pickup setting requires the replacement of the AC supply with a variable supply for gradual increase of voltage. As an approximation, pickup should be at steady-state voltage, whereas dropout should be at 80 percent.

6. For testing the fast trip of the AG-7 breaker, remove the control power and short the FTU contact in the DC breaker holding coil circuit to rule out the possiblity of a semi-high-speed trip by FTU dropout during testing. Insert the differential trip card(s) in the page. Apply the control-power, push the reset button on the fault monitor board, and close the DC breaker, applying jumpers as required in the closing circuit.

7. Testing of the differential trip board involves testing of two functions as follows:

   a. AC current exceeds DC current by 50% or more of rated current. This function is intended to trip the AC breaker by excitation of RLO relay.

   b. AC current exceeds the preset overcurrent rating (usually 120% of current limit) and is intended to trip the DC circuit breaker by the fast trip circuit.

   Connect the oscilloscope to observe rated current = 6 volts as indicated on the elementary diagram of the differential trip board. Increase the adjustable AC source mentioned in Part I Step 3, and observe the Led indicators on the differential board trip. Led 1 will light as the oscilloscope indication shows 3 volts peak or greater and RLO relay should pick up, which will light the breaker lockout light on the control door. If the adjustable source is decreased, all light indications remain..

   Connect the oscilloscope to the overcurrent point on the Test Point board as in Step 3 part 2. Increase the AC adjustable source and observe the Led lights again. The DC breaker will trip, Led 2 on the differential trip board and Led 1 on the Fault Monitor board will both light as the oscilloscope indication shows 6 to 7 volts and will remain lit as the adjustable source is lowered.

   Reduce the adjustable source to zero volts and preset the reset button on the TFMA board; all Led indicators should be off except Led 2 on the TFMA board and one or both Leds on the LO board.

8. The differential trip board also provides a fast trip function for the DC circuit breaker as DC current exceeds AC current by 50% of rated current. To test this function, connect an adjustable power supply, capable of about 5 volts at 1 ampere, across R_{DC} shown on the differential trip elementary with negative on common.

STEP 4 – DETERMINE THAT FIRING PULSES ARE DELIVERED TO EVERY THYRISTOR AT THE PROPER TIMES, THAT THEY ARE PROPER IN WAVE SHAPE AND ENERGY LEVEL, THAT THEY CAN BE ADJUSTED TO OCCUR OVER APPROXIMATELY 180 DEGREES RANGE FOR PHASE CONTROL PURPOSES AND THAT NO NOISE OR OTHER EXTRANEOUS SIGNALS ARE PRESENT IN THE CIRCUITS TO THE THYRISTOR GATES.

1. Apply control and excitation power and observe the 115 volt, 3-phase, 60 hertz in phase control power input to the signal and limit voltage transformers. It should be balanced 3-phase power, not less than 110 volts and not over 120, with phase rotation 1-2-3 as shown on the control power elementary diagram. It is quite important that the waveshapes be smooth, undistorted sine waves, without hash, spikes, or other noise effects present.

Refer to the AC control power circuit. Check the phase relationship between the incoming and outgoing voltages and assure that signal voltages and limit voltages are phased per the elementary diagram and phasor diagram.

2. Observe the output of the gate pulses at 60-degree intervals at the six pulse output of the Gate Pulse Generator and Buffer Amplifier outputs. These are both available at the Test Point Board at the pins indentified in the elementary diagrams for the gate pulse circuits.

   The wave form of the individual output pulses from the Gate Pulse Generator is a pulse that rises exponentially from zero volts to 12 or 14 volts and returns to zero in 200 microseconds. The wave form of the Buffer Amplifier is a square wave of about 180 microseconds duration but voltage amplitude between 5 and 7 volts; rise time is well below the microsecond time. Observe that all pulses are as described and similar in shape.

3. Impress a finely adjustable voltage signal $\pm$ 1.0 volt DC as a reference input at ER17 and ER18 input points after removing the interconnecting wires from Process Control, and observe phase control of the pulses over approximately 180 degrees range from full advance to cutoff. The following points may be helpful.

   a. This observation is at open loop and with a gain of about 50:1 in the amplifier, less than one volt is required to drive the pulses to full advance. Use the test supply board as a voltage source.

      Another convenient way to apply the 1.0 volt signal is to connect a rheostat of 100 ohms or more in series with the ohms scale of a conventional ohmmeter, using the brush arm as a dividing tap to obtain an adjustable reference signal.

   b. Note that pulses are produced in both gate pulse generators, F and R, with zero reference signal.

   c. Each output of the amplifier is driven by a separate integrated circuit operational amplifier; each is provided with a bias input signal which will make the amplifier output (-5.2) volts for a 60 hertz system. This negative output produces a phase retard of about 155 degrees in each gating circuit to provide a dead band effect to prevent circulating currents between the two rectifiers.

      With the Pre-Amplifier card withdrawn measure the output voltage. This should be between 5.2 and 5.4 for 60 hertz systems, between 6.25 and 6.45 for 50 hertz systems.

   d. With the Pre-Amplifier card withdrawn, to force 0 volts input to the amplifier, both sets of the gate pulses should be retarded about 155 degrees. If the Pre-Amplifier card is inserted and, as a test signal, is applied as in "a" above, one set of pulses should advance to the full advance stop — this should be a motion of 150 degrees. The other set of pulses should retard about 30 degrees and disappear. If the test signal polarity is reversed, the first set of pulses should disappear and the second set move to full advance stop.

   e. If misoperation is encountered on the above tests, it may be helpful in identifying the problem area to bypass the regulating amplifier module and drive each Gate Pulse Generator directly with a test reference of $\pm$ 6 volts ($\pm$ 8V for 50 Hz). In this mode, zero signal shifts

pulses to 90 degrees retard; 6 volts drives them to a full advance (7.2 volts at 50 Hz) and about (-) 5.5 volts retards them to cutoff. (- 6.8 for 50 Hz)

f. The pulses can be easily and quickly observed over the full range, but unless some apparent problem exists, it should not be considered necessary to make any detailed tracking tests over the full range.

STEP 5 — APPLY PRIMARY AC POWER TO THE CONVERTER TRANSFORMER WITH THE CONVERTER GATE PULSES REMOVED.

1. Remove all Buffer Amplifier and Gate Pulse Generator boards to remove all gate pulses.

2. With all other controls energized and operable, close the AC primary breaker, energizing the power circuit of the converter.

3. Measure and record transformer secondary voltage. Compare with the rating of the equipment.

4. For drives with AC line voltage above 600 volts, observe with a scope presence of 6 pulse (360 hertz on 60 hertz base) ripple across diode clipper output terminals 1-2 of VSS. The waveform will be similar to the DC bus voltage of a Silcomatic operating at full advance.

5. Observe DC bus with scope for absence of any voltage waveform to be sure no cells are shorted. No waveforms should exist.

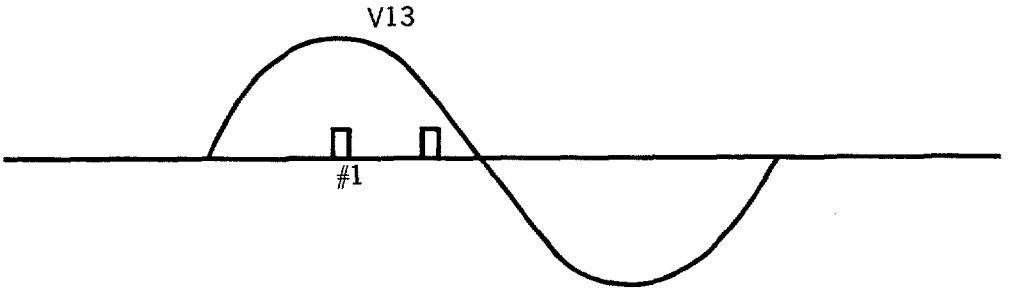
STEP 6 — PHASE OUT POWER CIRCUIT AND EXCITATION CONTROLS TO ENSURE THAT GATE PULSES WILL FIRE THE THYRISTORS IN PRECISELY CORRECT TIME RELATIONSHIP WITH MAIN AC POWER WAVE.

1. Remove the main power by opening the AC breaker, and remove in-phase control power.

2. Remove the Amplifier card and add a card with a jumper from pin 10 to pin 12 in the slot of the Amplifier board. Insert the Forward Gate Pulse Generator and both Buffer Amplifier printed circuits. ADD ONLY THE FORWARD GATE PULSE GENERATOR BOARD. This will produce zero volts at the control input to the Gate Pulse Generator which positions the forward gate pulses at 90 degrees retard.

3. Apply in-phase control power. Remove the gate lead plug on the X1-X2 terminals at the top leg assembly of one of the forward bridges. Connect a resistor of approximately 25 ohms from X1 to X2 of the PAT assembly with the oscilloscope connected across the resistor. This should display two pulses of about 7 volts, amplitude, 180 microseconds duration spaced 60 degrees apart. Position the first pulse (No. 1) at the middle calibration of the scope graticule.

4. Since the correct phase rotation and presence of all six consecutive gate pulses at 60-degree intervals has previously been checked, the relationship with the power wave is now easily determined. Refer to Figure 23B.



FIRING PULSES IN RELATION TO AC VOLTAGE
AT POWER BRIDGE

FIGURE No. 23B

5. Apply main power: Without changing any scope adjustment except vertical attenuation, observe the point at which power phase 1, with respect to 3 goes positive, giving care as to the polarity of the probe leads. Phase 1 is at the top facing the thyristor power module, feeding phase legs 1 and 4. Phase 3 is on the bottom feeding phase legs 5 and 2. It should become positive exactly 90 degrees before the index point for number 1 pulse. If not, check source relationship of power and control, phasing of the signal and limit transformers and phase connections in the rectifier.

6. Remove main power and in-phase power. Remove the Forward Gate Pulse Generator circuit boards and insert the Reverse Gate Pulse Generator board. ADD ONLY THE REVERSE GATE PULSE GENERATOR BOARD. On the jumper card in the Amplifier card slot, move the jumper between pins 10 and 12 to be between pins 16 and 17. This will produce zero volts at the control input to the reverse Gate Pulse Generator which will position the reverse gate pulses at 90 degrees retard.

   Compare pulse 1 of the Reverse Gate Generator with the voltage from line 1 to line 3 as in parts 3, 4 and 5 above.

   Remove main power and in-phase power. Remove all Gate Pulse Generators and replace the jumper card with the Amplifier board.

STEP 7 – OPERATE THE CONVERTER ON A RESISTIVE LOAD. OPERATE ONE POLARITY AT A TIME BY REMOVING PULSES OF THE OTHER PACKAGE. THEN OPERATE BOTH CONVERTERS BACK-TO-BACK.

1. Remove all power and connect a load resistor across the output DC bus, the resistor being rated for at least 5 percent rated amps at rated DC volts.

2. Connect a reference signal (the supply should be at least 15 volts DC, 10 MA) to the input

terminals. The interconnecting wires to Process Control should still be disconnected at this point.

3. Confirm that all printed circuit boards included in the elementary are in the page except the forward and reverse Gate Pulse Generator Cards.

4. Adjust the reference signal to zero and insert the forward Gate Pulse Generator board. Apply the control power, then the main power. Varying the reference from zero to 15 volts should vary the output from zero to approximately 1.25 x rated DC voltage with a light resistive load. Output DC voltage ripple will produce a saw-tooth like appearance on the waveform with 60 degree spacing (360 hertz on a 60 hertz base). These output waves should remain evenly balanced in amplitude and spacing. See Figures 23C, 24, 25, 26 and 27.

5. With the converter operating on the resistive load, recheck the suppression, lockout and other circuits as applicable, including voltage feedback, current balance, etc., for signal wave shapes. Be certain that all waves are present and that no extraneous signals or unexplained shapes are present. Basically the voltage and current waveforms are 360 hertz, evenly balanced similar to those in the main power circuit. Ascertain that the voltage signals at the loads of the auxiliary current transformers for the subject functions are correct for the circulating bus current; average DC measurements are satisfactory for this test.

   Lockout should occur with 2 to 5 percent current flowing in the forward converter. This function can be checked by noting the two outputs of the lockout board displayed on the Test Point board when the forward current is 2 to 5 percent of rated current.

6. With rated DC voltage on the converter output, confirm that the voltage sensor produces 6 volts feedback. At the same time,, determine that rated DC voltage is obtained from the converter with 15 volts DC reference applied. If this condition cannot be met, the value of DC bus divider resistors of the Input module may require replacement.

7. Adjust the reference supply to zero, remove all power, disconnect the forward Gate Pulse Generator, and repeat items (4) and (6) for the reverse direction.

8. With the lockout circuits functioning properly and with all thyristor power modules connected, the converter may be operated back-to-back by connecting all the Gate Pulse Generator boards.

9. Swing the reference from zero to full-on and observe the output for any irregularity. It should be completely smooth.

STEP 8 – OPERATE THE CONVERTER ON THE RESISTIVE LOAD. MEASURE THE RIPPLE.

Performance of the voltage regulator can be evaluated to a limited degree by observation of the ripple voltage appearing across the output of the regulating amplifier at point EC of test point card. Peak-to-peak amplitude of this ripple voltage at its maximum value has a definite relationship to the unity gain cross-over frequency of the voltage regulating loop. This relationship is such that approximately .5 volts peak-to-peak at maximum amplitude represents cross-over frequency of 1000 radians/second, the intended design value for 60 hertz applications. Ripple voltage is introduced into the regulating loop via the feedback.

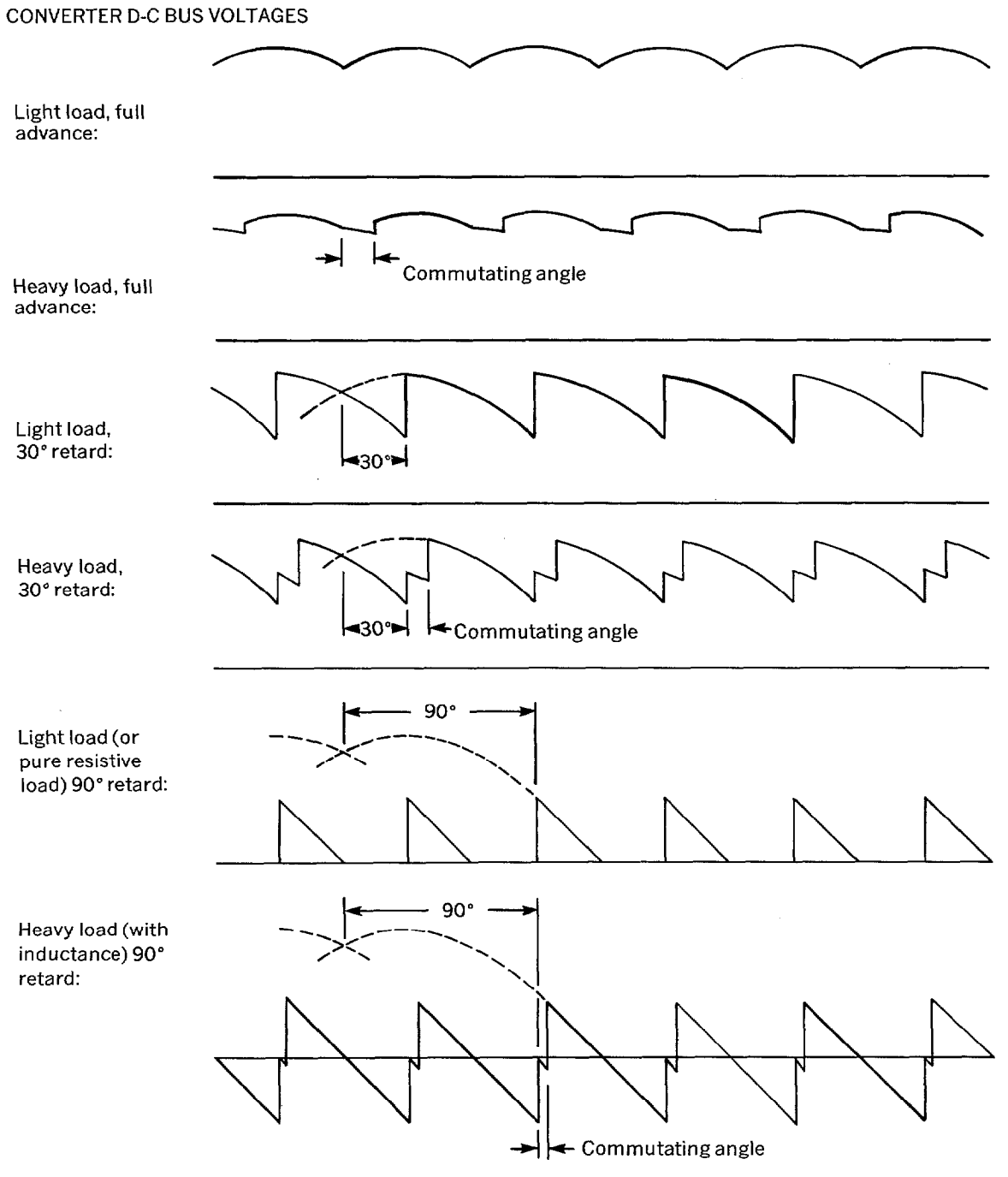


FIGURE No. 23C: DC BUS VOLTAGE WAVE FORMS – RECTIFYING

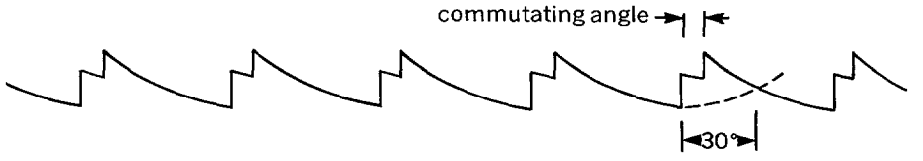
CONVERTER D-C BUS VOLTAGES

Light load, 30° advance

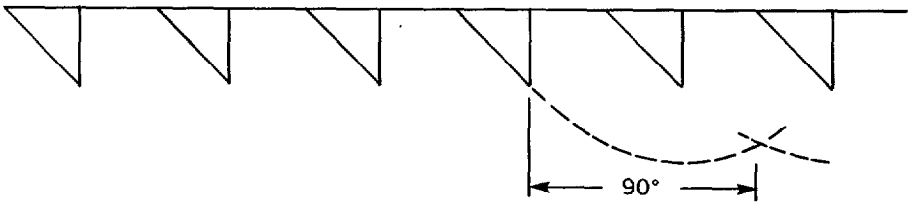
(Note-inverter cannot operate at 0° advance, and normally should not be operated at less than 20° advance)



Heavy load, 30° advance:



Light load, 90° advance:



Heavy load (with inductance), 90° advance:

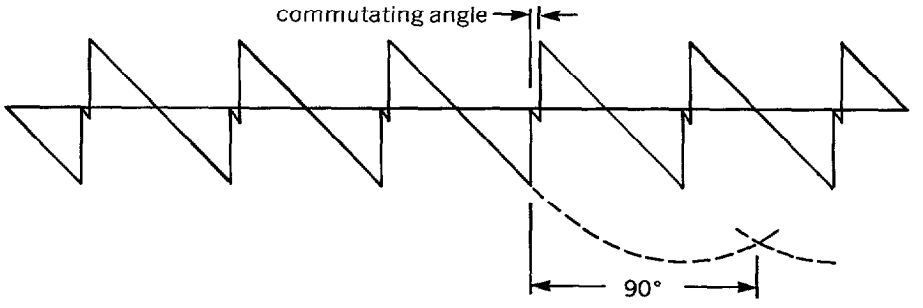
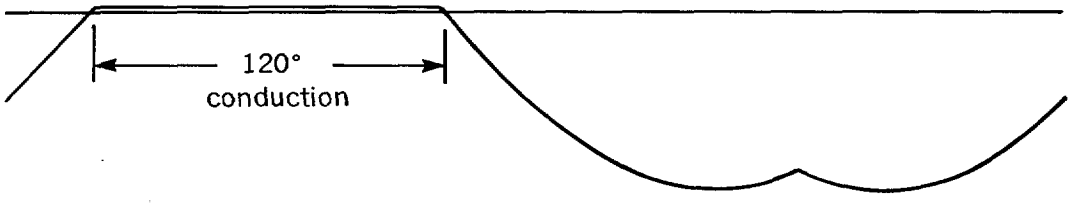


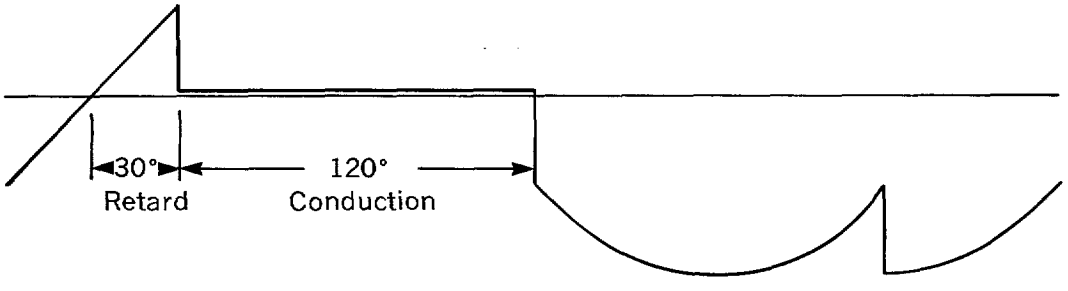
FIGURE No. 24: DC BUS VOLTAGE WAVEFORMS – INVERTING

SCR CELL VOLTAGES, ANODE TO CATHODE
(Bridge or Double-Wye Circuit)

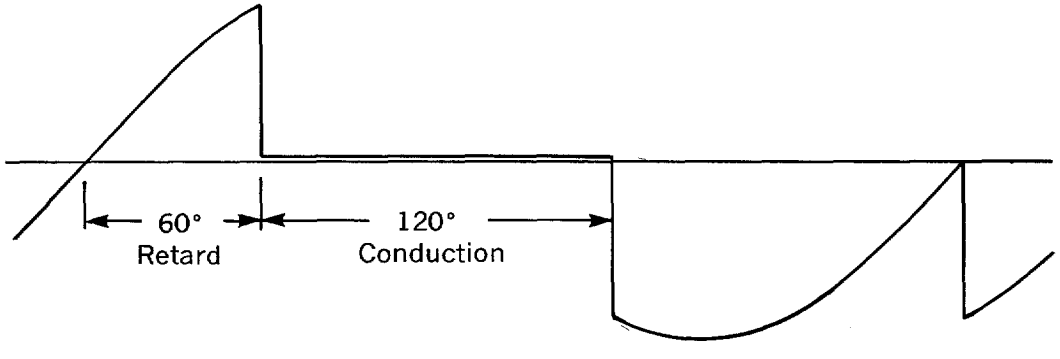
Full Advance (Rectifying):



30° Retard (Rectifying):



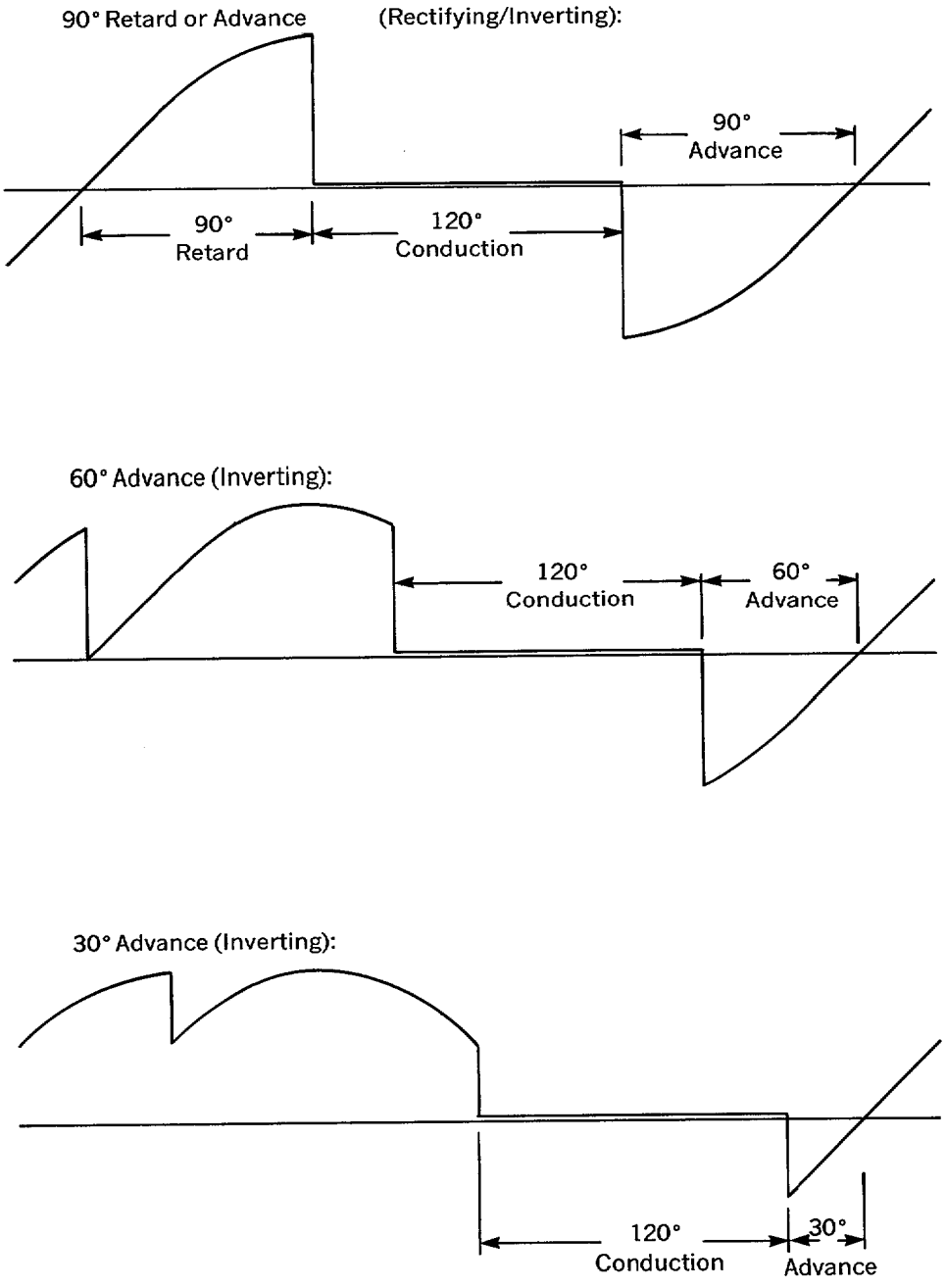
60° Retard (Rectifying):



THYRISTOR CELL VOLTAGES, ANODE TO CATHODE – RECTIFYING

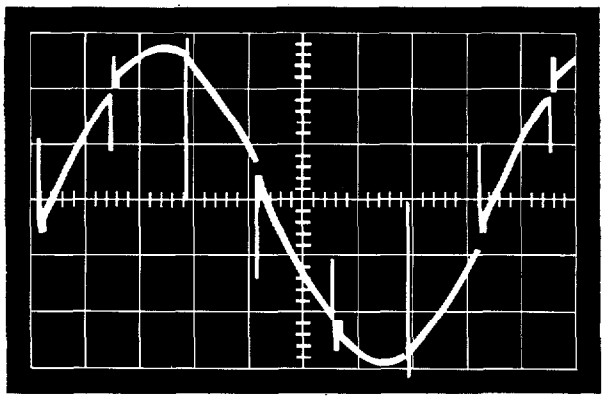
FIGURE No. 25

WAVEFORMS

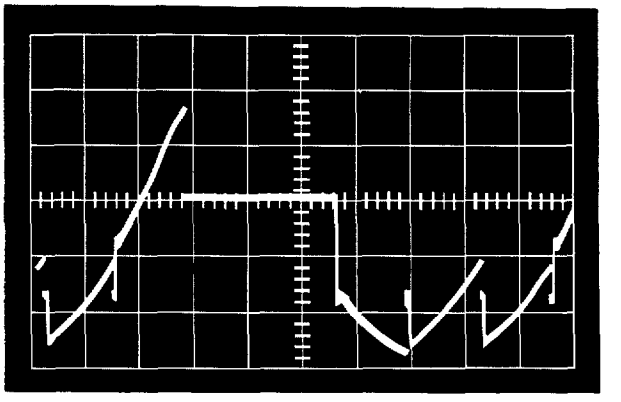


THYRISTOR CELL VOLTAGES, ANODE TO CATHODE – INVERTING

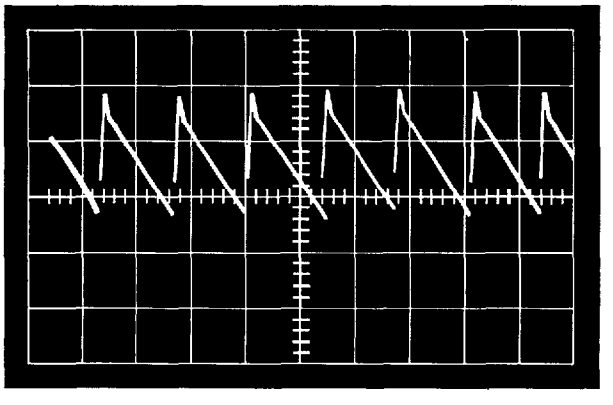
FIGURE No. 26



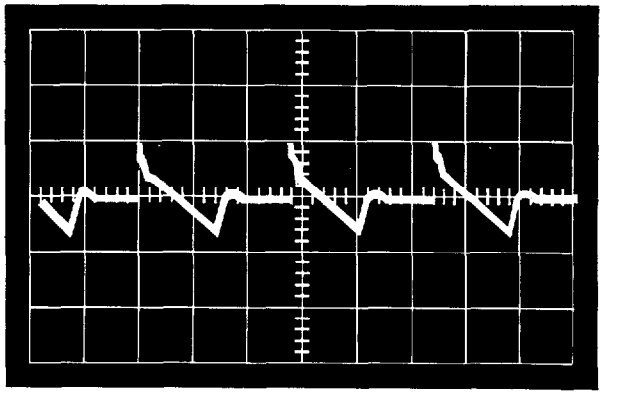
AC Secondary Line Volts
Scale: 2M sec/cm, 100 v/cm
Oper. Con. 100 V @ 100 A Inverting



SCR Cell Volts, Anode to Cathode
Scale: 2M sec/cm, 100 v/cm
Oper. Cond. 220 V @ 100 A Rectifying



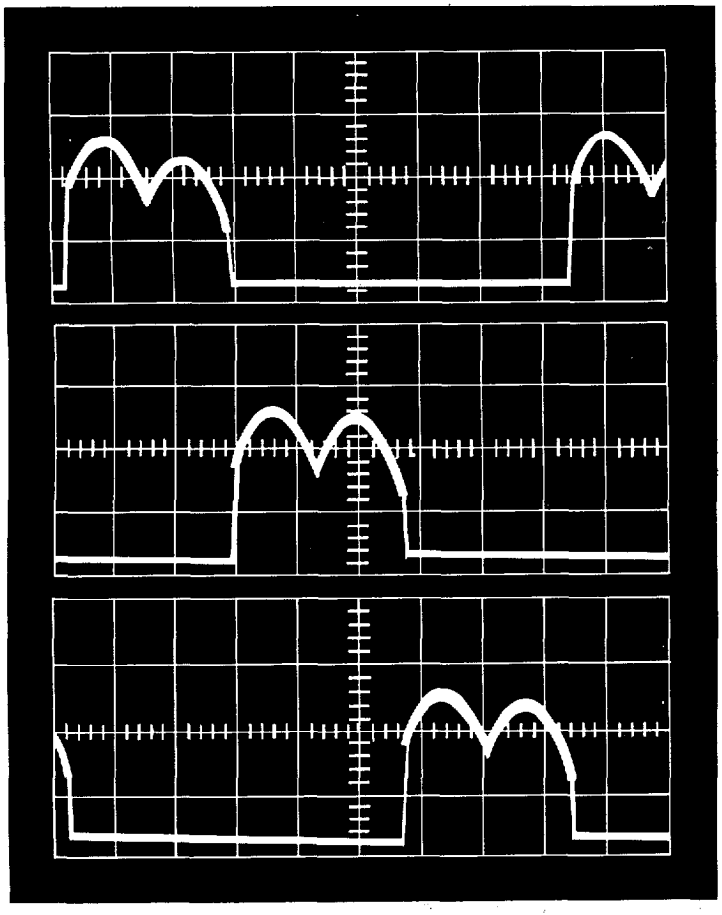
Rectifier DC Bus Volts
Scale: 2M sec/cm, 100 v/cm
Oper. Cond. 95 V @ 125 A Rectifying



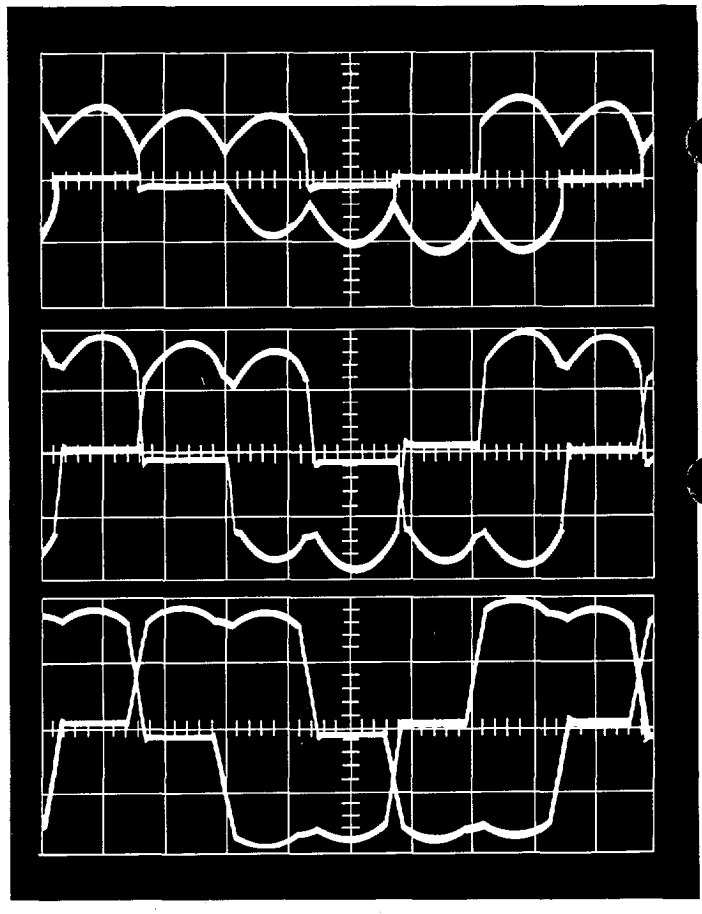
Rectifier DC Bus Volts
Scale: 1M sec/cm, 100 v/cm
Oper. Cond. +V @ +A, Motor Load, Running Light

TYPICAL OSCILLOSCOPE PHOTOGRAPHS OF
CONVERTER WAVEFORMS

FIGURE No. 27



Rectifier Cell Current
Phase Legs 1, 3, 5 Top to Bottom
Oper. Cond: @ 25% Rated Motor Load
Calib.: 25 A/CM, 2 MS/CM Sweep



AC Secondary Line Current
Phases A and C
Oper. Cond.:
Top Trace: 25% Motor Load
Center Trace: 50% Motor Load
Bottom Trace: 100% Motor Load
(Bottom Trace Amps/CM is 2X Calib. of Top and Center)

CONVERTER CURRENT WAVEFORMS

Peak-to-peak amplitude of amplifier signal ripple voltage varies as a function of the percent ripple on converter DC bus volts. Converter bus ripple varies from a minimum of about 5 percent of E_{do} at full advance (operation as diode) to a maximum (theoretical) of 24 percent at zero DC volts with continuous current. Therefore, ripple volts on the regulating amplifier signal also increases as phase retard results in decreased converter volts, so long as the current is continuous.

In the actual bridge converter, current becomes discontinuous (current ceasing between successive conduction periods) at between 60 and 90 degrees retard, depending on the amount of inductance in the circuit. Ripple appearing on the regulating amplifier signal decreases with further phase retard after current becomes discontinuous, due to the increasing zero-current time in each conduction period. Therefore, the maximum value of ripple voltage on the regulating amplifier signal occurs at the lowest converter DC volts at which current remains continuous for any given converter and load circuit inductance. From about 25 percent rectifier volts to zero, there is little change in the percent ripple volts; therefore, the amplitude of ripple on regulating amplifier signal can be measured at any point between these values, provided current is continuous, for determination of cross-over frequency of the voltage regulator.

Measuring the ripple is relatively simple with an oscilloscope, with the following points being observed:

a. Load current out of the rectifier at least 5 percent or more is necessary for waveshapes to to assume proper values for accurate measurements.

b. When measuring the ripple using a resistive load, it must be recognized that discontinuous current occurs at higher rectifier volts (approaching 50 percent of E_{do}) than with the more inductive motor circuit. Therefore, peak-to-peak amplitude of approximately 1.5 volts on resistive load corresponds to 1000 radians/second cross-over compared to 2 volts on typical motor loads.

c. The easiest method of obtaining load is by operating at stalled current on the motor once it is checked out and available. However, on initial installation procedure note that the motor is not at this point ready to be operated and it may be advisable to make this measurement on resistive load if possible.

d. The point at which the current becomes discontinuous can be recognized by the flat portion of the wave (zero volts) which appears between successive conduction periods as rectifier voltage is decreased. This flat section represents CEMF volts on a motor load. Continuous current is present as soon as this flat segment disappears, and the Regulating Amplifier ripple is maximum at this point.

e. Swing the reference rapidly and observe for any irregularity at the regulating amplifier output.

Successful operation above indicates that the converter equipment is operating normally and that it is now ready to power the motor.

STEP 9 – OPERATE THE CONVERTER ON THE MOTOR LOAD WITH MANUAL CONTROL OF THE TEST REFERENCE.

NOTE: For reversing converter, operate one direction at a time during the following tests by disconnecting the appropriate Gate Pulse Generator board.

1. Remove all power.

2. The test reference as described to this point can still be applied in these tests; however, the most suitable reference source, if available, would be the Process Control final amplifier for the reference signal to the Silcomatic (via wires R17-R18). Since the final amplifier may be an integrator, it may be necessary to transform the amplifier to a straight gain amplifier by by-passing the amplifier capacitor with a resistor. By disconnecting the regulating reference into this amplifier and substituting a manually controlled signal by a potentiometer, the normal characteristics of reference into the Silcomatic are duplicated. Furthermore, the amplifier output clamps can be used to advantage.

3. Before proceeding with the operation of the motor, which is sensitive to voltage changes and is capable of producing a regenerative fault, the suppression, overcurrent, and differential trip functions should be tested to the fullest extent. The first requirement along this line is to apply a short circuit to the converter output by means of cable or bus bar of adequate size to carry converter rated current. On drives with reactors, the shorting cable or bus bar should be applied between the reactor and the motor so as to benefit from the current-limiting characteristic of the reactor in the event of difficulty. Adjust the reference to zero and apply control and main power. Adhere to the duty cycle during these tests.

    a. On main drive applications, close the DC breakers, short out the tap of the DC output of the converter bridges in the secondary of the auxiliary current transformers applicable to the Differential Trip to the tap of the fault monitor on the Current Sensor and apply an oscilloscope across resistor R_d. Gradually increase the current, noting the symmetry at the waveform at the same time. Lack of proper waveforms could mean incorrect polarities on the bus saturable reactor. The DC breaker should trip when the voltage across the resistor is approximately 6 volts, which in most cases will represent 100 percent current. Adjust the reference to zero and remove the jumper.

        Close the DC breaker, connect the oscilloscope across resistor tap, and the jumper resistor RDC. Increasing the bus current, the AC and DC breakers should trip by means of pickup of relay RLO when the peak voltage across the differential resistors is 3 volts. Again adjust the reference to zero and remove the jumper.

    b. For confirmation of the overcurrent trip settings, independently test each 6-pulse converter by disconnecting the remaining Gate Pulse Generator. Close the breaker(s) and connect the oscilloscope across the fault current tap of the Current Sensor.

        Increasing the bus current to the overcurrent trip setting; the voltage across the tap should be 6 volts and the DC breaker should trip. Confirm the operation of the overcurrent trip by means of the suppression red indicating light and reset the TFMA card. Adjust the reference to zero.

Repeat the foregoing procedure for the other direction. Then remove all power and disconnect the converter output short.

4. Connect the motor load, apply all power, and close the DC breaker(s). Accelerate the motor cautiously while observing the performance with an oscilloscope connected to the DC bus and/or the amplifier output, taking note of smoothness and regularity of the ripple volts, the symmetry of the six consecutive 360 hertz loops, and the presence of any extraneous signal, especially during acceleration.

   For the reversing converter, the 360 hertz ripple waveforms, as seen across the amplifier output, probably provide the most meaningful signal information that can be observed. The ripple should remain smooth, symmetrical, 360 hertz half-sine waveloops, without gaps or extraneous shapes. It should be recognized that the usual flat zero portion shows during any mode where current is discontinuous.

   A vertical spike sometimes appears at the firing point of the thyristors, varying somewhat in magnitude, but on the order of $\pm$ 1 volt. Any other voltage waveform which deviates from the smooth 360 hertz ripple should be investigated to determine the cause. Maximum amplitude on the ripple should be slightly higher than on the resistive load, but not more than 2 volts peak-to-peak absolute maximum.

   Repeat the foregoing operation in the other direction using the same approach as above.

   Connect the gating controls for both directional converter packages and operate them back-to-back. Note that inversion occurs when reducing the reference following motoring in either direction.

   CAUTION: Some caution is to be observed in manipulating the reference pot to avoid imposing excessive current.

   Adjust the reference only at a rate which will hold currents within normal operating limits and will not exceed the commutation abilities of the motor.

STEP 10 – OPERATE THE CONVERTER WITH THE PROCESS CONTROL REGULATOR, MAKING FINAL TUNEUP ADJUSTMENTS.

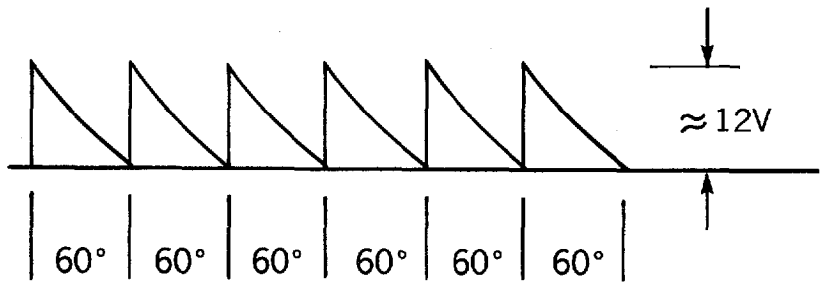
1. For the reversing converter, operate one converter package only until it is ascertained that operation is stable on the complete system regulator. Otherwise, unnecessary suppressions will occur if instability is present. Instability can be recognized readily by appearance of 180 hertz, or even 120 hertz if severe enough, components in the output ripple waveform of the Regulating Amplifier.

2. Make all final settings and adjustments as required for the application. Refer to Figure No. 27 for additional oscilloscope photographs of rectifier waveforms.

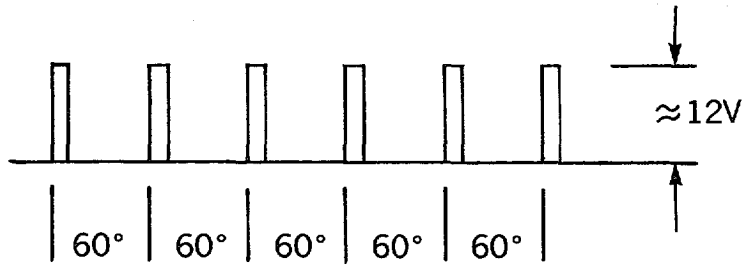
TEST POINT CARD TTFAI WITH APPLICATION INFORMATION

This Test Point Card is used in the Silcomatic Plus in a 6-pulse page or in the master and slave page in 12-pulse, 2 or 4 package systems. This allows observation of pulses as seen on the Gate Pulse Generator and Buffer Amplifier cards as well as control signals applied to these cards. Listed below is an explanation and typical wave shapes. The card is provided with a double set of functions for use with FWD and REV packages.

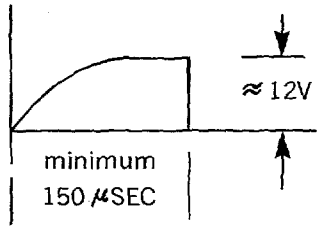
BA: Buffer Amplifier; to observe the output of the Buffer Amplifier showing all pulses at one time.



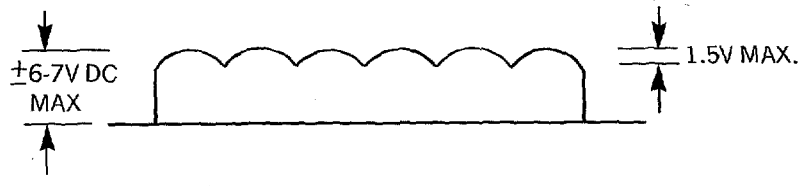
GP: (Pin 8) Gate Pulse Generator; to observe the output of the Gate Pulse Generator showing all pulses at one time.



GP: (Pin 11) Gate Pulse Generator; to observe the phase 1 pulse output of the GPG.



EC: Control Voltage; to observe the Gate Pulse Generator Control Voltage (Error Signal)



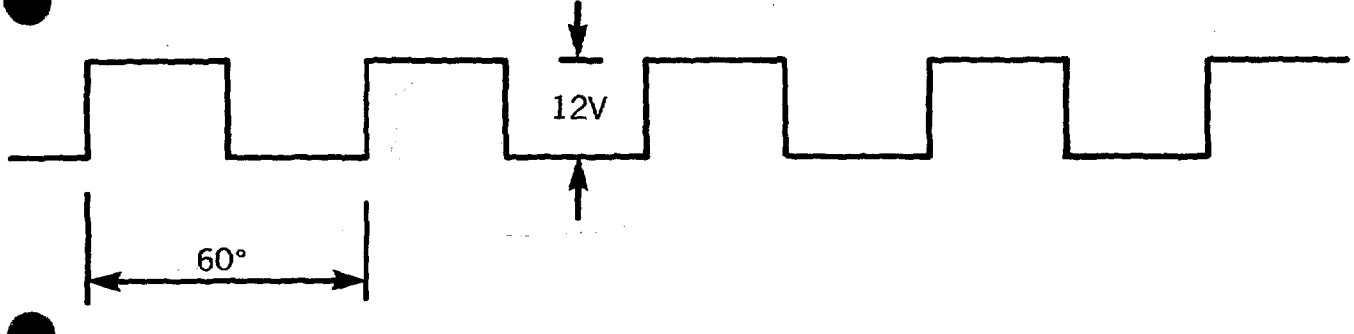
RS: Retard Stop; to observe the state of the retard stop signal.



OC: Overcurrent; to observe the overcurrent signal provided by the current sensor assembly.



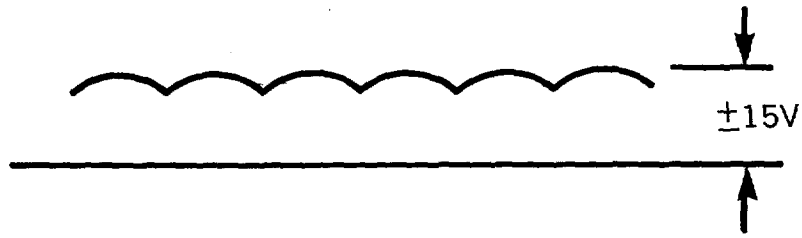
LO: Lockout; to observe the Lockout signal as applied to the Buffer Amplifier.



TEST SUPPLY BOARD TTFCI WITH APPLICATION INFORMATION

This card is used in the Silcomatic Plus 6 or 12-pulse master pages for observation of signals provided by the Current Sensor assembly as well as for insertion of voltage levels to observe simulated operation of differential trip and fault monitor circuits. Listed below is an explanation of test card nameplate terminology and typical waveshapes.

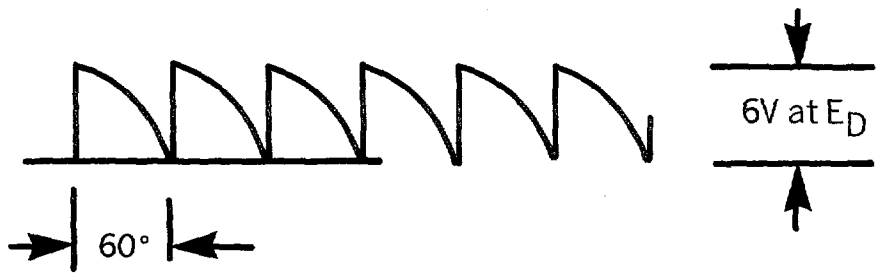
RE: Reference; to observe the system reference voltage.



AC FL: AC Full Load current; to observe the signal representing AC full load current provided by the CSA for testing differential trip and fault monitor operation. Wave shape similar to FOC, 6 V at 100% current.

DC FL: DC Full Load current; same purpose as the full load signal, 6 V at 100% current.

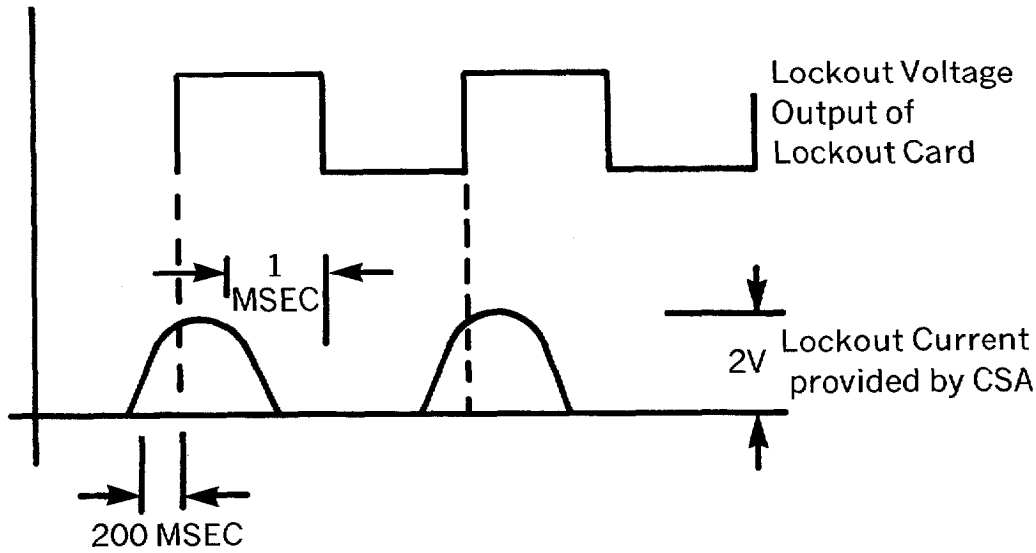
VS: Voltage Feedback monitor



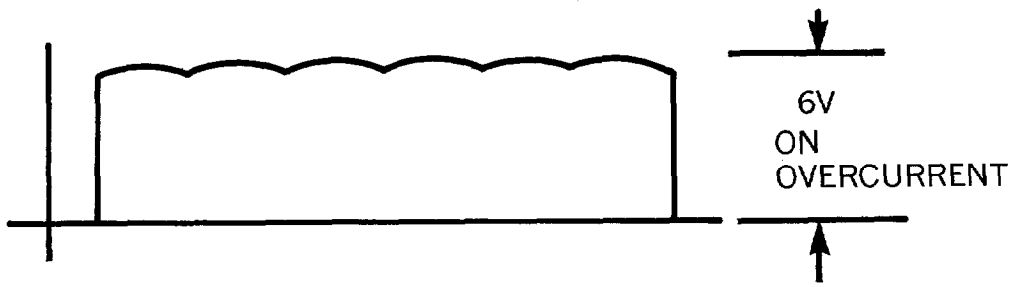
DC BK: DC Breaker; to observe the relay driver output which picks up the instantaneous overcurrent relay IOC on overcurrent. (Similar signal trips DC breaker, 12-volt signal on fault, 0 volts under normal conditions).

AC BK: AC Breaker; to observe the relay driver output which drops out Lockout relay RLO which in turn trips out AC Breaker. 12-volt signal under normal condition, 0 volts on fault.

ILO: Lockout Current sensing; to observe the Lockout signal provided by the current sensor assembly or to inject the 2V lockout voltage normally provided by the CSA for testing Lockout operation.



FOC: Forward overcurrent; to observe the overcurrent signal provided by the current sensor or to inject the 6V level normally provided by the CSA for testing overcurrent operation.



ROC: Reverse Overcurrent; same purpose as the forward overcurrent, same signal.

This board is provided with a 10 turn pot to establish a DC level between (+) and (–) 15 volts for use as a test reference. A +7.5V level is provided for differential trip, current balance, and overcurrent operation and –2.4V is provided for Lockout operation.

The basic power circuit used in the Silcomatic power converter is the three-phase bridge or six-phase double-way circuit. This circuit is made by connecting two simple three-phase converter circuits in series. In order to fully understand the operation of the three-phase bridge circuit, it is necessary to review the operation of the simple three-phase converter circuit. Refer to Figure No. 28 which is a simple three-phase circuit with a resistive-inductive load, for the initial analysis commutation and phase control will not be considered.

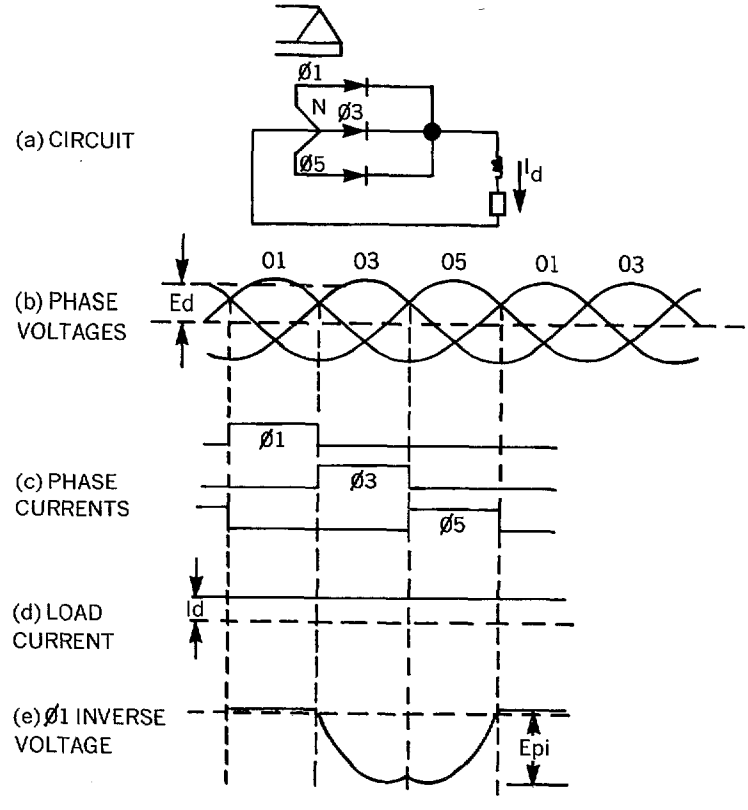


Fig. 28 Three-phase Converter Circuit

The rectifying devices conduct for 120 degrees and the average resulting d-c voltage referred to as E_d* is shown in Figure 28 (b). Each rectifying device conducts for 120 degrees every 360 degrees. The average DC current is referred to an Id and is shown in Figure 28 (d). Figure 28 (e) shows the voltage that appears across the silicon rectifying device in Phase 1. During conduction (120 degrees), this voltage is very small, in the order of one volt per cell. Upon completion of conduction, the voltage across the rectifying device in Phase 1 will be the voltage between Phases 1 and 3 for the next 120 degrees and the voltage between Phases 1 and 5 for the next 120 degrees. The peak voltage during this period is referred to a peak inverse voltage or E_{pi}.

As previously stated, the three-phase bridge circuit is obtained by connecting two, three-phase circuits in series. This connection is shown in Figure 29 (a). Also shown in Figure 29 are the following:

---

*By definition the voltage is E_{do} (no commutation, phase retard, or resistance drops) but for this analysis we will refer to this voltage as E_d.

(b) — Phase Voltages
(c) — Phase Voltages
(d) — Total

Figure 29 utilizes two separate transformer windings. By using a single transformer winding as shown in Figure 30, the six-phase double-way or three-phase bridge circuit is obtained. Also shown in Figure No. 30 are the following:

(b) — Phase Voltages
(c) — DC Voltage Ed
(d) — DC Current Id
(e) — Secondary Winding (a) Current

In the preceding review,only diode or full advance operation of the rectifying devices was considered. Commutation, arc drop, and phase retard are additional factors that must be considered. Figure 31 illustrates the condition of commutation which is simply transferring of current from Phase 1-N to Phase 3-N at a time when Phase 3-N voltage is higher than Phase 1-N voltage. Due to the circuit inductance (represented by reactance X_c), the current cannot transfer from one rectifying device to the other instantaneously. For a certain time the two rectifying devices carry current simultaneously and the DC output voltage is the average of the two transformer line-to-neutral voltages.

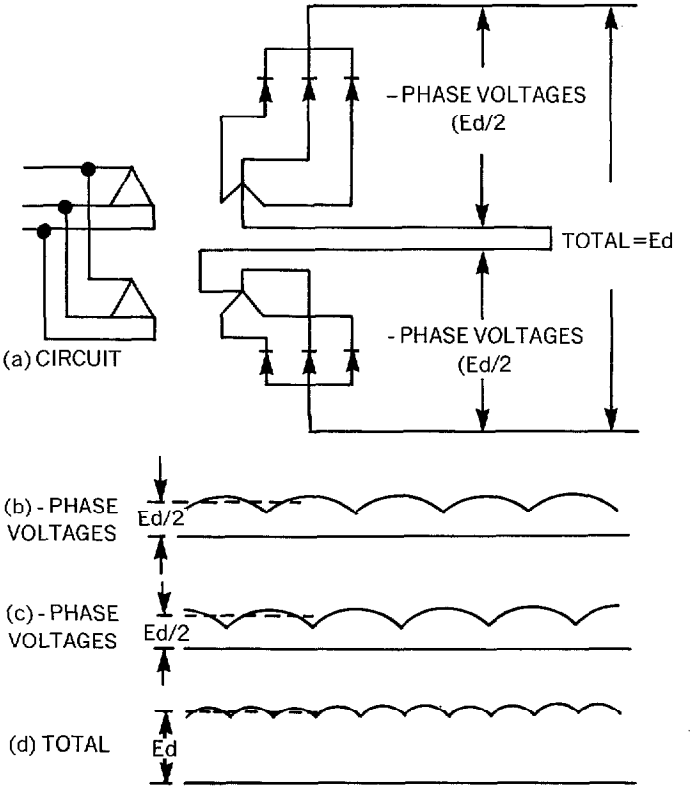


Fig. 29 Series Connection of two, Three-phase Converter Circuits

SIX-PHASE DOUBLE-WAY OR
THREE-PHASE BRIDGE CONVERTER CIRCUIT

FIGURE No. 30

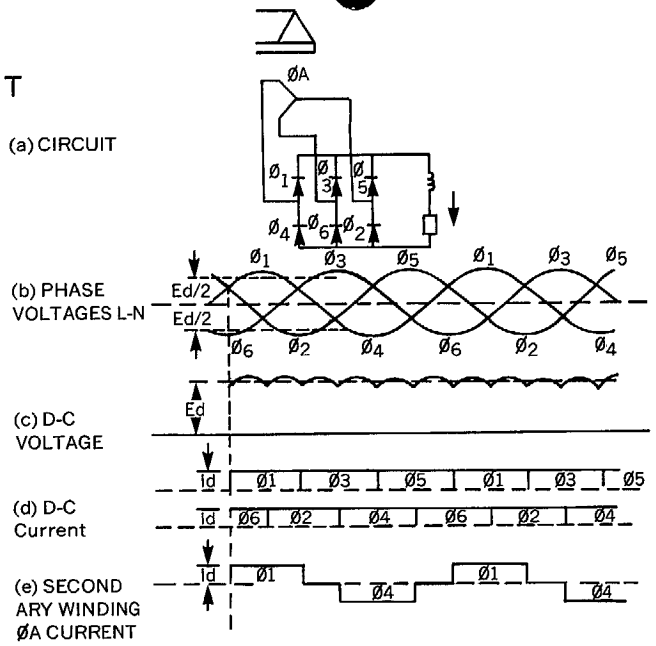


Fig. 30 Six-Phase Double-Way or Three-Phase Bridge Converter Circuit

CONVERTER COMMUTATION

FIGURE No. 31

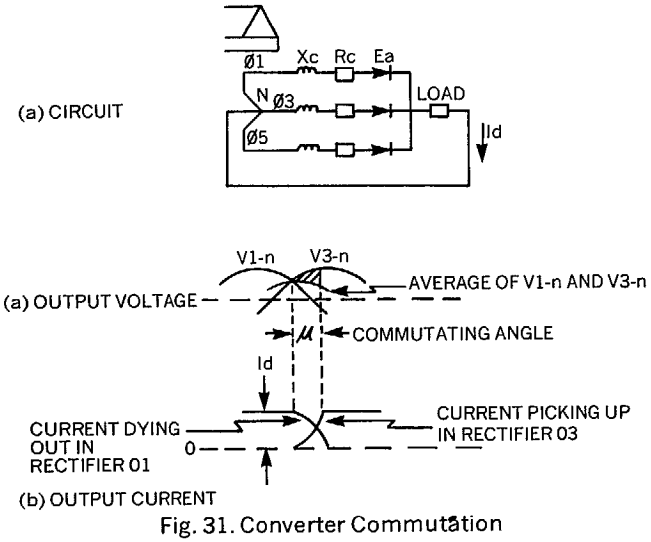


Fig. 31. Converter Commutation

CONVERTER VOLTAGE WAVEFORM
(DIODE OR FULL ADVANCE OPERATION)

FIGURE No. 32

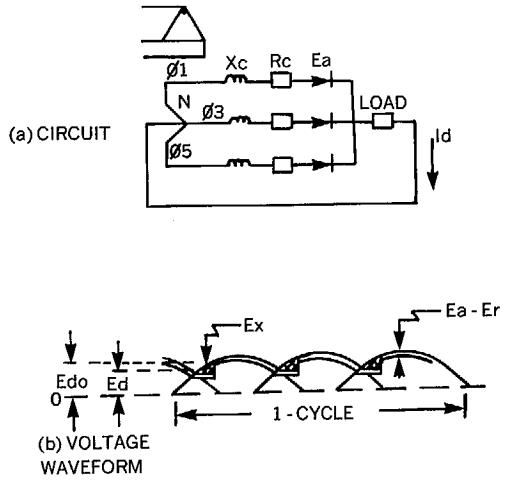


Fig. 32. Converter Voltage Waveform (Diode or Full Advance Operation)

CONVERTER PHASE CONTROL
AND COMMUTATION

FIGURE No. 33

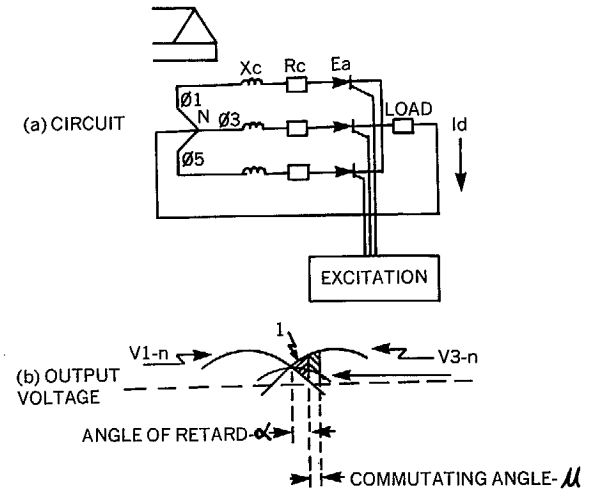


Fig. 33 Converter Phase Control and Commutation

The angle u is the commutating angle and the average value of the shaded portion of the curve is referred to as E_x. (Voltage regulation due to reactance is X_c). Arc drop and resistance drop in the circuit are self-explanatory. The resulting average voltage is the difference between the no-load DC average voltage E_{do} and the voltage drops discussed above or:

$$E_d = E_{do} - E_r - E_a - E_x$$

This is illustrated in Figure 32.

The SCR rectifying device has a property of blocking forward conduction until excited or "fired" by an external circuit. After being fired, this type of device continues to conduct until the rectifier in the next succeeding phase leg is fired, or until zero current condition is known as phase retard. Phase retard is expressed in degrees and is referred to as the angle a. The result of phase retard is the reduction of average DC voltage represented by the shaded area (1) in Figure 33. The resulting average voltage can now be represented by the equation below:

$$E_d = E_{do} \text{ Cos } a - E_r - E_a - E_x$$

From the previous discussion, it is observed that the three-phase bridge circuit produces six-phase ripple voltage on the DC bus.

To obtain a reversing power supply, two sets of non-reversing converters are connected back-to-back as shown in Figure 34 and by utilizing proper logic and excitation, the requirements for four quadrant operation are met. Quadrants two and four are referred to as regenerating and require inverting capability of the conversion equipment. Fundamental rectifying conditions can be extended to cover the fundamentals of the inverting operation.

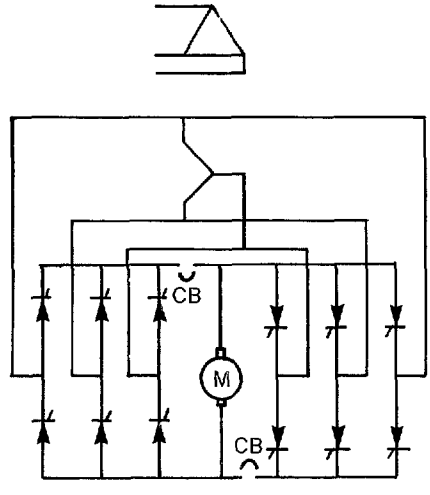


Fig. 34 Motor Drive System Using Back-to-Back Connected Three-Phase Bridge Converters

Figure 35 shows a commutation interval while inverting. The condition of transferring current from a phase of lower voltage to a phase of higher voltage still stands and the commutating angle u still applies. The equation defining average DC voltage is still:

$$E_d = E_{do} \text{ Cos } a - E_r - E_a - E_x .$$

As stated in this equation, the voltage is controlled by changing the firing retard angle a. This control is smooth and continuous with the polarity and magnitude of voltage being determined by the function Cos a. With a less than 90 degrees, Cos a is positive; with a greater than 90 degrees, Cos a is negative.

As stated previously, the maximum positive voltage is obtained with a equal to zero. The Cos a term also has a maximum negative value at a equal to 180 degrees. The limit condition as a approaches zero gives operation of a diode rectifier; the limit condition as a approaches 180 degrees gives shoot-through. With a shoot-through, the average DC voltage is established at zero. For a resistive-inductive load such as a motor field, this loss of voltage causes the current to decay on its natural time constant instead of being forced down. Any time that the proper firing angle is re-established, the forcing voltage is re-established.

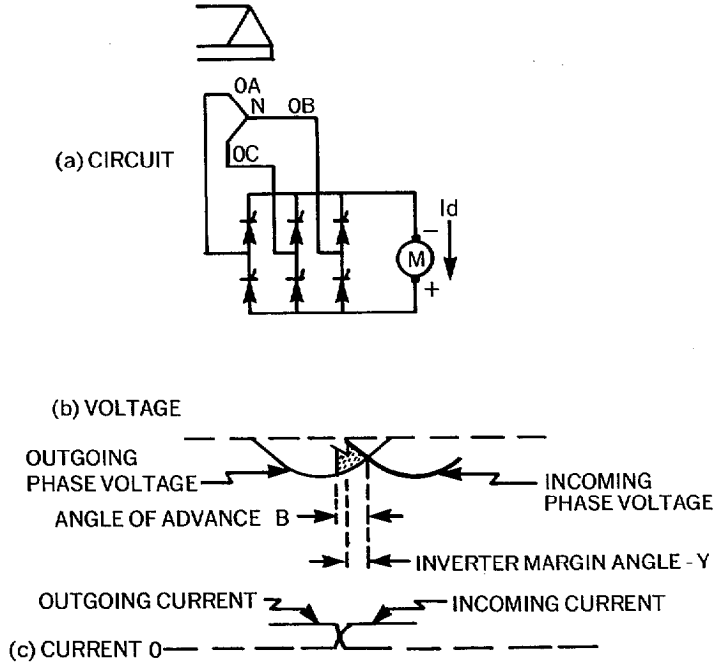


Fig. 35. Converter Phase Control and Commutation While Inverting

However, with a counter EMF load such as a motor armature, current builds up rapidly to fault magnitude and must be removed by opening a protective device such as a d-c breaker.

To focus attention on the limitation which must be observed in designing a phase controlled rectifier for inverter use, an angle of advance b is used where:

$$b = 180 - a.$$

Substituting for a in the voltage equation gives:

$$E_d = E_{do} \text{ Cos } b + E_r + E_a + E_x,$$

or recognizing separately that E_d is a negative number the equation may be written.

Referring again to Figure 35, it can be seen that a voltage margin must be designed into a power supply to insure that commutation can be completed with a margin angle remaining for the recovery of the blocking property of the SCR.

Figure 36 illustrates three-phase bridge rectifier waveform while inverting. The following waveforms are plotted:

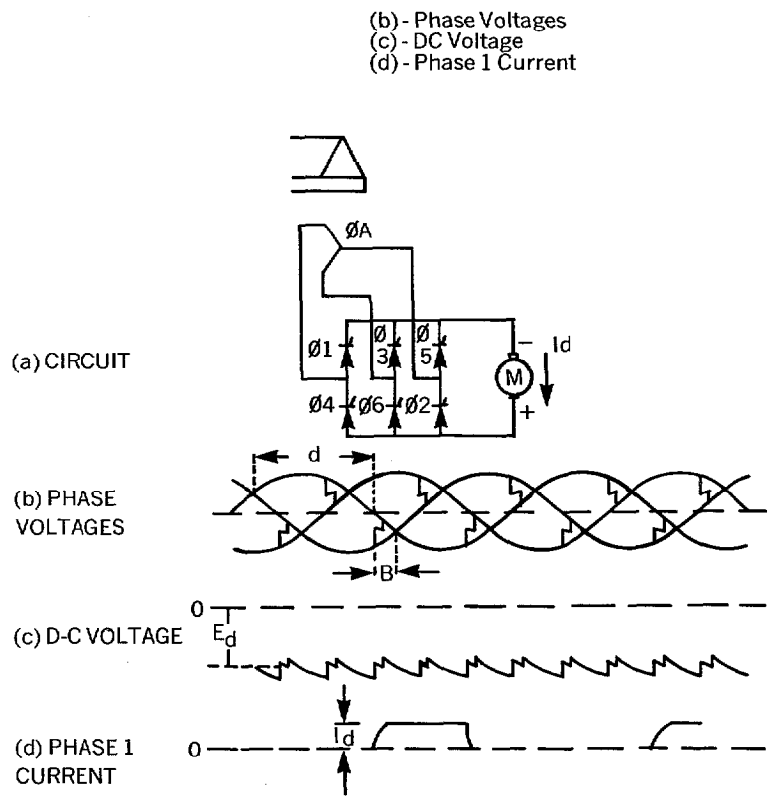


Fig. 36. Three Phase Bridge Converter Waveform While Inverting

The reversing power converter utilizes two non-reversing power converters back-to-back and thus permits four-quadrant operation. Figure 37 illustrates motor drive reversal in detail. Figure 37 (a) is a plot of power supply voltage, motor CEMF, and current versus time for a drive reversal. Forward operation is from the t_1 to t_2. At time t_2, the process control initiates a drive reversal and the power supply voltage starts to decrease. The load current also starts to decrease. At time t_3, power supply voltage and motor CEMF are equal resulting in zero current. From time t_1 to t_3 the motor was motoring as illustrated in Figure 37 (b) and quadrant number one of Figure 37 (e).

From time t_3 (current zero) to time t_4, the motor is regenerating. Assuming the process control has current limit at time t_4, the system is operating at the current limit setting. It will continue to do so until time t_6. From time t_3 to time t_5, the motor is regenerating as shown in Figure 37 (c) and is operating in quadrant number two, Figure 37 (e).

At time t_5, zero motor speed condition is reached; thus the motor has stopped. From time t_5 to t_6, the motor accelerates in current limit. At time t_6, the desired speed has been reached and the system comes out of current limit. This motoring condition is illustrated in Figure 37 (d) and 30 (e) quadrant number three.

In order to fully appreciate the back-to-back operation of the converter during this motor reversal, consider Figure 37 (f). Here is a block diagram of a full reversing Silcomatic power supply. The reference comes from the process control and is compared with the voltage feedback signal. The error signal is fed into the regulating amplifier, which consists of two amplifiers, and two bias circuits as shown. The output of this module is then fed to the gate pulse generator and from the gate pulse generator to the power module as shown.

The characteristics of this circuit are illustrated in Figure 37 (g), which is a plot of rectifier volts in percent of rated versus regulator amplifier output voltage and Figure 37 (h) which is a plot of a full reversing Silcomatic phase retard in degrees versus DC control signal in volts.

Going back to Figure 37 (a) as a reference and using Figure 37 (g) and 37 (b), the operation of the reversing Silcomatic converter can be explained. Points in time are shown on Figure 37 (g) and 37 (h). By utilizing the bias, logic and high-response voltage regulator, the back-to-back converter operation is achieved. At time $t_1 - t_2$, the forward converter is carrying current and the reverse converter is locked out. At time t_3, zero current, the lock-out circuit in the reverse rectifier permits it to phase on and accept current. The bias functions to prevent both forward and reverse converters firing at the same point in time. The high-response voltage regulator enables the rectifier to switch from the forward to the reverse units with minimum off time and therefore does not interfere with the normal mill operation.

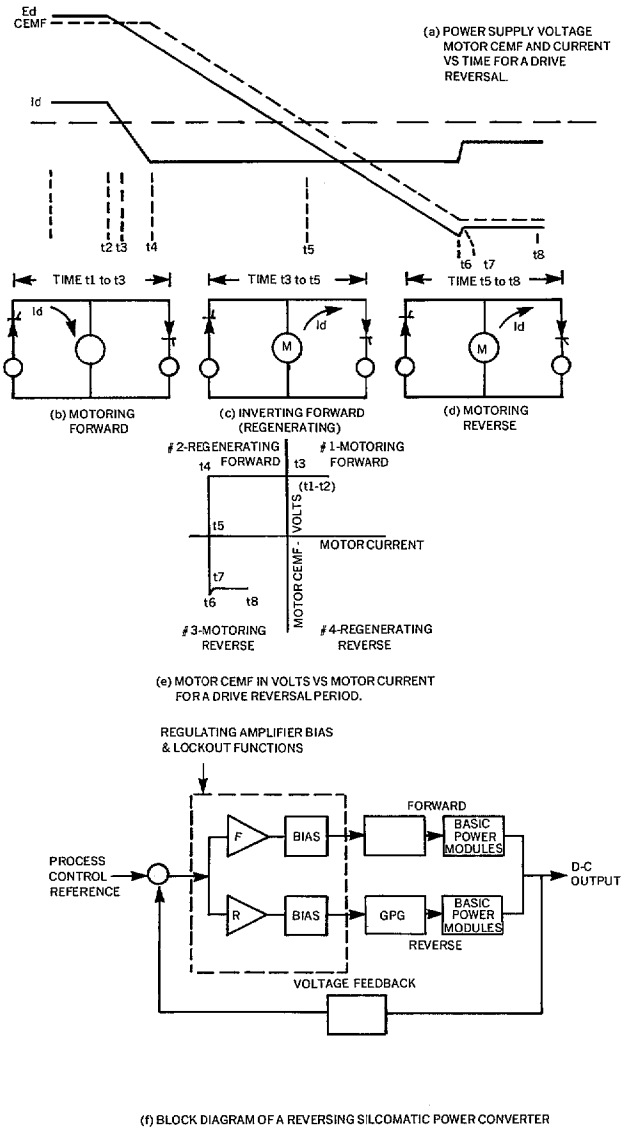
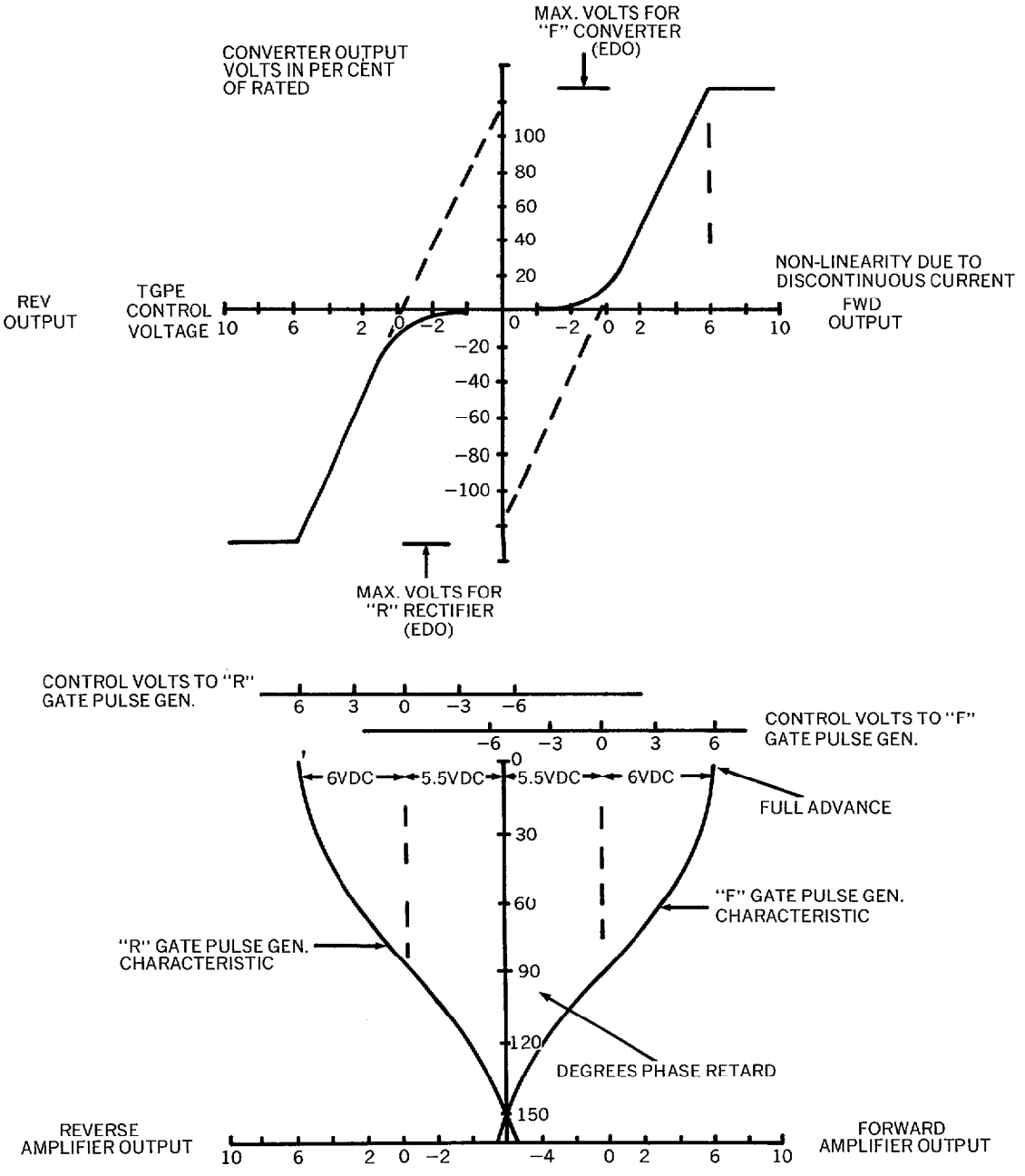


Fig. 37. Motor Drive Reversal

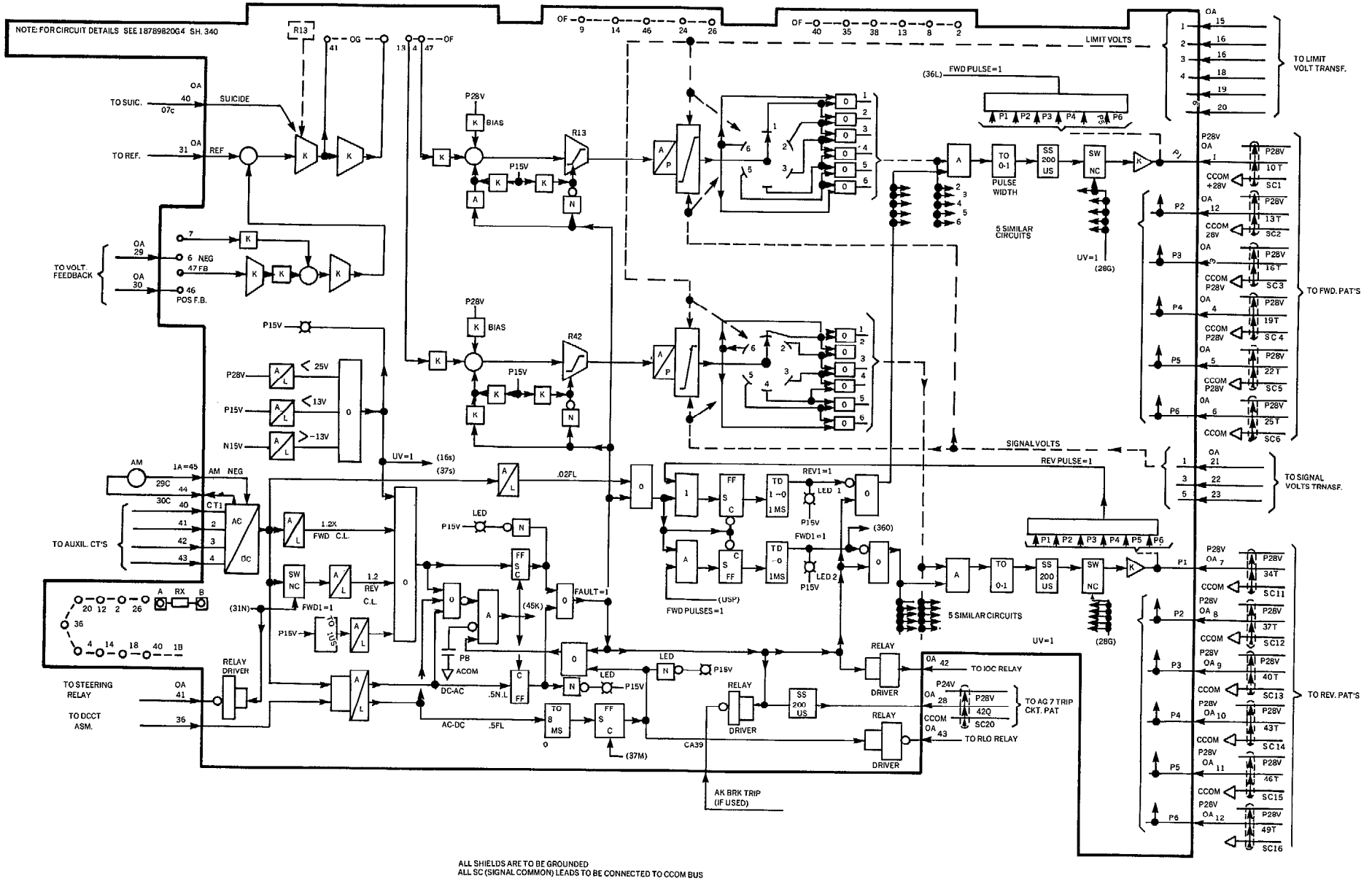
MOTOR DRIVE REVERSAL

FIGURE No. 37



CONVERTER AND GATE PULSE GENERATOR CHARACTERISTICS

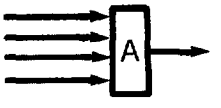
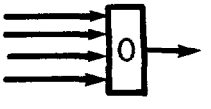
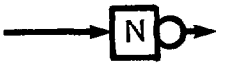
FIGURE No. 38

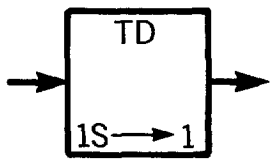
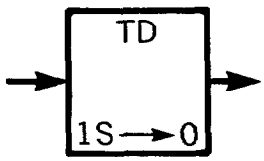
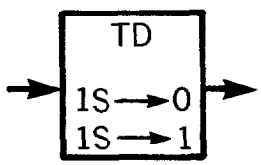

NOTE: FOR CIRCUIT DETAILS SEE 18789820G4 SH. 340
TO SUIC.
SUICIDE
TO REF.
REF
TO VOLT. FEEDBACK
NEG
POS F.B.
BIAS
R13
R42
P15V
P28V
25V
13V
-13V
N15V
UV=1
(16s)
(37s)
.02FL
AM
AM NEG
TO AUXIL. CT'S
1.2X
FWD C.L.
1.2
REV
C.L.
FWD1=1
(31N)
LED
(45K)
FAULT=1
PB
ACOM
DC-AC
AC-DC
.5N.L
.5FL
TO STEERING RELAY
TO DCCT ASM.
RELAY DRIVER
TO 8 MS
(37M)
CA39
AK BRK TRIP (IF USED)
LIMIT VOLTS
FWD PULSE=1
(36L)
PULSE WIDTH
5 SIMILAR CIRCUITS
UV=1
(28G)
TO LIMIT VOLT TRANSF.
TO FWD. PAT'S
SIGNAL VOLTS
REV PULSE=1
REV1=1
FWD1=1
LED 1
LED 2
FWD PULSES=1
(USP)
(360)
TO IOC RELAY
TO AG 7 TRIP CKT. PAT
TO RLO RELAY
TO SIGNAL VOLTS TRNASF.
TO REV. PAT'S
ALL SHIELDS ARE TO BE GROUNDED
ALL SC (SIGNAL COMMON) LEADS TO BE CONNECTED TO CCOM BUS

EXPLANATION OF SYMBOLS
As Used In Firing Circuit Diagram, Page 77

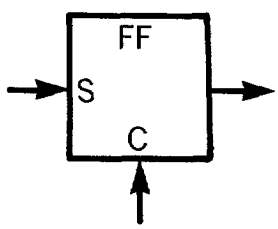
To fully understand the operation of the Silcomatic Plus equipment, a complete understanding of the firing circuit is essential. It shows the symbol diagram for a typical 6-pulse reversing main drive with a high speed DC breaker. The diagram describes functionally how the firing circuit works. It is NOT a circuit diagram!

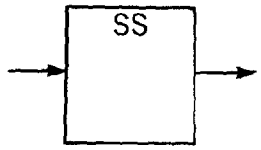
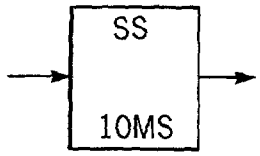
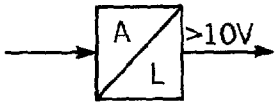
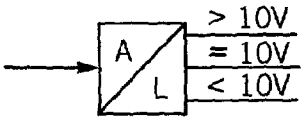
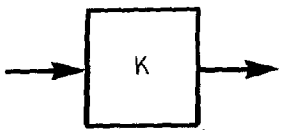
In order to read the symbol diagram it is important that the reader understands how to interpret the basic logic and analog symbols used.

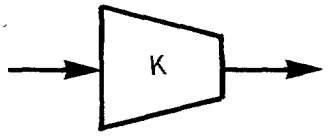
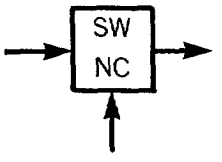
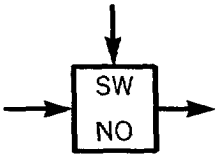
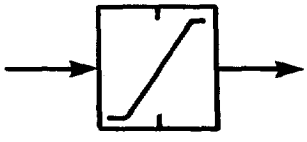
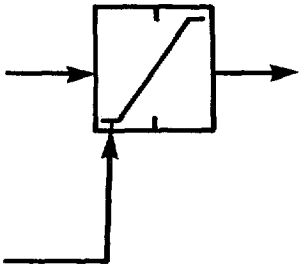
SYMBOL	DESCRIPTION
BASIC LOGIC ELEMENTS	
1. AND 	This symbol represents a logic "AND" element. The symbol, as shown, is to be interpreted as follows: "1's" are required on all four (4) inputs to obtain a "1" output. Conversely, a "0" on one or more inputs will result in a "0" output
2. OR 	This symbol represents a logic "OR" element. The symbol as shown is to be interpreted as follows: A single "1" at any one of the four (4) inputs is all that is required for a "1" output. Of course, more than one "1" input will also result in a "1" output. Conversely, "0's" on all four (4) inputs will result in a "0" output.
3. NOT (Element)  OR 	This symbol represents a logic "NOT" element. In other words, this is a logic inversion element where the input and output are of opposite states. Although the two symbols perform identical functions, their true states are read differently. The first symbol reads "0" input gives a "1" output. The second reads "1" input gives a "0" output. The selection of which symbol to use is dependent upon which true state makes the diagram more readable.

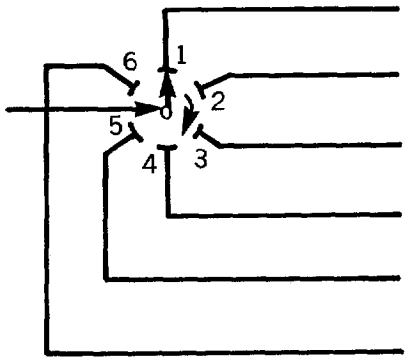
SYMBOL	DESCRIPTION
4. TIME DELAY	This symbol represents a logic "time delay" element. The first two (2) symbols, as shown, are to be interpreted as follows:
	One (1) second after application of a "1" input will result in a "1" output, as indicated by the expression, IS —— 1.
Other Examples	The time delay may be in the direction of an output "0", as indicated by the expression, IS —— 0, denoting a one (1) second time delay to "0" output.
	The following abbreviations of the units of time delay should be used:
	us = Microsecond ms = Millisecond s = Seconds m = Minutes h = Hours
	The numeric value and time unit is determined by its application.

Special Purpose Logic Functions: These are frequently used logic functions which, as a general rule, can be constructed from basic logic elements.

5. FLIP-FLOP	This symbol represents a logic "flip-flop" which is a logic device that has memory or storage properties. By definition, it is a two (2) input device. The S (or "set") input inserts a signal which is stored on output. The C (or "clear") input removes the stored signal output. Furthermore, a "clear" signal input will be preferential over a "set" signal input. The symbol, as shown, is to be interpreted as follows:
	A "1" on the S (or "set") input will cause a "1" on the output, which will remain "1" when the S input signal is returned to a "0". A "1" on the C (or "clear") input will cause the output to go to a "0", which will remain "0" when the C input signal is returned to a "0". Should a "1" appear simultaneously on both the S and C input terminals, then the output will remain a "0" since the C input always has preference over S input.

SYMBOL	DESCRIPTION
6. SINGLE SHOT	This symbol represents a logic single shot, a logic device that emits a pulse output. The symbol, as shown, is to be interpreted as follows:
	A "1" input will cause a "1" output pulse. Changing the input to "0" will have no affect on the output.
	For applications where pulse width is important, its duration should be specified as shown. Use the same units of time for TIME DELAY logic function.
7. ANALOG TO LOGIC CONVERTER	This represents a logic function that converts an analog input to a logic output in accordance with the expression shown above its output. The first symbol, as shown, is to be interpreted as follows:
	An analog signal input greater than 10 volts will cause a "1" output.
	The second symbol as shown is to be interpreted as follows:
	There will be three (3) logic outputs. One will be a "1" when input greater than 10V; another, a "1" when input equal to 10V; and another, a "1" when input less than 10V.
	The following signs should be used to express the relationship to a specified analog input: = Equal to. Not equal to. Less than. Greater than. = Less than or equal to. = Greater than or equal to.
8.	This represents a linear transfer function K in which there is no inversion between input and output.
	Examples: resistors, resistance bridges, non-inverting amplifiers, etc.

	SYMBOL	DESCRIPTION
9.		Same as above except that there IS an inversion between input and output.
10.	 	This represents a switch that gates the output signal with respect to its input signal. The input is on the left and the output is on the right. The signal for operating the switch may be shown along the upper or lower side of the box. The letters NC imply that the switch would be closed if the switch operating signal were removed. The letters NO imply that the switch is open under the same conditions. Examples: Interlocks, SCR's, transistors, etc.
11.	 	Represents a linear transfer function in which the output has positive and negative limits. This example shows that the negative limit is controlled by another signal.

	SYMBOL	DESCRIPTION
12.		This symbol represents the generation of 6 pulses which appear sequentially 1 through 6.

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